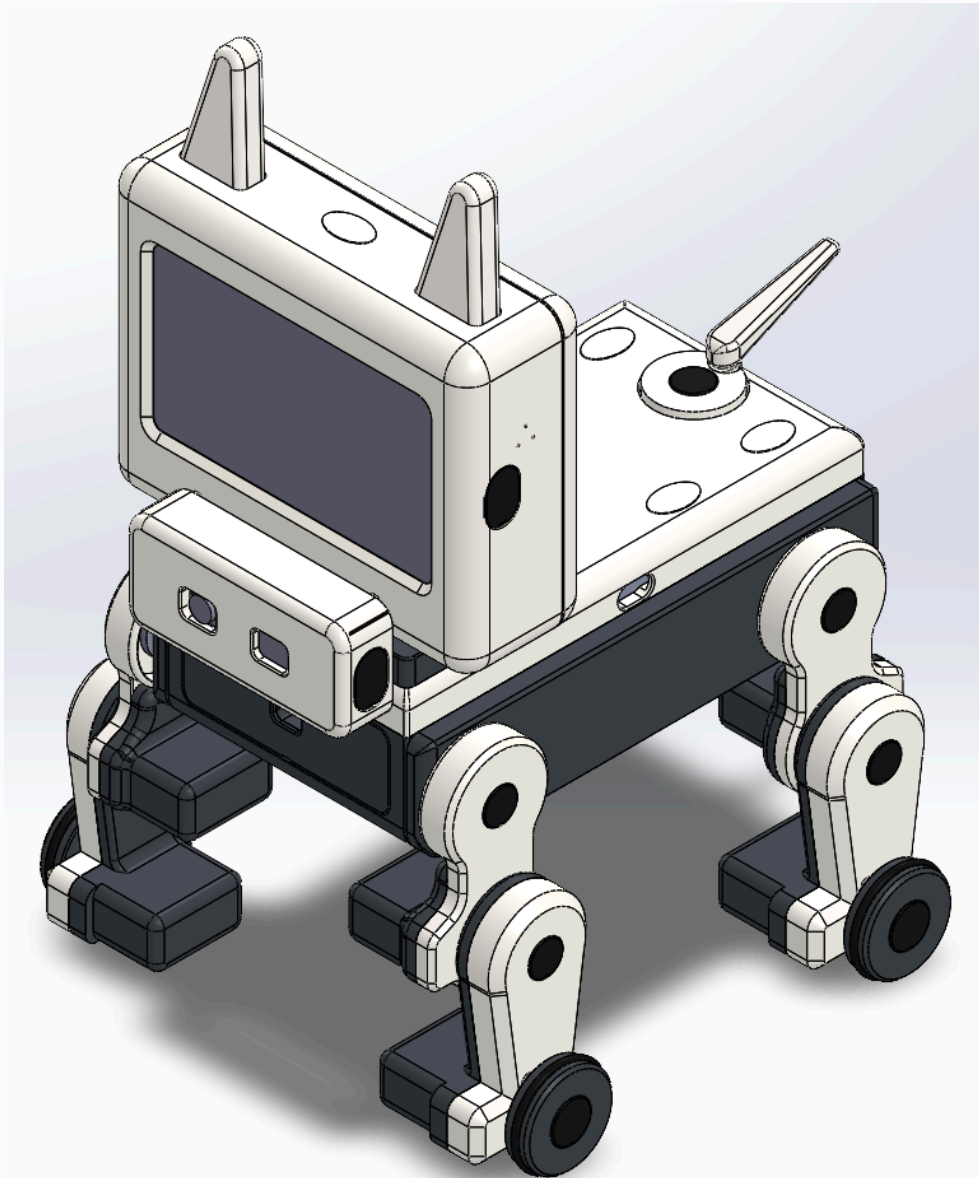


BUILD MANUAL

Desk Buddy

Assembly Guide



v3.1

October 2026 · first edition

A four-legged desk robot dog with wheels on its paws, a 4.3 inch face and a camera. This guide takes you from printed parts and a box of electronics to a robot that stands up.

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1 CHAPTER 1 Before you start

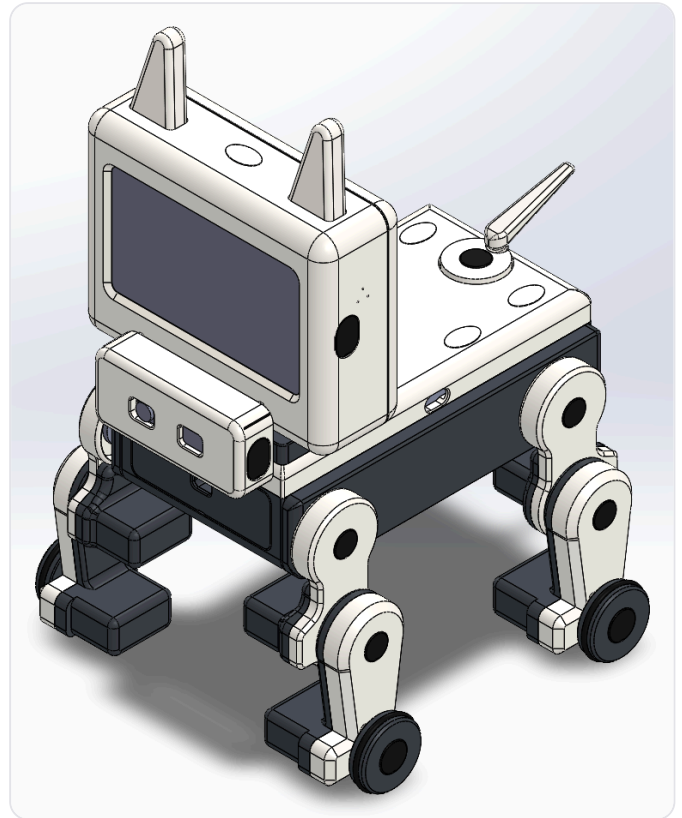
Read this chapter once, all the way through, before you touch a soldering iron. It lists the tools, the safety rules for the battery and the iron, and how every step in this guide is laid out.

1.1 What you are building

Desk Buddy is a small robot dog about 17 cm wide. Each of its four legs has a hip servo and a knee servo, and each paw has a wheel servo, so it can walk slowly or roll. A tail servo wags. Its face is a 4.3 inch touch screen, its eyes are a camera in the muzzle, and it hears you through two microphones.

You will build it in this order: print the parts, check every bought part, build the wiring harness, test everything on the desk, assemble the body, lid, head, legs, tail and ears, and finally switch it on and calibrate it.

Plan on roughly two days of printing at the maker space, an evening for the harness, an evening for the bench test and a weekend for the assembly. Nothing here is hard, but almost everything is small, so work slowly and in good light.



1.2 Tools

You already bought most of these. The ones marked **get** are cheap and worth having before you start: buy them at Jaycar or Bunnings.

Tool	Used for	Status
☐ ES02 soldering kit and digital multimeter	Every soldered joint; measuring voltages and checking for shorts	bought (EU plug: use the plug adapter)
☐ Solder, 0.7 to 0.8 mm, rosin core	Thin solder is easier on small pads	get if the kit has only thick solder
☐ Flux pen	Makes solder flow onto old or oxidised pads	get (Jaycar)
☐ Small Phillips screwdrivers, PH0 and PH1	All the M2 screws	at home
☐ Flush cutters (PLATO 170)	Trimming wires and cleaning prints	bought
☐ Wire stripper	About 40 wire ends; cutters nick the strands and nicked wires snap later	get
☐ 2.0 mm drill bit (used by hand, no drill)	Opening the servo horn holes so M2 screws pass	get
☐ Digital caliper	Measuring the servos before printing	bought
☐ Superglue, cable ties, foam tape, heat-shrink	Mics, foil pads, slack loops, boards, joints	bought

Tool	Used for	Status
☐ Breadboard and jumper wires	The bench test	bought
☐ Masking tape and a fine marker	A name flag on every lead	at home
☐ Laptop with PlatformIO, a USB-C data cable, a 5 V 2 A phone charger	Uploading the test sketch; charging	at home
☐ A mug or small box	Propping the robot up so its wheels are off the desk	at home

1.3 Safety

SOLDERING AND FUMES

The iron tip is about 350 degrees C. Always put it back in its stand, never on the bench. Work by an open window or with a small fan pulling the smoke away from your face, not towards it. Wash your hands after soldering and before you eat. Wear glasses when you trim wire ends, because they fly.

18650 CELLS

These cells hold a lot of energy. Never let anything metal touch both ends of a cell: no loose cells in a pocket or a bag with keys or coins. Never use a cell with a torn or dented wrapper. Check the + and - marks twice before a cell goes into the holder, every single time. Never mix the Samsung cells with the spare Tinker cells, and always change both cells together.

USE THE FUSE

The protection board only cuts off at about 20 A, which is enough to melt thin wire inside a plastic shell. The 7.5 A fuse is what protects the robot. Never bridge it, and never replace it with a bigger one.

POWER OFF BEFORE YOU TOUCH THE WIRING

Switch the robot off before you plug or unplug anything. For any work inside the tub, also take the cells out. Keep your fingers away from the legs while the robot is switched on: servos are strong enough to pinch.

HEAT

The printed parts are PLA, which softens at about 55 degrees C. Never leave the robot in a car or in direct sun behind a window.

1.4 How to read this guide

Every assembly step looks the same. Here is what each part of a step means.

A2 The step number and title. The letter is the section: A Body, B Lid, C Head, D Legs, E Tail, F Ears and G Close up.

The picture. The part you fit in this step has an **orange outline**. A bought part it holds, like a servo or a sensor board, has a **blue outline**. Everything already built is faded. Orange arrows show which way a part moves to go on.

You need. The parts and the screws for this step. Screws are colour-coded by length, the same way every time:

-  M2 x 4
-  M2 x 6
-  M2 x 8
-  horn screw

The text. Read it all before you pick anything up. Then do it in order.

LEFT, RIGHT, FRONT AND BACK

Left and right always mean the robot's own left and right, as if you were sitting on it facing forward. The front is the end with the head. So when you face the robot, its left side is on your right.

PART NAMES

Printed parts are named by their file, for example `Thigh_FrontLeft_Outer`. Bought parts use the names in the Parts check chapter.

COLOURS

Printed parts come in three colours:  **cream**,  **brown** and  **black** (the small caps and covers). Most pictures in this edition were rendered before the colours were chosen, so they show the parts in off-white and dark grey; the plainer views drawn straight from the print files already use the new colours. Print the off-white parts in cream and the dark grey parts in brown. The small caps and covers stay black.

Check tells you how to know the step went right. **Common mistake** is the thing people most often get wrong at that step.

SCREWS INTO PLASTIC

All screws are M2 self-tapping; they cut their own thread in the printed hole. Drive each screw in once before the part goes on, then take it out (this is called pre-threading). Tighten until the head touches, then a quarter turn more, and stop. Over-tightening strips the thread.

2 CHAPTER 2 Printing

The robot is 55 printed parts on nine print jobs, about 386 g of PLA and about \$43 at the Deer Park Library maker space. Print the small test plate first, check the fits with the real parts, and only then print the rest.

2.1 Measure one servo first

The printed parts are designed around a few servo measurements the sellers do not publish. When the servos arrive, measure one MG90S with the caliper and compare it with the table. If any number is off by more than about 0.2 mm, stop: the numbers go into the CAD script (`cad/build_print_parts_v3_1.py`), the parts are rebuilt, and you print from the new files.

Measure on the servo	Design value	Your value
Underside of the mounting tabs to the top of the rectangular gear case	4.5 mm	
Underside of the tabs to the top of the round boss the spline comes out of (boss diameter 11.8)	7.5 mm	
Case bottom to the tab underside	21.0 mm	
Tab underside to the spline top	14.4 mm	
Tab thickness	2.8 mm	
Distance between the two tab holes	28.15 mm	
Tab ear corners: round (R1.1 or more) or square? If square, file the two corners that face down to 1 x 45 degrees	round	
The three servo wires twisted into a round bundle: must pass a 2.6 mm hole	2.4 mm	
Cross horn from the servo bag: centre to tip / arm thickness / arm width / how far its top sits above the spline	11.5 / 1.8 / 6.6 / 1.0	
The tyre O-ring's thickness (the 31 mm OD ring from the assortment)	2.4 mm	

2.2 What to ask at the maker space

Book the UltiMaker 2+ Connect (0.4 mm nozzle, PLA) and take the STL files from `cad/print_v3_1_stl` on a USB stick. You can read this out to the staff:

THE REQUEST

"I'd like to print nine small PLA jobs for a robot, plus one test plate first. The settings are 0.2 mm layers, 2 walls, 0.8 mm top and bottom, 15 percent gyroid infill and no supports. Please use a 5 mm brim on the two long flat plates, G1 and W1, and a skirt on the rest. The colours are cream, brown and black PLA. Every plate is under 60 grams and fits the bed; Cura's Arrange All lays each one out. Could you tell me which filaments you have in those colours, and whether the 60 gram price band applies per job?"

Slicer settings (UltiMaker Cura)

Layer height	0.2 mm
Line width	0.4 mm
Wall line count	2 (0.8 mm)
Top / bottom thickness	0.8 mm (4 layers)
Infill	15 %, gyroid
Supports	none (see note A)
Adhesion	brim 5 mm on G1 and W1, skirt on the rest

Why 2 walls: the hidden walls are 1.6 mm, exactly four lines of a 0.4 mm nozzle, so two walls per side print them solid. Three walls

NOTE A - ROUND EDGES ON THE BED

The lid, the head front and the head back print with their visible rounded face on the bed, so the first 1 to 2 mm of that edge grows outward over nothing. PLA usually prints this with a slightly soft first millimetre. If the first lid shows a drooping edge, reprint that plate with "Support: touching build plate" at 60 degrees, and sand the edge lightly.

would add about 37 g and take the robot over its 400 g print budget.

ORDER OF PRINTING

Print T1 first and test it (2.3). If a fit is wrong, the design is corrected and T1 is reprinted before anything else. Then print B1 (cheap, and the caps are needed early), then the body plates, then the legs.

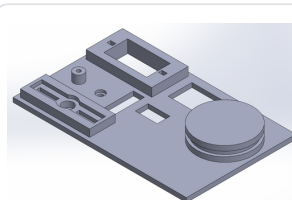
2.3 The test plate, T1

T1 costs about \$4 and checks every fit that matters with the real parts, before you spend \$43 on the robot. Test each coupon like this, and write down anything that does not fit.

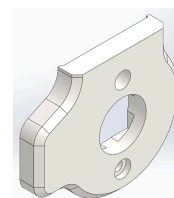
Fit_Test_Coupon. Push an MG90S into the servo slot: it should slide in with light finger pressure and not rattle. Clip the power switch into its cut-out. Press a screen screw into the boss. Hold a laser board behind the windows. Stretch a 26 mm O-ring into the groove. Lay a cross horn in the horn pocket.

Coupon_Knee_Thigh and Coupon_Knee_Shin. Screw a real MG90S to the thigh coupon with two M2x6. Fit a horn into the shin coupon, press it onto the spline, and press a 12 mm cap into its seat. Thread a servo lead through the lead chamber and swing the joint back and forth 500 times: the lead must not rub through or jam.

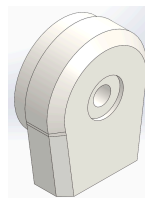
Coupon_Snap_White and Coupon_Snap_Graphite. Click the hook into the catch: it should click in firmly and release with a thumbnail. Press a 12 mm and a 14 mm cap into the two seats: snug, and removable with a fingernail in the notch.



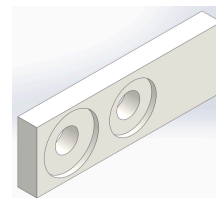
Fit_Test_Coupon
servo slot, power-switch cut-out, screen boss, sensor windows, O-ring groove, horn pocket



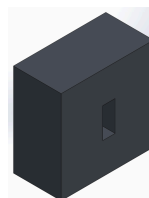
Coupon_Knee_Thigh
knee servo tabs and case (screw a real MG90S to it)



Coupon_Knee_Shin
knee cross horn, cap and lead chamber



Coupon_Snap_White
one snap hook, a 12 mm and a 14 mm cap seat



Coupon_Snap_Graphite
the snap catch for the hook

2.4 The print jobs

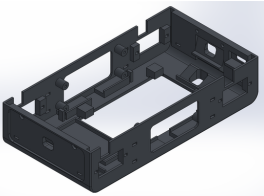
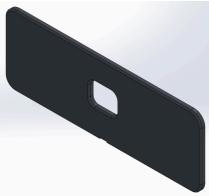
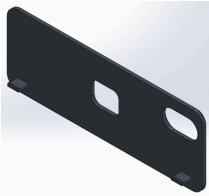
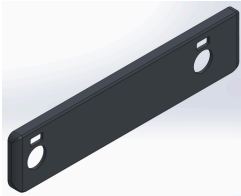
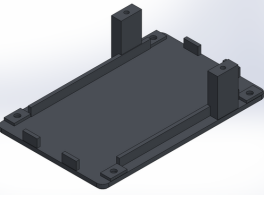
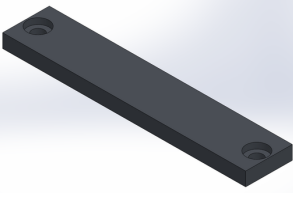
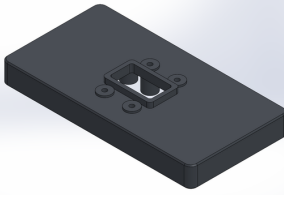
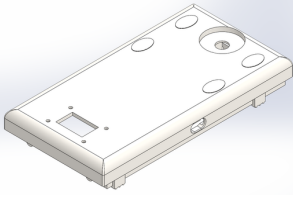
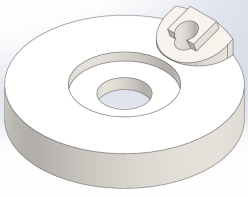
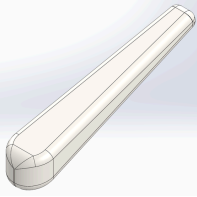
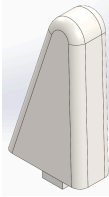
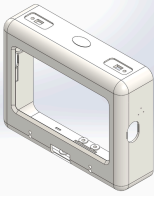
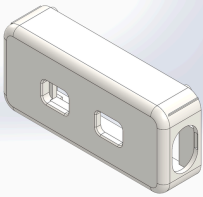
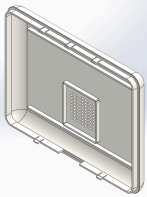
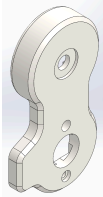
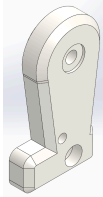
One filament colour per plate. Both price columns are shown: the maker space's price bands (\$1 up to 10 g, \$2 up to 20 g, \$4 up to 40 g, \$6 up to 60 g) and the \$1.10 per 10 g rate. Skirts and brims add a gram or two, so G1, W1 and W4 may land just over 60 g.

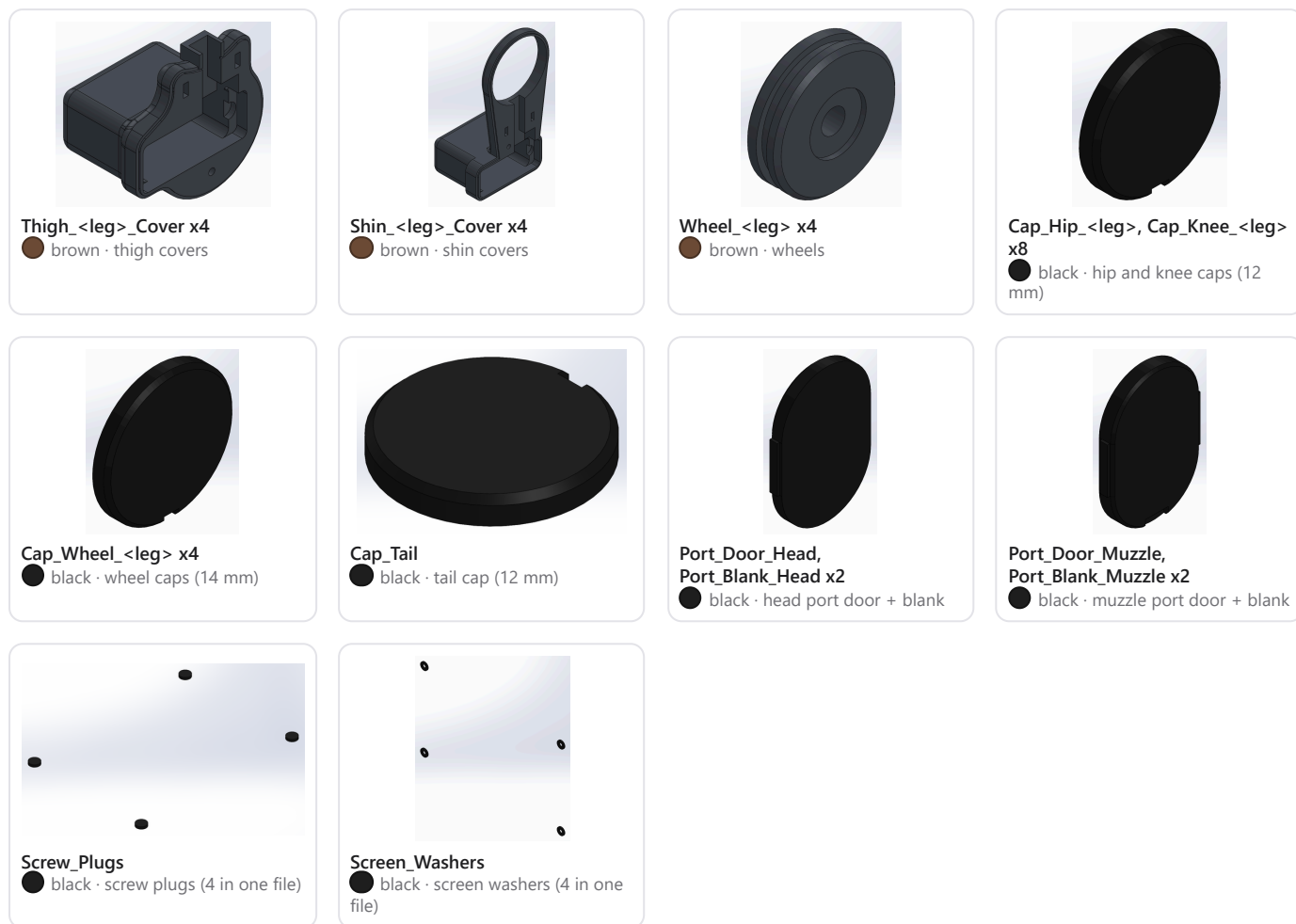
Order	Plate	Filament	Parts	Grams	Band price	\$1.10 / 10 g
1	T1	any	Fit_Test_Coupon, both knee coupons, both snap coupons, 1 x 12 mm + 1 x 14 mm cap	26.5	\$4.00	\$2.92
2	B1	● black	13 caps, 4 port doors and blanks, screw plugs, screen washers	4.6	\$1.00	\$0.51
3	G1	● brown	tub + 2 bands	56.6	\$6.00	\$6.23
4	W1	● cream	lid + 2 ears + tail root + tail	53.7	\$6.00	\$5.91
5	G2	● brown	2 side panels + battery door + clamp + collar	54.6	\$6.00	\$6.01
6	W2	● cream	head front + muzzle	47.2	\$6.00	\$5.19

Order	Plate	Filament	Parts	Grams	Band price	\$1.10 / 10 g
7	W3	cream	head back + 1 thigh outer	38.3	\$4.00	\$4.21
8	W4	cream	4 shin outers + 3 thigh outers	57.9	\$6.00	\$6.37
9	G3	brown	4 thigh covers + 3 wheels	37.5	\$4.00	\$4.12
10	G4	brown	4 shin covers + 1 wheel	35.3	\$4.00	\$3.88
Robot, plates 2-10				385.7	\$43.00	\$42.43
With the test plate				412.2	\$47.00	\$45.35

2.5 Every printed part

The 26 kinds of printed part, with their colour and what each one holds. Parts marked x4 are one per leg. The front-left and back-right thighs are the same shape turned round, and so are the front-right and back-left. The two left shins match, the two right shins match, and all four wheels match.

 Torso_Bottom brown · the tub	 Band_Chest brown · chest band	 Band_Rump brown · rump band	 Side_Panel_Left / _Right x2 brown · side panels
 Battery_Door brown · battery door	 Battery_Clamp brown · battery clamp	 Collar brown · collar (neck)	 Torso_Lid cream · the lid
 Tail_Root cream · tail root	 Tail cream · tail	 Ear_Left / Ear_Right x2 cream · ears	 Head_Front cream · head front
 Muzzle cream · muzzle	 Head_Back cream · head back	 Thigh_<leg>_Outer x4 cream · thigh outer halves	 Shin_<leg>_Outer x4 cream · shin outer halves



2.6 Printed parts checklist

Tick each part as it comes off the printer and passes a quick look: no drooping edges, no strings in the slots, every snap hook whole.

T1 test prints, any colour

- ☐ Fit_Test_Coupon
- ☐ Coupon_Knee_Thigh
- ☐ Coupon_Knee_Shin
- ☐ Coupon_Snap_White
- ☐ Coupon_Snap_Graphite

B1 ● black

- ☐ Cap_Hip_FrontLeft
- ☐ Cap_Hip_FrontRight
- ☐ Cap_Hip_BackLeft
- ☐ Cap_Hip_BackRight
- ☐ Cap_Knee_FrontLeft
- ☐ Cap_Knee_FrontRight
- ☐ Cap_Knee_BackLeft
- ☐ Cap_Knee_BackRight
- ☐ Cap_Wheel_FrontLeft
- ☐ Cap_Wheel_FrontRight
- ☐ Cap_Wheel_BackLeft
- ☐ Cap_Wheel_BackRight
- ☐ Cap_Tail
- ☐ Port_Door_Head
- ☐ Port_Blank_Head
- ☐ Port_Door_Muzzle
- ☐ Port_Blank_Muzzle
- ☐ Screw_Plugs (4)
- ☐ Screen_Washers (4)

G1 ● brown

- ☐ Torso_Bottom
- ☐ Band_Chest
- ☐ Band_Rump

W1 cream

- ☐ Torso_Lid
- ☐ Ear_Left
- ☐ Ear_Right
- ☐ Tail_Root
- ☐ Tail

G2 brown

- ☐ Side_Panel_Left
- ☐ Side_Panel_Right
- ☐ Battery_Door
- ☐ Battery_Clamp
- ☐ Collar

W2 cream

- ☐ Head_Front
- ☐ Muzzle

W3 cream

- ☐ Head_Back
- ☐ Thigh_FrontLeft_Outer

W4 cream

- ☐ Thigh_FrontRight_Outer
- ☐ Thigh_BackLeft_Outer
- ☐ Thigh_BackRight_Outer
- ☐ Shin_FrontLeft_Outer
- ☐ Shin_FrontRight_Outer
- ☐ Shin_BackLeft_Outer
- ☐ Shin_BackRight_Outer

G3 brown

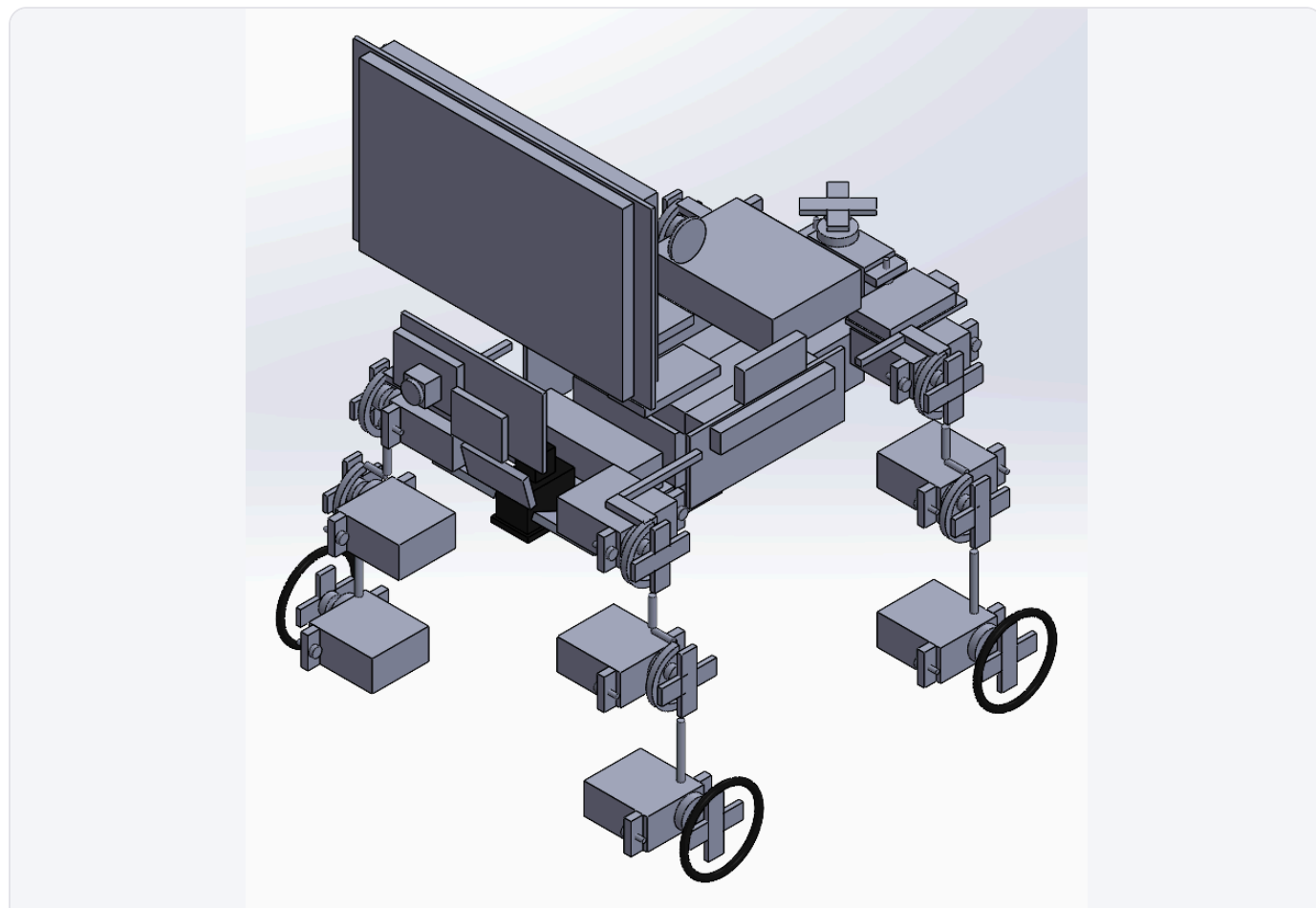
- ☐ Thigh_FrontLeft_Cover
- ☐ Thigh_FrontRight_Cover
- ☐ Thigh_BackLeft_Cover
- ☐ Thigh_BackRight_Cover
- ☐ Wheel_FrontLeft
- ☐ Wheel_FrontRight
- ☐ Wheel_BackLeft

G4 brown

- ☐ Shin_FrontLeft_Cover
- ☐ Shin_FrontRight_Cover
- ☐ Shin_BackLeft_Cover
- ☐ Shin_BackRight_Cover
- ☐ Wheel_BackRight

3 CHAPTER 3 Parts check

Unpack every bag the day it arrives, tick it off here, and test the electronics that same week (chapter 5). If something is missing or dead, AliExpress disputes are much easier while the order is new.



Every bought part in its place inside the robot (the printed shell hidden).

3.1 Bought parts checklist

The quantity is what you should have. "Where it goes" tells you where the part ends up in the robot. The store links are in the Appendix.

Part	Qty	Where it goes	Store and listing
☐ Guition ESP32-S3 4.3" touchscreen board (JC4827W543C)	1	Head front, behind the screen window	AliExpress
☐ ESP32-S3 N16R8 camera board + OV5640 camera	1	Muzzle (lens in the left window)	AliExpress
☐ ESP32-S3-DevKitC-1 N16R8	1	Not fitted - bench spare	AliExpress
☐ MG90S metal-gear servo, 180 degrees	10	4 hips, 4 knees, tail, 1 spare	AliExpress
☐ MG90S metal-gear servo, 360 degrees (continuous)	4	Wheels, one in each paw	AliExpress
☐ PCA9685 16-channel servo board	2	Tub, one on each side wall	AliExpress
☐ VL53L0X laser distance sensor	8	7 fitted (chest, rump, 3 desk-edge in the tub floor, 2 sides in the lid) + 1 spare	AliExpress

Part	Qty	Where it goes	Store and listing
☐ MS3625 I2S microphone	2	Mic rings on each side of the head front	AliExpress
☐ Cavity speaker 3 W 4 ohm (1.25 mm plug)	1	Head back, between its fences	AliExpress
☐ MPU6050 motion sensor (GY-521)	1	Under the lid, in its fence	AliExpress
☐ APDS-9960 gesture sensor	1	Muzzle, on its shelf behind the right window	AliExpress
☐ MPR121 touch board	1	Under the lid, in its fence	AliExpress
☐ Type-C 2S boost charging board	1	Tub, raised ledge at the rump; socket in the rump opening	AliExpress
☐ 2S 20 A protection board (with balance)	1	Tub floor, between its fences	AliExpress
☐ 2 x 18650 battery holder (series, wired)	1	Battery door	AliExpress
☐ Mini560 buck converter, fixed 5 V 5 A	2	On the perfboard hub under the lid	AliExpress
☐ KCD1 rocker switch	1	Front of the belly, flush in its recess	AliExpress
☐ Mini blade fuse holder, 7.5 A, 16 AWG	1	On top of the protection board	AliExpress
☐ M2 self-tapping screw kit (500)	1	Everywhere - see the screw chart	AliExpress
☐ Dupont jumper wires (120)	1	Signal wiring	AliExpress
☐ MX1.25 4-pin pre-wired plugs (10)	1	Plugs for the screen board ports P1, P3, P4	AliExpress
☐ Perfboard 5 x 7 cm (5)	1	The power and sensor hub under the lid	AliExpress
☐ 3M double-sided foam tape, 20 mm	1	Sensor boards and the hub	AliExpress
☐ Heat-shrink tubing (127)	1	Every soldered joint	AliExpress
☐ 830-point breadboard	1	Tool: bench test	AliExpress
☐ ES02 soldering kit + digital multimeter	1	Tool	AliExpress
☐ PLATO 170 flush cutters	1	Tool	AliExpress
☐ Digital caliper 150 mm	1	Tool: measure the servos before printing	AliExpress
☐ Samsung 35E 18650 cell, 3500 mAh, button top	2	In the battery holder	TinkerTech AU
☐ Tinker 25R 18650 cell, 2500 mAh (spare pair)	2	Spare pair - never mix with the Samsungs	TinkerTech AU
☐ Heavy-duty hook-up wire 7.5 A, red (WH3040)	2 m	Battery and servo power (+)	Jaycar
☐ Heavy-duty hook-up wire 7.5 A, black (WH3041)	2 m	Battery and servo power (-)	Jaycar
☐ 1000 uF 16 V capacitor (RE6220)	2	Hub: one across each Mini560 output	Jaycar
☐ 2.2 kOhm resistors (RR0580)	1 pack	Hub: wheel-stop pull-up on the OE line	Jaycar
☐ 10 kOhm resistors (RR0596)	1 pack	Hub: battery-level divider (top)	Jaycar
☐ 4.7 kOhm resistors (RR0588)	1 pack	Hub: battery-level divider (bottom)	Jaycar
☐ NBR O-ring assortment, OD 21-35 mm x 2.4 mm CS	1	Wheel tyres: use the 31 mm OD rings (about 26.2 mm inside), 4 + spares	AliExpress
☐ Superglue 3 g (6 pack)	1	Only where the guide says	Bunnings
☐ Cable ties 100 mm (25)	1	Hub anchors, slack loops	Bunnings

ALSO GET THESE BEFORE YOU BUILD

A flux pen, thin rosin-core solder (0.7 to 0.8 mm), a wire stripper and a 2.0 mm drill bit (see Tools, 1.2).

A USB-C data cable. Test yours: plug your phone into the laptop with it. If the phone shows up as a device, the cable can program the boards.

WHEEL TYRES

The tyres come from the AliExpress NBR O-ring assortment (outside diameter 21 to 35 mm, all 2.4 mm thick). Take out four rings of the **31 mm outside diameter** size, plus a spare: each is about 26.2 mm across the inside, which is what the wheel groove is designed for. Check the thickness with the caliper (2.4 mm), and try one on a wheel (or the groove on the Fit_Test_Coupon) before you fit all four.

3.2 Check the listings match

These look-alike parts cost less and are easy to get wrong. Check each one against the part in your hand.

- ☐ **Screen board:** the variant must be "C", which means capacitive touch (the GT911 chip). The "N" and "R" variants look the same.
- ☐ **2S charging board:** it must be the "boost" type with 5 V input. A look-alike "buck" board (TP5100 type) cannot charge from USB.
- ☐ **Battery holder:** the SERIES version, with one red and one black lead and the two cells end to end in series.
- ☐ **Microphones (MS3625):** each must have an I2S interface and an L/R pin.
- ☐ **Speaker:** the "1.25 terminal" variant, so it plugs straight into the screen board.
- ☐ **Servos:** ten 180 degree MG90S and four 360 degree MG90S. The 360 ones keep turning when you push the pulse off centre; flag them "360" the day they arrive, because they look the same.

3.3 Screw chart, actual size

PRINT THIS PAGE AT 100 %

If you print this page, turn off "fit to page". Check the ruler below measures exactly 50 mm with the caliper before you trust the outlines.

All the assembly screws are M2 self-tapping screws from the 500-piece kit. Lay a screw on an outline to find its length (measured under the head), and sort them into three cups before you start. The horn screws come in the servo bags; keep them in their bags.

M2 x 4

15 in the robot



lay the screw on the outline

M2 x 6

58 in the robot



lay the screw on the outline

M2 x 8

25 in the robot



lay the screw on the outline

Check ruler, 50 mm:



Screws per chapter

Count these out into a small cup for each chapter before you start it. If a cup is empty at the end of a chapter and nothing is left over, every screw went in.

Chapter	M2x4	M2x6	M2x8	What for
A Body	-	12	2	hip servos 8 x M2x6, servo boards 4 x M2x6, battery clamp 2 x M2x8
B Lid	4	2	4	tail servo 2 x M2x6, touch pads 4 x M2x4, collar 4 x M2x8
C Head	1	4	8	screen 4 x M2x6, head touch pad 1 x M2x4, muzzle 4 x M2x8, head to collar 4 x M2x8
D Legs (all four)	8	32	8	per leg: hip horn 2 x M2x4; knee servo, wheel servo, knee horn and wheel horn 2 x M2x6 each; 2 covers 1 x M2x8 each
E Tail	2	-	1	tail horn 2 x M2x4, tail to root 1 x M2x8
F Ears	-	-	2	one M2x8 per ear
G Close up	-	4	-	lid to tub 4 x M2x6
7 First power-on	-	4	-	battery door 4 x M2x6
Total	15	58	25	plus 13 horn screws from the servo bags

4

CHAPTER 4 Wiring

You build the whole wiring harness on the desk first: the battery pack, the power switch and fuse, and the hub with its two 5 V modules. The Bench test in chapter 5 then runs on this real harness, and the Assembly chapter only has to plug it in.

This chapter comes before the bench test because every test needs the battery and the 5 V modules. Read the ten rules first. They come from the parts audit, and each one prevents a way of destroying a board.

4.1 The ten wiring rules

RULE 1

Never use the screen board's BAT port. That port is for one 3.7 V cell, and the 7.4 V pack would destroy the board. Feed 5 V into port P1, using only its 5 V and GND wires. Check each wire against the printed labels with the multimeter first.

RULE 2

Sensors get 3.3 V only. Every sensor, and the logic pin (VCC) of each servo board, runs on 3.3 V. Only the green V+ screw terminal of the servo boards gets 5 V. Before you connect any module, measure its SDA and SCL pins: they must read 3.3 V or less.

RULE 3

Build the battery pack in this order. Take the cells out of the holder. Solder a short wire to the metal link between the two cell positions: this is the middle wire the protection board needs. Check that the two cells are within 0.05 V of each other. Connect negative first, then the middle wire, then positive. Plug the charger in once to wake the protection board up.

RULE 4

The fuse comes straight after the protection board. It sits before both the charger and the switch, so a short anywhere in the robot blows the fuse instead of cooking a wire.

RULE 5

Charge with the robot switched off. Use a 5 V, 2 A phone charger and a USB-A to USB-C cable. This charger cannot run the robot while it charges.

RULE 6

Unplug the 5 V leads before you plug in the laptop. Before you connect a USB cable to the screen board or the camera board, unplug that board's 5 V lead from the hub. Two power sources on one board can damage it or the laptop port.

RULE 7

Wheel stop. The servo boards' OE pin goes to IO14, with the 2.2 kOhm resistor to 3.3 V. Do not use 10 kOhm, and never use IO46. The firmware turns on the watchdog, which restarts the board if the code freezes; while it restarts, OE is pulled high and every wheel stops. Run the first tests with the robot propped on a block.

RULE 8

Battery limits in the code. The firmware warns at 7.0 V, and at 6.8 V it parks the robot and switches the servos off.

RULE 9

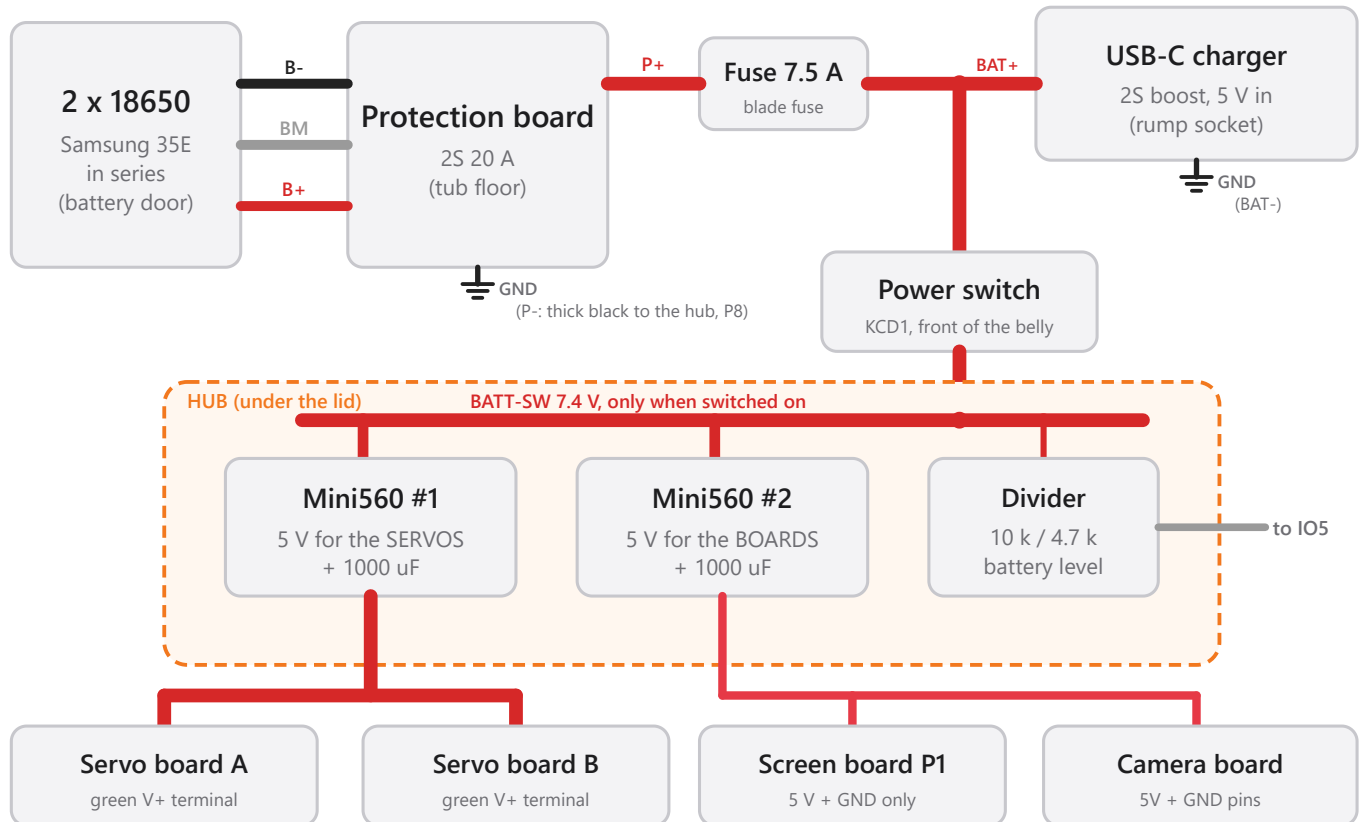
Speaker. It plugs into the "Speak" port. Never connect either speaker wire to ground.

RULE 10

Servo setup. Centre each leg servo before you fit its horn. Each wheel servo's stop point is found in software.

4.2 The power path

Red = + (thick where the servo current flows) Black = ground White = battery level



Every GND symbol is the same wire: the thick black ground that starts at the protection board's P- and meets at the hub.

Power flows one way. The two cells in series make about 7.4 V. The protection board stops over-charge, over-drain and shorts, and the 7.5 A fuse comes straight after it, before both the charger and the switch. The switch feeds the hub, where two Mini560 modules each make 5 V: module #1 only for the servos, module #2 only for the screen and camera boards. They are kept apart so the servos' current spikes cannot reset the brain. A 1000 uF capacitor across each output smooths the spikes. A 10 k and 4.7 k resistor pair lets the screen board read the battery level on IO5.

The charger connects after the fuse and before the switch, so it charges the cells whether the robot is on or off. Switch the robot off to charge.

4.3 Build the battery pack and power leads

CELLS OUT

Do all of this with the cells out of the holder. They go in for the first time in the bench test (5.3), and come out again before you assemble the robot.

- 1. Middle wire.** Inside the cell holder, find the metal link that joins the two cell positions. Put a touch of flux on it, and solder on a 10 cm thin wire. Be quick: the holder's plastic melts. This wire goes to the protection board's BM (middle) pad.
- 2. Holder to protection board.** Shorten the holder's red and black leads to about 12 cm: long enough for the battery door to come off and sit beside the robot while it is still wired. Solder black to B-, the middle wire to BM, and red to B+.
- 3. Fuse.** Solder one lead of the fuse holder to P+. Cut two thick red wires: 8 cm for the charger and 12 cm for the switch. Twist both onto the fuse holder's other lead, solder the joint, and cover it with heat-shrink.
- 4. Charger.** Solder the 8 cm red wire to the charger's BAT+ (sometimes marked OUT+). Solder an 8 cm thick black wire from P- to the charger's BAT-.
- 5. Switch.** Solder the 12 cm red wire to one switch terminal and a 25 cm thick red wire to the other; that one goes to the hub's BATT-SW pad. Solder a 25 cm thick black wire to P- as well; it goes to the hub's GND.

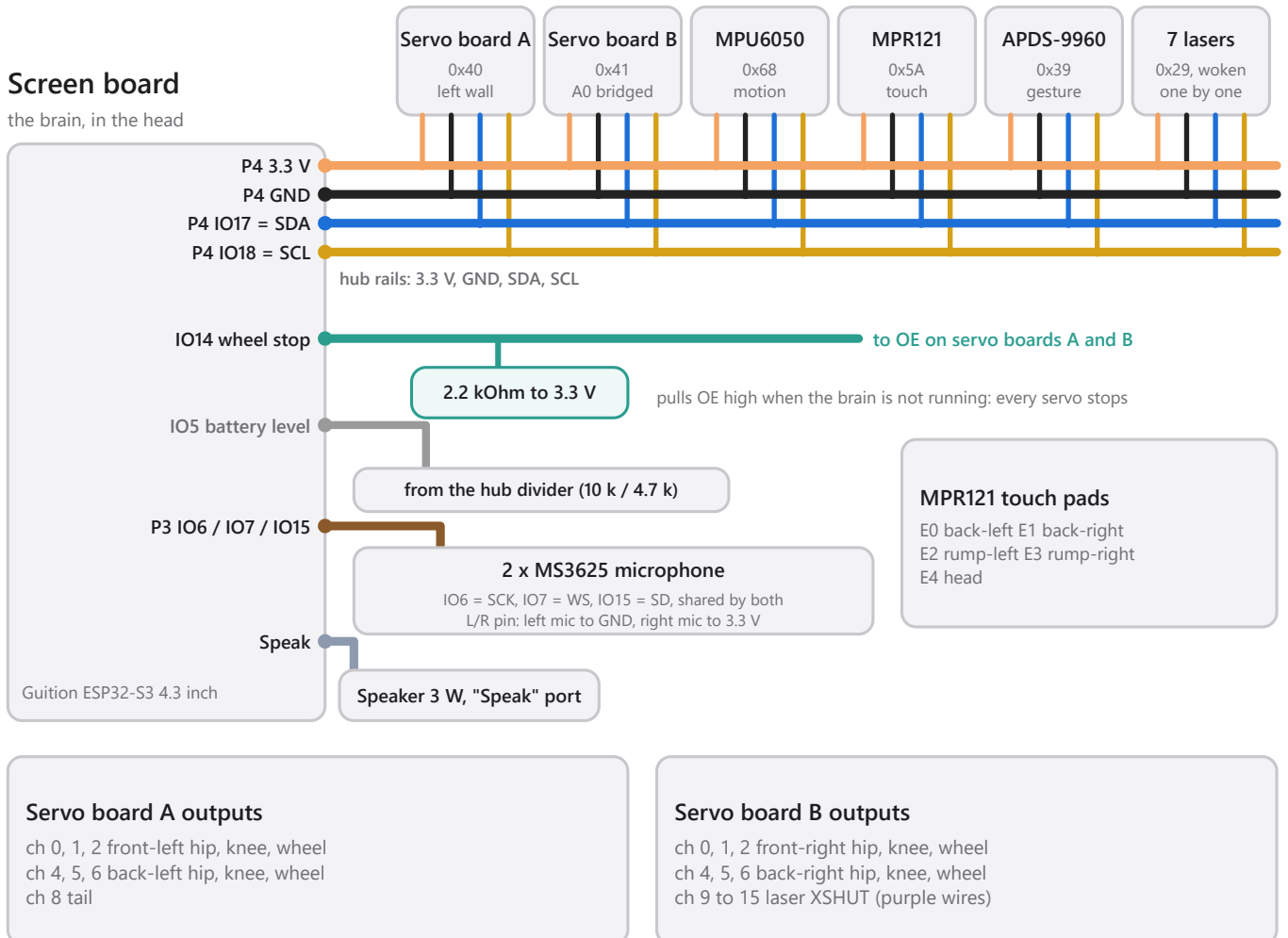
6. **Finish.** Slide heat-shrink over every joint before you solder it, then shrink it. Flag every loose end with tape: CHG+, CHG-, SW, HUB+, HUB-.

CHECK

With the multimeter on beeper mode, P+ to P- must NOT beep, and CHG+ to CHG- must NOT beep. The two switch wires beep only when the switch is ON.

4.4 The signals

One I2C bus from the screen board to every sensor and both servo boards, wired as a star at the hub.



Every sensor, and the VCC pin of each servo board: 3.3 V.

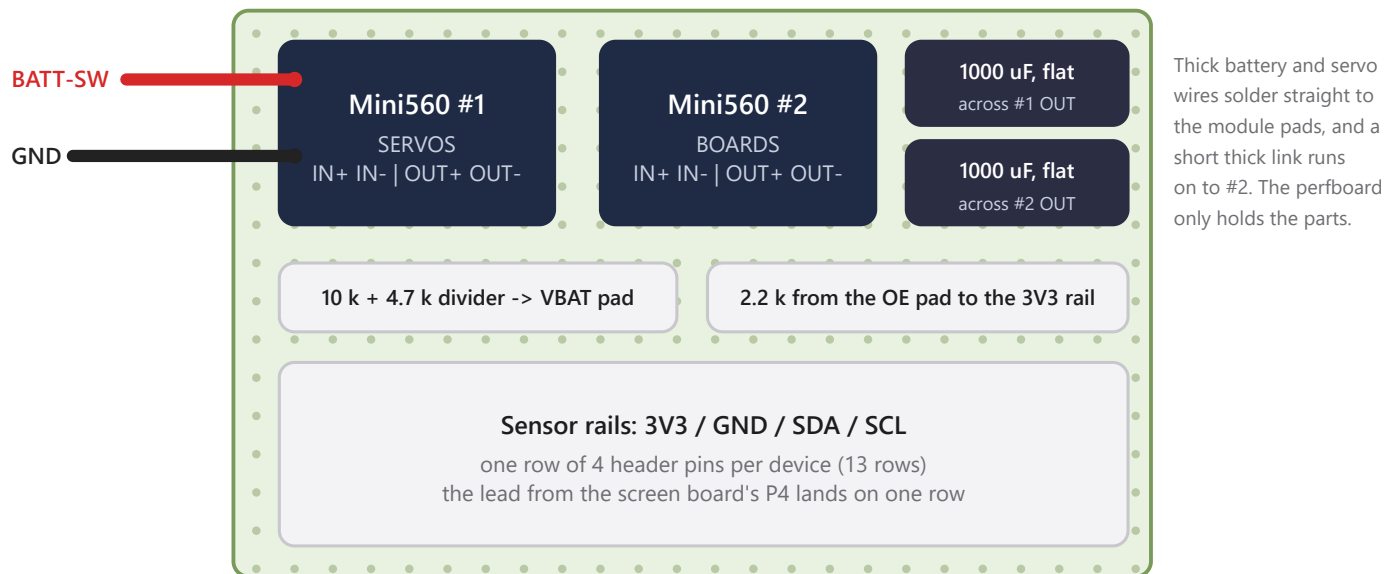
Only the servo boards' green V+ screw terminal gets 5 V.

The screen board is the brain. One I2C bus, on its port P4 (IO17 for data, IO18 for clock, plus 3.3 V and GND), goes down through the collar to the hub, and from there a short lead of four wires goes to every sensor and both servo boards. Each device has its own address, shown in the diagram. The seven lasers all start at the same address, 0x29; at start-up the firmware wakes them one at a time with their XSHUT wires (from servo board B) and gives each a new address.

IO14 is the wheel stop. It drives the OE pin of both servo boards, and a 2.2 kOhm resistor pulls it up to 3.3 V. While the brain runs, it holds IO14 low and the servos work. If the brain crashes or restarts, the resistor pulls OE high and every servo output stops, so the wheels stop.

The two microphones share three wires from port P3. The speaker plugs into the "Speak" port. The camera board only needs 5 V and GND: it talks to the laptop over Wi-Fi.

4.5 Build the hub



- 1. Board.** Cut a perfboard to 50 x 35 mm: score both sides along a row of holes with a knife and snap it over the edge of the table.
- 2. Modules.** Place the two Mini560 modules side by side, pads up. Solder the thick BATT-SW red wire to module #1's IN+ and run a short thick red link on to module #2's IN+. Do the same with the thick GND black wire to both IN- pads. The perfboard only holds the modules; the current runs in the thick wires.
- 3. Capacitors.** Lay each 1000 uF capacitor on its side, and solder it across one module's OUT+ and OUT-. The stripe on the capacitor marks its minus leg; it goes to OUT-.
- 4. Servo power.** From module #1's output, solder a 20 cm thick red and black pair for each servo board's green terminal. From module #2's output, make a red and black lead for the screen board (on an MX1.25 plug: use only its 5 V and GND wires, and cut and insulate the other two) and a Dupont red and black pair for the camera board.
- 5. Resistors.** Solder the 10 k from BATT-SW to a spare pad (VBAT), and the 4.7 k from VBAT to GND; add the white wire from VBAT, for IO5. Solder the 2.2 k from another pad (OE) to the 3V3 rail, and the green wires from OE: one to IO14, one to each servo board.
- 6. Sensor rails.** Run four bare wires along the rest of the board as rails: 3V3, GND, SDA and SCL. Solder a row of four header pins across the rails for each device (use the loose header strips that came with the sensor boards). The input from the screen board's P4 lead lands on one of these rows.

CHECK BEFORE IT EVER SEES POWER

Beeper mode: BATT-SW to GND must not beep. Each module's OUT+ to OUT- must not beep. The 3V3 rail to GND must not beep. SDA to SCL must not beep.

4.6 Servo boards: address and leads

Board B's address. Each servo board has six small solder pads marked A0 to A5. On the board that will go on the robot's right wall, bridge the A0 pad with a blob of solder across its two halves. That board is now "board B", address 0x41. Leave board A (left wall) as it came, address 0x40. Mark both boards with tape.

Leads. On the six-pin header at the side of each board, connect GND (black), OE (green), SCL (yellow), SDA (blue) and VCC (orange, 3.3 V). Leave the header's V+ pin empty: servo power goes only on the green screw terminal, V+ and GND.

VCC IS 3.3 V

VCC is the chip's own supply, 3.3 V from the hub rail. Putting 5 V on VCC makes the board's data pins 5 V and can damage the screen board.

4.7 Wire colours and lengths

Use one colour for one job everywhere. When a colour is missing from the jumper kit, use any colour and flag both ends with tape. Lengths are estimated from the 3D model, with slack for the lid to stand beside the tub while you work. Cut long, then trim; never cut a servo lead.

-  **red, thick (7.5 A)** - battery + after the fuse, switched battery, 5 V to the servo boards
-  **black, thick (7.5 A)** - battery - and power ground
-  **red (Dupont)** - 5 V to the screen board and camera board
-  **black (Dupont)** - every signal ground
-  **orange** - 3.3 V to the sensors and the servo boards' logic
-  **blue** - I2C data (SDA)
-  **yellow** - I2C clock (SCL)
-  **green** - wheel stop (servo boards' OE)
-  **white** - battery level (to IO5)
-  **purple** - laser on/off (XSHUT)
-  **grey** - touch pads
-  **brown** - microphone lines (mark each one with tape)

4.8 Every connection

Tick each row when it is soldered, and again when it is plugged in during Close up (step G1).

Power

	From	To	Wire	Length	Note
P1	Cell holder, black lead (-)	Protection board B-	holder's own lead	as supplied	long enough for the door to sit beside the robot
P2	Metal link between the two cells	Protection board BM (middle)	thin red or white, soldered	10 cm	the "middle wire" - solder it with the cells OUT
P3	Cell holder, red lead (+)	Protection board B+	holder's own lead	as supplied	connect last (negative, middle, positive)
P4	Protection board P+	Fuse holder, lead 1	red, thick	5 cm	7.5 A blade fuse fitted
P5	Fuse holder, lead 2	USB-C charger BAT+ (OUT+)	red, thick	8 cm	the fuse protects the charger too
P6	Fuse holder, lead 2 (same joint as P5)	Power switch (KCD1) terminal 1	red, thick	12 cm	
P7	Protection board P-	USB-C charger BAT- (OUT-)	black, thick	8 cm	
P8	Protection board P-	Hub GND pad	black, thick	25 cm	slack so the lid lifts off
P9	Power switch (KCD1) terminal 2	Hub BATT-SW pad	red, thick	25 cm	slack so the lid lifts off
P10	Hub BATT-SW / GND	Mini560 #1 (servos) IN+ / IN-, Mini560 #2 (boards) IN+ / IN-	red / black, thick	on the hub	7.4 V in
P11	Mini560 #1 OUT+ / OUT-	PCA9685 A and B, green V+ / GND screw terminal	red / black, thick	20 cm each	5 V servo power only
P12	1000 uF capacitor #1	Across Mini560 #1 OUT (+ leg to OUT+)	capacitor legs	on the hub	the stripe marks the - leg
P13	Mini560 #2 OUT+ / OUT-	Screen board port P1: 5 V / GND only	red / black (MX1.25 plug)	30 cm	through the collar; P1 only, never BAT
P14	Mini560 #2 OUT+ / OUT-	Camera board 5V / GND pins	red / black (Dupont)	30 cm	through the collar
P15	1000 uF capacitor #2	Across Mini560 #2 OUT (+ leg to OUT+)	capacitor legs	on the hub	

Signals

	From	To	Wire	Length	Note
S1	Screen board port P4: 3.3 V, GND, IO17, IO18	Hub 3V3, GND, SDA, SCL rails	orange, black, blue, yellow (MX1.25 plug)	30 cm	through the collar; IO17 = SDA, IO18 = SCL (the servo sketch checks)
S2	Hub 3V3 / GND / SDA / SCL	PCA9685 A and B: VCC, GND, SDA, SCL	orange, black, blue, yellow	20 cm each	VCC is 3.3 V logic - NOT 5 V
S3	Hub 3V3 / GND / SDA / SCL	MPU6050 (VCC, GND, SDA, SCL)	orange, black, blue, yellow	10 cm	under the lid
S4	Hub 3V3 / GND / SDA / SCL	MPR121 (3.3V, GND, SDA, SCL)	orange, black, blue, yellow	10 cm	under the lid
S5	Hub 3V3 / GND / SDA / SCL	7 laser sensors (VIN, GND, SDA, SCL)	orange, black, blue, yellow	10-25 cm	2 in the lid, 5 in the tub
S6	Hub 3V3 / GND / SDA / SCL	APDS-9960 gesture sensor (VCC, GND, SDA, SCL)	orange, black, blue, yellow	30 cm	through the collar to the muzzle
S7	Screen board IO14	Hub OE pad -> PCA9685 A OE and B OE	green	30 cm	2.2 kOhm from the OE pad to 3V3 (wheel stop)
S8	Hub BATT-SW -> 10 kOhm -> VBAT pad -> 4.7 kOhm -> GND	Screen board IO5 from the VBAT pad	white	30 cm	reads the battery; after the switch, so it never drains the pack when off
S9	PCA9685 B channels 9-15 (PWM pin)	XSHUT pin of each laser (see the laser table)	purple	10-25 cm	turns the lasers on one at a time at boot
S10	MPR121 E0, E1, E2, E3, E4	Foil pads: back-left, back-right, rump-left, rump-right, head	grey	5-30 cm	the wire is clamped under the pad screw
S11	Screen board port P3: IO6, IO7, IO15	Both mics: SCK (IO6), WS (IO7), SD (IO15); 3.3 V and GND	brown (MX1.25 plug)	10 cm	both mics share all three lines
S12	Left mic L/R pin	GND (right mic L/R pin to 3.3 V)	black / orange	3 cm	tells the two mics apart
S13	Speaker (1.25 mm plug)	Screen board "Speak" port	speaker's own lead	as supplied	never connect a speaker wire to ground
S14	13 servo leads	PCA9685 headers (see the servo table); brown wire on the GND row	servo's own lead	do not cut	wheel lead needs about 200 mm, knee lead about 135 mm

4.9 Servo and laser channels

Each servo plugs into the servo board on its own side of the robot, so no lead has to cross the tub. Board A is on the left wall and board B on the right wall. The laser on/off (XSHUT) wires all go to board B. The same map is written into the servo test sketch, and the robot firmware must use it too. Every pin and channel is also listed in one table in the Appendix (9.4).

	Servo	Servo board	Channel	Lead route
☐	Hip, front-left	A (left wall, 0x40)	0	straight through the left tub wall
☐	Knee, front-left	A	1	front-left leg, through the side-panel slot
☐	Wheel, front-left (360)	A	2	front-left leg, through the side-panel slot
☐	Hip, back-left	A	4	straight through the left tub wall
☐	Knee, back-left	A	5	back-left leg, through the side-panel slot
☐	Wheel, back-left (360)	A	6	back-left leg, through the side-panel slot
☐	Tail	A	8	down from the lid
☐	Hip, front-right	B (right wall, 0x41)	0	straight through the right tub wall
☐	Knee, front-right	B	1	front-right leg, through the side-panel slot

Servo	Servo board	Channel	Lead route
☐ Wheel, front-right (360)	B	2	front-right leg, through the side-panel slot
☐ Hip, back-right	B	4	straight through the right tub wall
☐ Knee, back-right	B	5	back-right leg, through the side-panel slot
☐ Wheel, back-right (360)	B	6	back-right leg, through the side-panel slot

Laser	Where	XSHUT to board B channel
☐ Front obstacle	behind the chest band window	9
☐ Rear obstacle	behind the rump band window	10
☐ Desk edge, front-left	tub floor incline, left of the switch	11
☐ Desk edge, front-right	tub floor incline, right of the switch	12
☐ Desk edge, rear	tub floor incline at the rump, centred	13
☐ Side, left	shelf under the lid, left wall	14
☐ Side, right	shelf under the lid, right wall	15

5 CHAPTER 5 Bench test

Test every servo, board and sensor on the desk before any of it goes inside the robot. A dead sensor found now costs five minutes. The same sensor found after assembly means taking the lid, the side panels and a leg off again.

5.1 Get ready

Header pins. Solder the header pins onto every sensor board: the eight laser boards, the motion sensor, the touch board, the gesture sensor and the two microphones. Push the pins into the breadboard to hold them straight, lay the sensor on top, and solder. This is good practice before the harder joints.

Software. The servo test sketch is in `guide/tools/servo_center`. PlatformIO is already installed in the laptop project's Python environment. Every command below is typed in a terminal (PowerShell) in that folder:

```
cd D:\Claude-Projects\RobotCompanion\guide\tools\servo_center
$pio = "D:\Claude-Projects\RobotCompanion\software\laptop\.venv\Scripts\pio.exe"
& $pio run                # build (checked: builds cleanly, 28 Sep 2026)
& $pio run -t upload       # upload to the screen board
& $pio device monitor      # open the Serial Monitor (Ctrl+C to leave)
```

One power source per board: while a board is on the laptop's USB cable, unplug its 5 V lead from the hub (for the screen board, the P1 plug).

5.2 Screen board on its own

Plug the screen board into the laptop with the USB-C cable, and nothing else. It should light up and run the seller's demo. Touch the screen: the demo should react. That proves the screen, the touch chip and the board.

Before anything overwrites it, back up the factory firmware, so you can always put the demo back:

```
& $pio pkg exec -p tool-esptoolpy -- esptool.py --chip esp32s3 read_flash 0 0x4000000 guition_factory_backup.bin
```

Keep `guition_factory_backup.bin` somewhere safe. Do the same for the camera board, and test the camera with the seller's example sketch, using the seller's pin map, not the AI-Thinker one.

5.3 Power check with the multimeter

Connect the battery pack and switch you built in 4.3 to the hub, with nothing plugged into the hub's outputs. Put the cells in the right way round (black lead to middle wire first), wake the protection board by plugging the USB-C charger in for a few seconds, and switch on.

Measure (DC volts)	Should read	Reading
BATT-SW to GND on the hub	6.0 to 8.4 V	
Mini560 #1 (servos) OUT+ to OUT-	4.9 to 5.2 V	
Mini560 #2 (boards) OUT+ to OUT-	4.9 to 5.2 V	
The red and black wires at the screen board's P1 plug, red probe on red	4.9 to 5.2 V, not minus	
VBAT pad to GND	about 0.32 x the battery voltage (1.9 to 2.7 V)	

Only when every reading is right, switch off and plug in the servo boards' green terminals, the P1 plug and the camera board. Switch on and measure the two module outputs again: they must still read 4.9 to 5.2 V.

5.4 Centre every servo

A servo's horn must go on while the servo sits at its centre, 90 degrees, or the leg will stand crooked. The test sketch does this for all thirteen servos at once, and keeps them there.

1. Unplug P1. Connect the laptop to the screen board and upload the sketch (`& $pio run -t upload`). Open the Serial Monitor.
2. Plug the servos into the servo boards by the channel table (4.9), brown wire on the GND row. For the bench test they can lie loose on the desk, with no horns fitted.
3. Switch the battery on. Every 180 degree servo moves to its centre and holds; the 360 degree wheel servos stop or creep slowly.
4. Type `n` and press Enter. The first servo in the table is selected and wiggles, so you know which one it is. Put its name flag on its lead. Repeat `n` for all thirteen. Type `t` to sweep the selected servo from 60 to 120 degrees and back: it should move smoothly without grinding.
5. Switch off. From now on, do not turn a servo's shaft by hand, or it is no longer centred. If you do, run the sketch again.

At start-up the sketch tries SDA on IO17 first and swaps to IO18 if no servo board answers, lists every device on the bus by name, and says "found" or "NOT FOUND" for each servo board. Type `h` for the list of commands.

5.5 Test the sensors

Plug each sensor into a row of the hub, one at a time at first. After each one, type `s` in the Serial Monitor and check it appears in the list.

Device	Address in the scan	Extra test
Servo board A / B	0x40 / 0x41 (0x70 may also show at start-up: ignore it)	Servos centre at start-up
MPU6050 motion sensor	0x68	
MPR121 touch board	0x5A	
APDS-9960 gesture sensor	0x39	
VL53L0X lasers	none until woken	Type <code>l</code> : each of the seven must say "OK, answers at 0x29"
Battery level	-	Type <code>v</code> : within 0.2 V of the multimeter

Cheap copies of these sensors sometimes report a different chip ID. If a sensor does not appear at all, check its wires first; if it appears at an unexpected address, write the address down rather than calling the part dead. Test the spare laser too, and keep it for later.

5.6 Wheel stop points and a servo strength test

Wheel stop points. Select each wheel servo with `n`, then press `+` or `-` until it stops completely. Write the number into the calibration table in chapter 7.

Strength test (spare servo). The knees work hard. Hang about 450 g (a full water bottle) from the spare 180 degree servo's horn, 25 mm from the shaft, with the servo held at centre by the sketch, for 5 minutes. Feel for heat and listen for buzzing. Warm is fine. Too hot to hold, or a constant loud buzz, means the knees need stronger servos: tell the lead before assembly.

Afterwards, switch off and take the cells out again. They stay out until chapter 7.

BENCH TEST PASSED WHEN

All thirteen servos centre and sweep, both servo boards and every sensor show in the scan, all seven lasers answer, the battery reading matches the multimeter, and the camera board streams a picture.

6 CHAPTER 6 Assembly

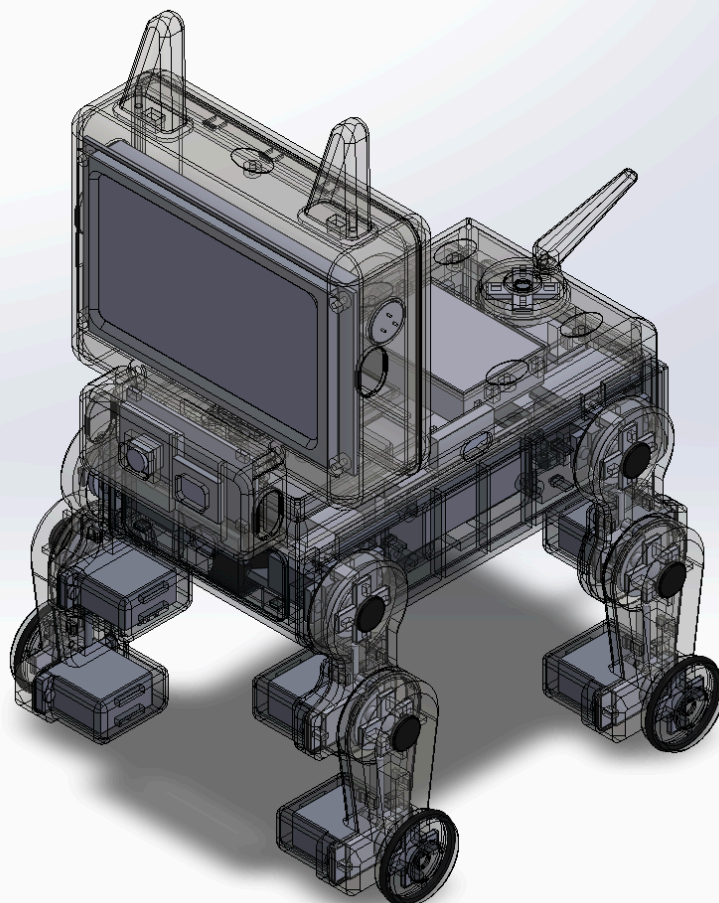
Seven short sections, in this order: the body (tub), the lid, the head, the four legs, the tail, the ears, and closing it all up. The body, lid and head are built as separate pieces; the legs go on the body, and the lid with the head closes it at the end.

Before you start each section, count its screws into a cup (3.3), lay out its parts, and read all its steps once.

Every servo must be centred before its horn goes on (5.4). If you turned a shaft by hand since then, run the servo sketch again first.

Keep the cells out of the robot for the whole of this chapter. They go in during chapter 7.

Section	Steps	Screws
A Body (tub)	A1-A9	12 x M2x6, 2 x M2x8
B Lid	B1-B7	4 x M2x4, 2 x M2x6, 4 x M2x8
C Head	C1-C7	1 x M2x4, 4 x M2x6, 8 x M2x8
D Legs, x4	D1-D11	8 x M2x4, 32 x M2x6, 8 x M2x8, 12 horn
E Tail	E1	2 x M2x4, 1 x M2x8, 1 horn
F Ears	F1	2 x M2x8
G Close up	G1-G4	4 x M2x6

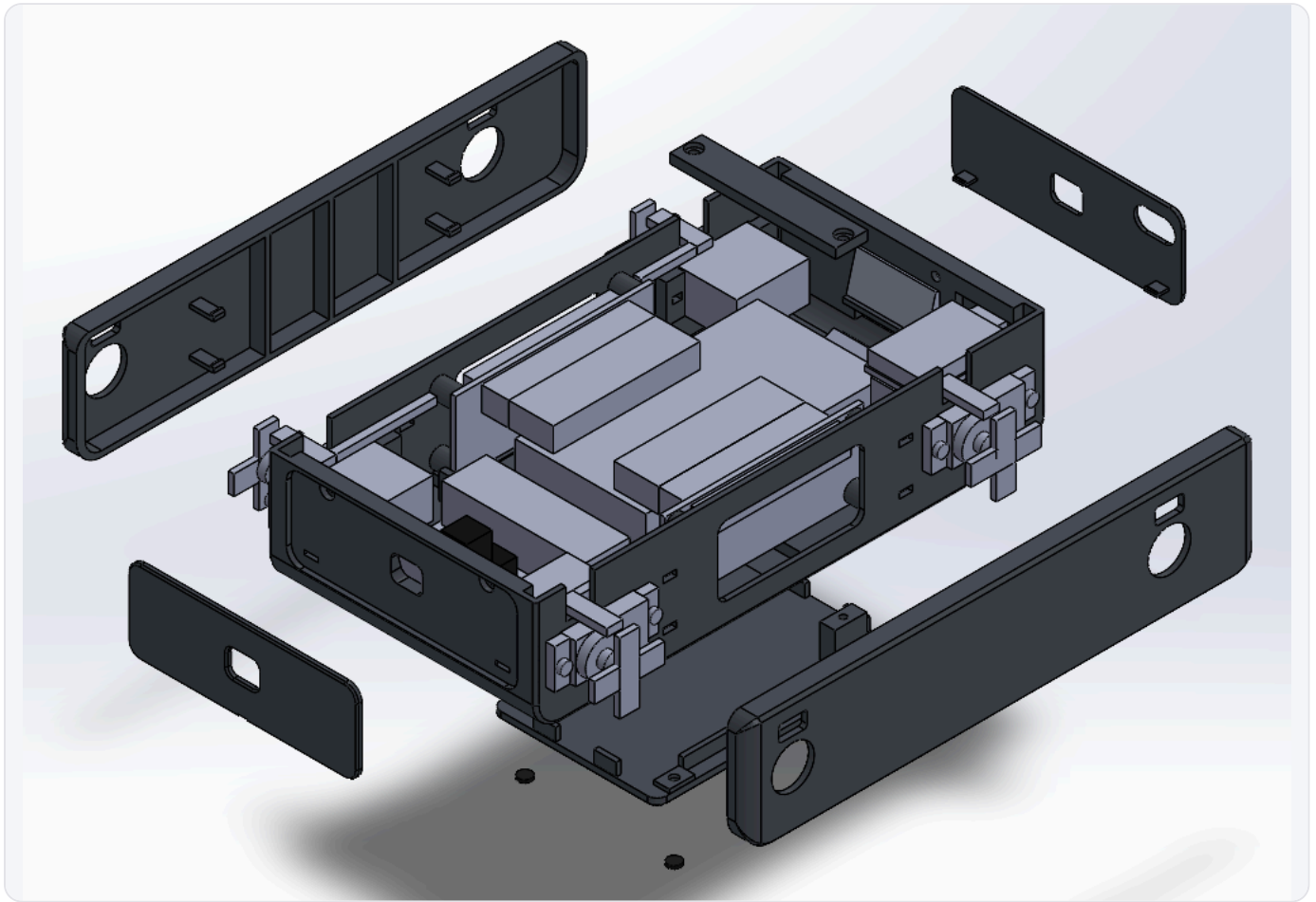


The finished robot with its shells drawn see-through: every board, servo and sensor in its place.

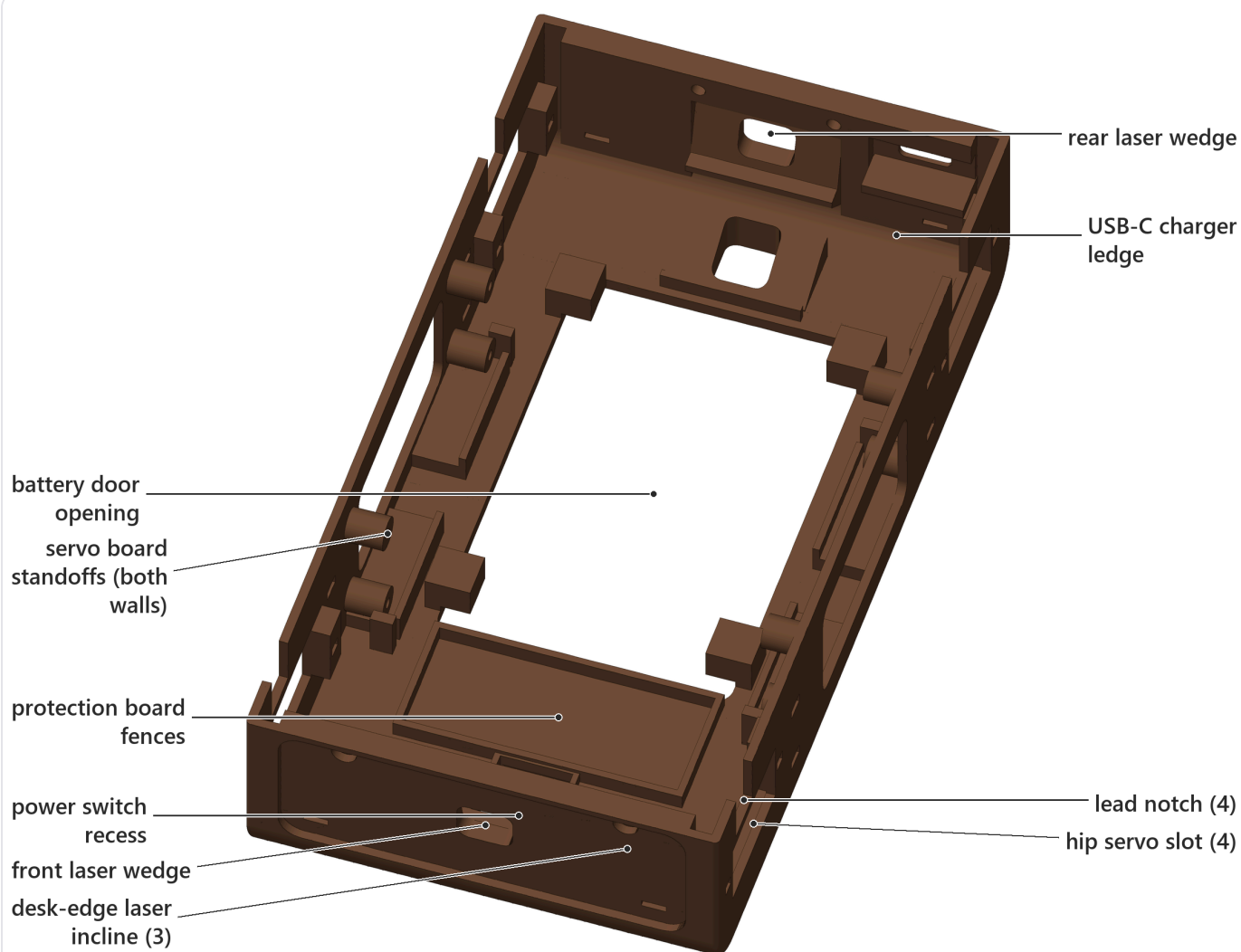
SECTION A

A Body

The tub holds the four hip servos, the two servo boards, the protection board and fuse, the charger, the power switch and five of the seven lasers. The battery door carries the cell holder. The side panels close the sides.



A1 Get to know the tub



YOU NEED

the tub (Torso_Bottom)

The tub is the robot's belly. Almost every electronic part sits in it, so spend two minutes finding each place in the picture before you start. The front of the robot is where the power switch recess is.

Clean the print first: pull off any brim, snip stray strings with the flush cutters, and make sure every slot and window is open.

Pre-thread the screw holes you will use in this chapter. Drive an M2x6 screw into each one and take it out again. The first time is the stiff one, and doing it now, with nothing fitted, means you never have to lean on a finished part.

CHECK

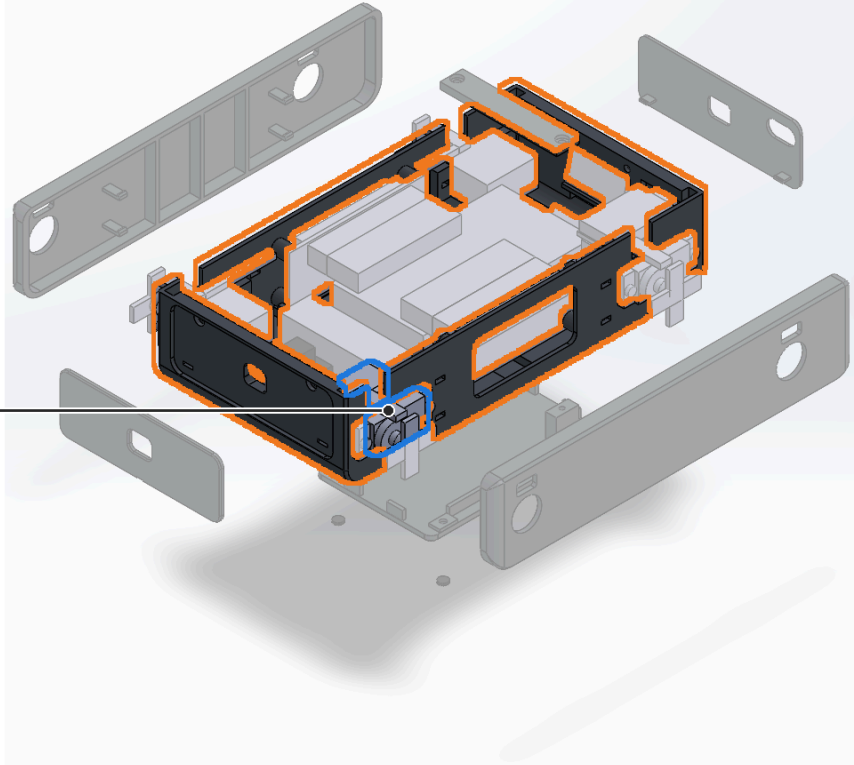
All four hip slots are clean, and a servo slides into each one from outside without forcing.

COMMON MISTAKE

Driving a screw into an un-threaded hole on a thin wall. Go slowly, straight, and stop as soon as the head touches.

A2 Hip servos into the tub

hip servo, tabs
outside the wall



YOU NEED

4 x MG90S 180 degree servo,
centred (Bench test, step 5.4)



8 x M2 x 6

Mark each servo lead with a tape flag before you fit it: HFL, HBL, HFR and HBR, for hip front-left, back-left, front-right and back-right. Left and right are always the robot's own left and right.

Push each hip servo through its slot in the tub wall from the outside, plug and lead first. The two mounting tabs stay outside, flat against the wall, and the round output boss points outward.

Fix each servo with 2 x M2x6 through its tabs. The screws that came in the servo bag also fit.

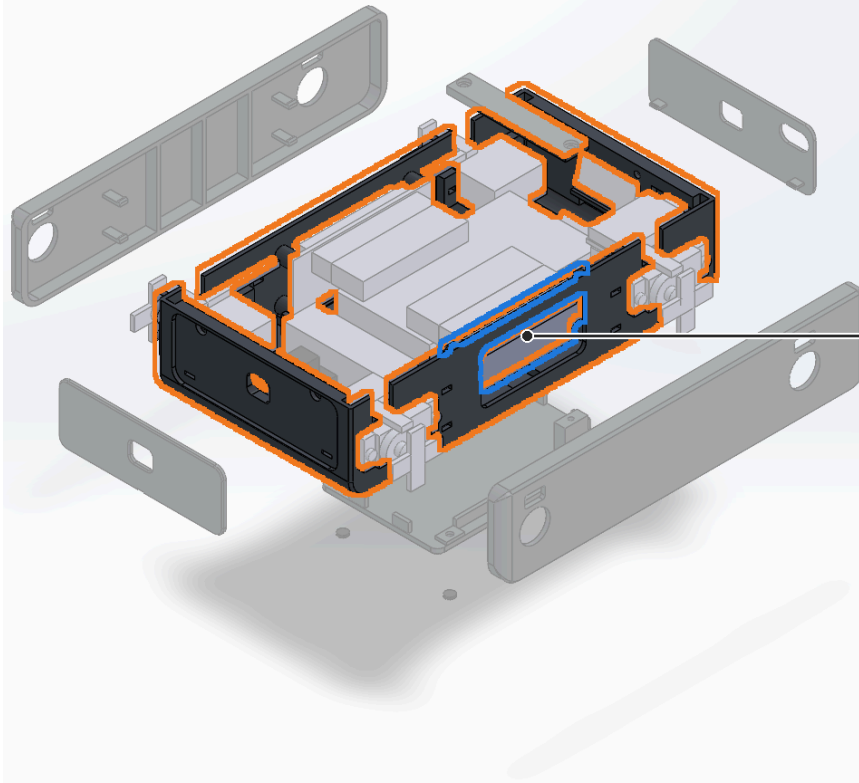
CHECK

Each servo's tabs sit flat on the wall with no gap, and its output shaft sits in the middle of the round boss hole when you hold a side panel against it.

COMMON MISTAKE

Trying to fit the servo from inside the tub. The tabs must be outside, so the lead goes in first.

A3 Servo boards on the side walls



servo board A,
left wall

YOU NEED

2 x PCA9685 servo board (A0 pad bridged on board B)



4 x M2 x 6

Each servo board stands upright against a side wall, on four printed standoffs. The servo header rows run along the top edge and the green V+ screw terminal sits in the gap at the bottom middle.

Board A goes on the robot's left wall. Board B, the one with its A0 pad bridged with solder (Wiring, step 4.6), goes on the right wall.

Screw each board through its two upper holes into the upper standoffs, 2 x M2x6 per board. The two lower standoffs are spacers only: a screwdriver cannot reach them square, so leave them empty.

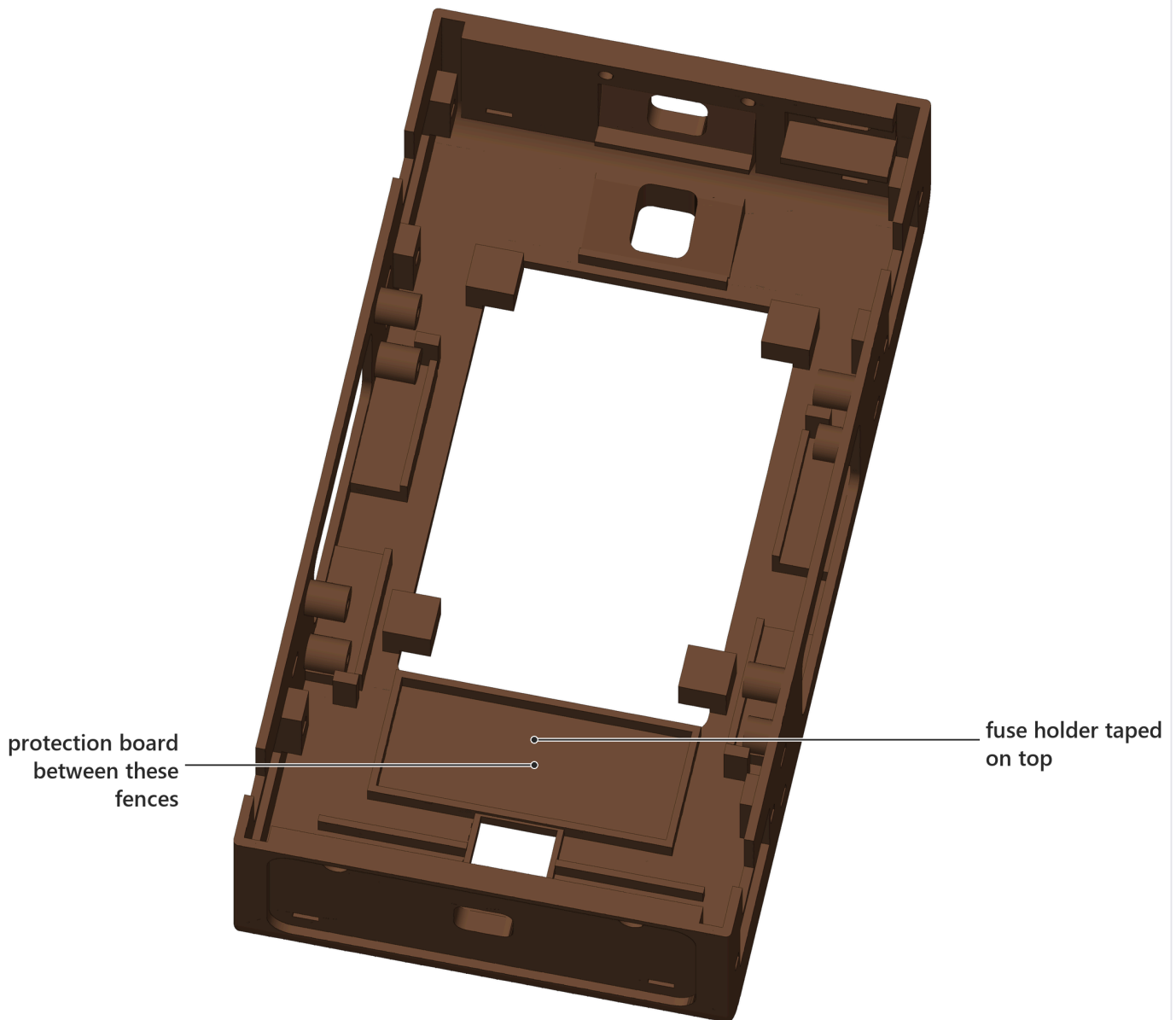
CHECK

The boards do not rock, and the servo headers face the middle of the tub.

COMMON MISTAKE

Swapping the boards. If the right board is not the bridged one, both boards answer at the same address and neither works.

A4 Protection board and fuse



YOU NEED

battery pack harness from
Wiring step 4.3 (protection
board, fuse holder, switch leads)
foam tape

Lay the protection board flat on the tub floor, between its four fences, near the front. Its wires were soldered on in the Wiring chapter.

Put a strip of foam tape on top of the protection board and stick the fuse holder to it. The 7.5 A fuse stays in.

Lead the three battery wires (black, middle and red) back towards the battery door opening, and the red and black power wires towards the rump, where the charger goes.

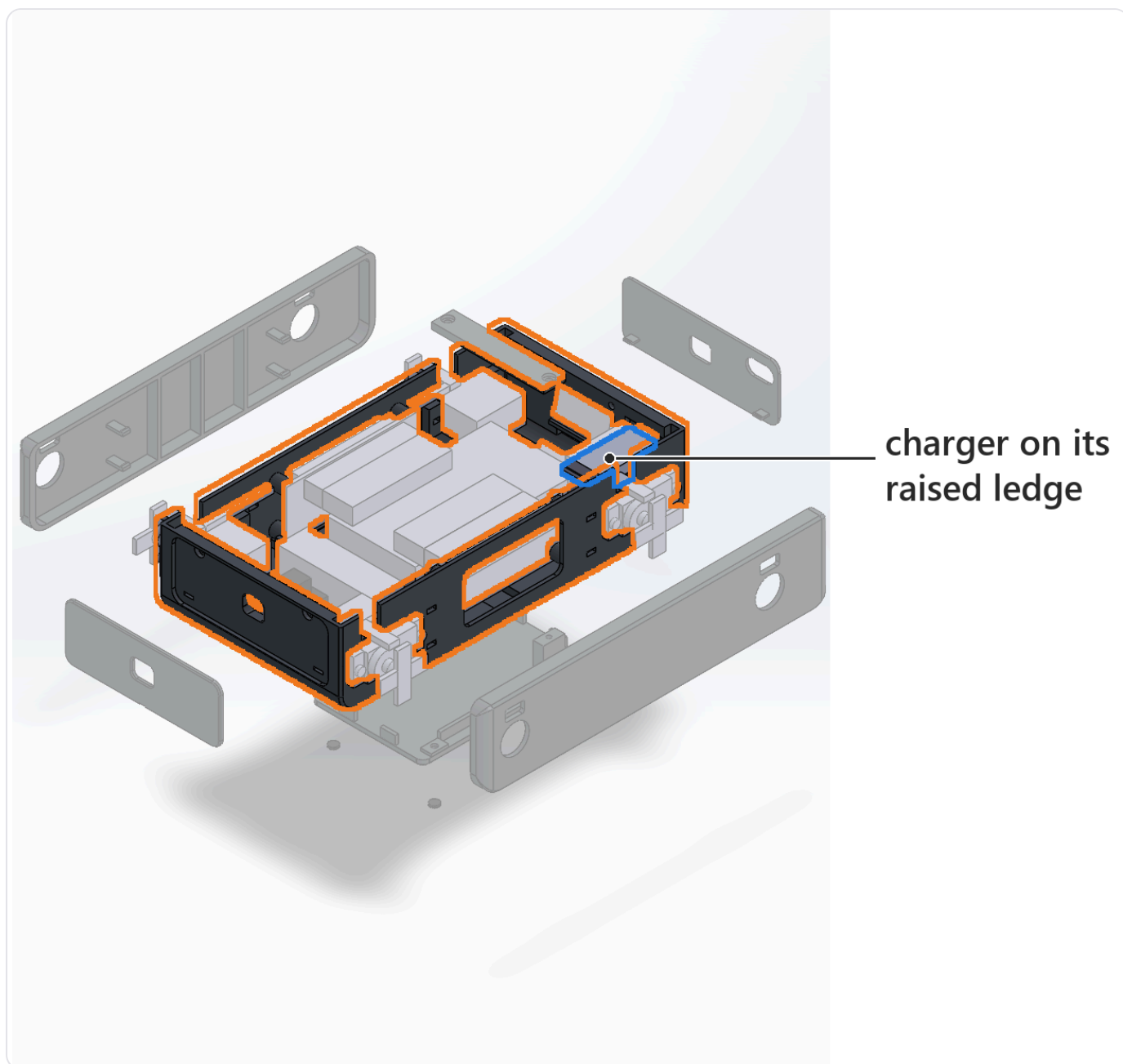
CHECK

The board sits flat and nothing metal on it touches a screw tip or another board.

COMMON MISTAKE

Leaving the cells in the holder while you work. Keep them out until the First power-on chapter.

A5 USB-C charger at the rump



YOU NEED

Type-C 2S charging board
(already wired, Wiring step 4.3)
foam tape

The charger sits on a raised ledge inside the rump wall, above the rear-left hip servo, with its USB-C socket in the opening of the rump wall.

Put a small piece of foam tape on the ledge, slide the board in socket first until the socket face is just behind the opening, and press it down.

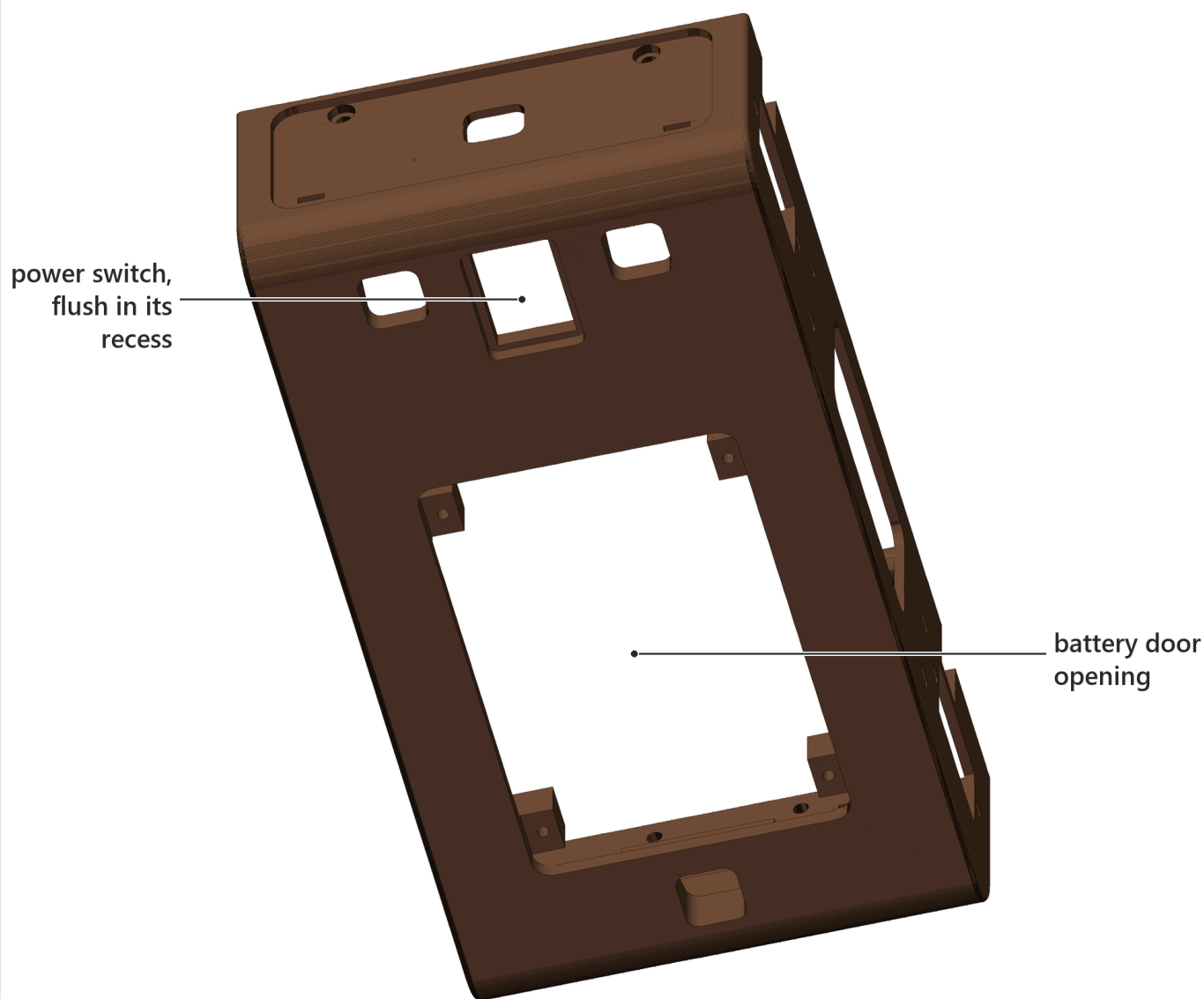
CHECK

Plug a USB-C cable in from outside: it goes all the way in without pushing the board back.

COMMON MISTAKE

Taping the board too far back, so the plug cannot reach the socket once the rump band is on.

A6 Power switch in the belly



YOU NEED

KCD1 rocker switch (already wired, Wiring step 4.3)

Turn the tub upside down. The power switch clips into the rectangular cut-out at the front of the belly, from outside, turned so the rocker runs front to back.

Feed its two wires in first, then press the switch in until both side clips snap behind the wall. The rocker sits flush in its recess.

Decide now which way is ON and mark it with a tiny dot of paint or a sticker. The switch is the only way to turn the robot off quickly.

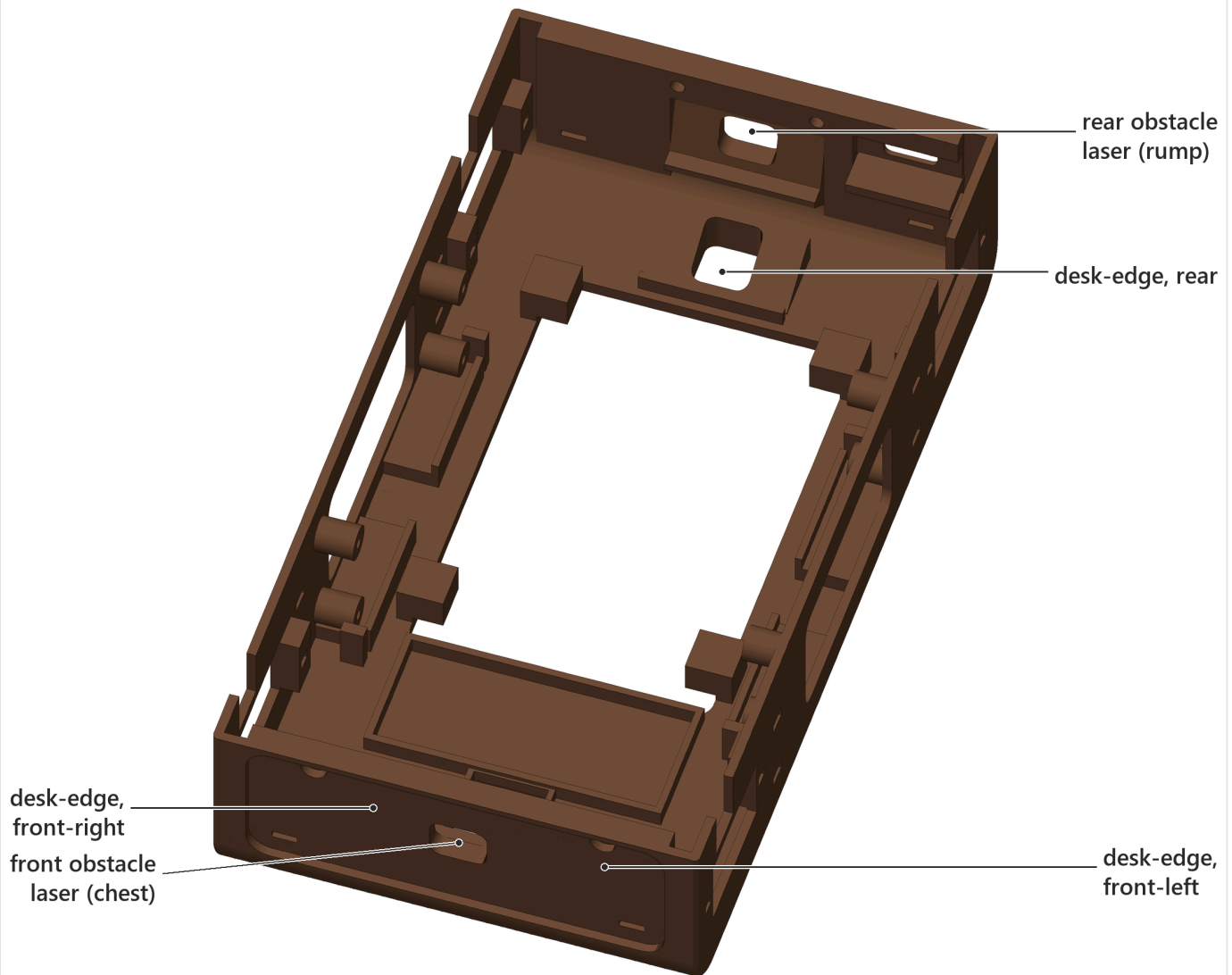
CHECK

The switch does not move when you press the rocker hard in both directions.

COMMON MISTAKE

Soldering the switch wires after it is clipped in. Solder first (Wiring chapter), then clip it in.

A7 Five laser sensors in the tub



YOU NEED

5 x VL53L0X laser sensor, tested (Bench test)
foam tape

Five of the seven laser sensors live in the tub. Two look forward and backward for obstacles, through the chest and rump windows. Three look down at the desk to stop the robot before an edge: two at the front, either side of the power switch, and one at the rear.

Peel the protective film off each sensor chip. Stick each board to its incline or wedge with a small square of foam tape, chip facing the window, and centre the chip on the hole.

Flag each sensor lead with its name. Their XSHUT wires go to servo board B, channels 9 to 13 (Wiring, laser table).

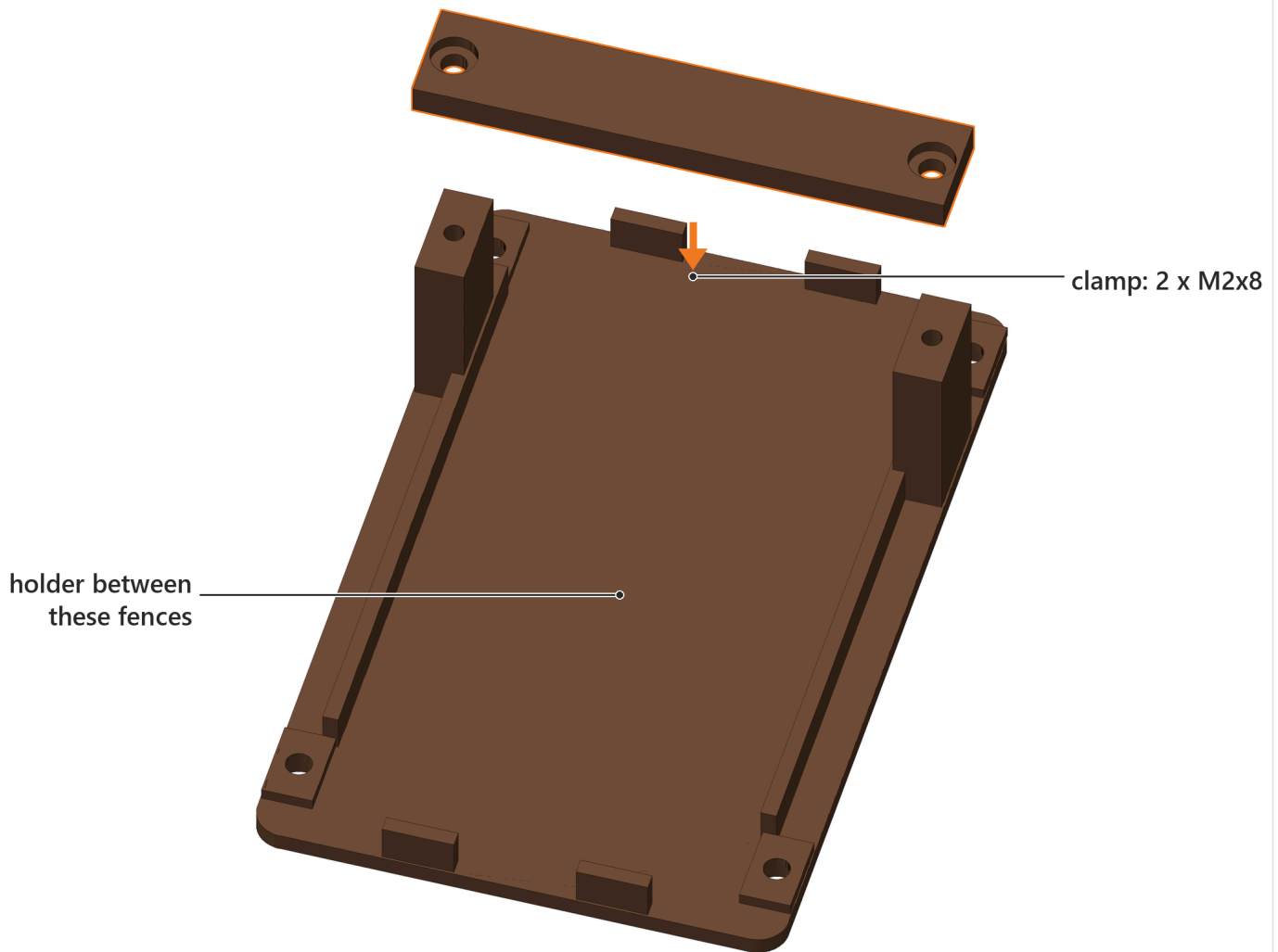
CHECK

Look through each window from outside: you can see the small black chip in the middle of the hole.

COMMON MISTAKE

Leaving the clear film on the chip. The sensor then reads a short, wrong distance all the time.

A8 Battery holder onto the battery door



YOU NEED

2 x 18650 holder (wired, cells OUT)

battery door and clamp



2 x M2 x 8

Lay the empty cell holder between the fences on the inside of the battery door. Its wires point to the front.

Put the clamp bar across the holder and screw it to the two posts, 1 x M2x8 each. Tighten until the holder cannot slide, no further.

Do not fit the door to the tub yet. The cells go in, and the door goes on, in the First power-on chapter.

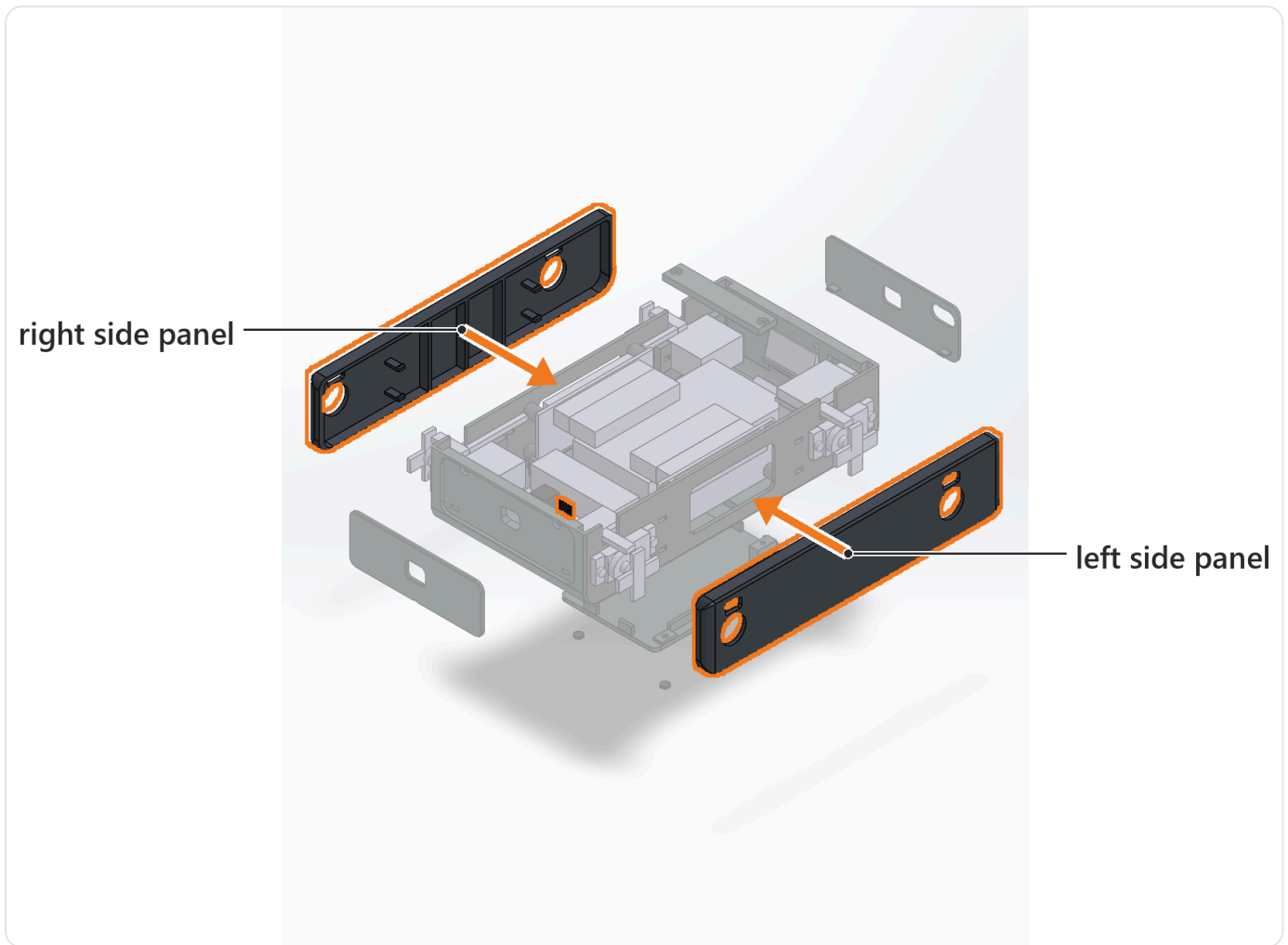
CHECK

The holder does not move when you shake the door.

COMMON MISTAKE

Over-tightening the clamp and bending the holder, which then loosens the cell contacts.

A9 Side panels on



YOU NEED

left and right side panels (no screws)

Each side panel covers the two hip servos on its side. It has four snap hooks and no screws.

Hold the panel so its two round holes line up with the round bosses of the hip servos, then press it straight onto the wall until all four hooks click. Do the left panel on the left, the right panel on the right: the slot for the leg leads is at the top of each round hole.

Fit the side panels before the legs, because the thigh cups help hold the panels in place.

CHECK

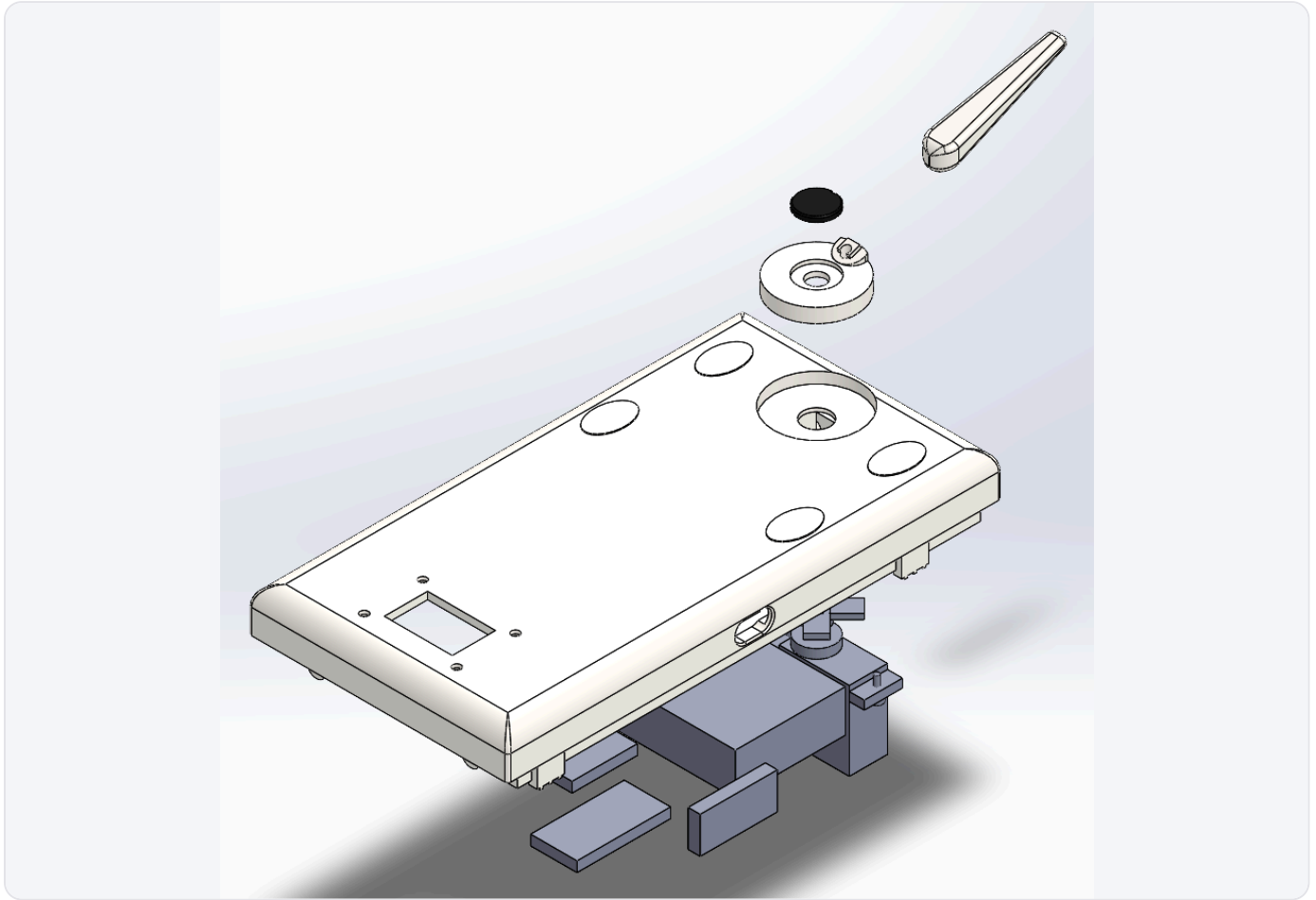
The panel sits flush all round, and each servo boss is centred in its round hole.

COMMON MISTAKE

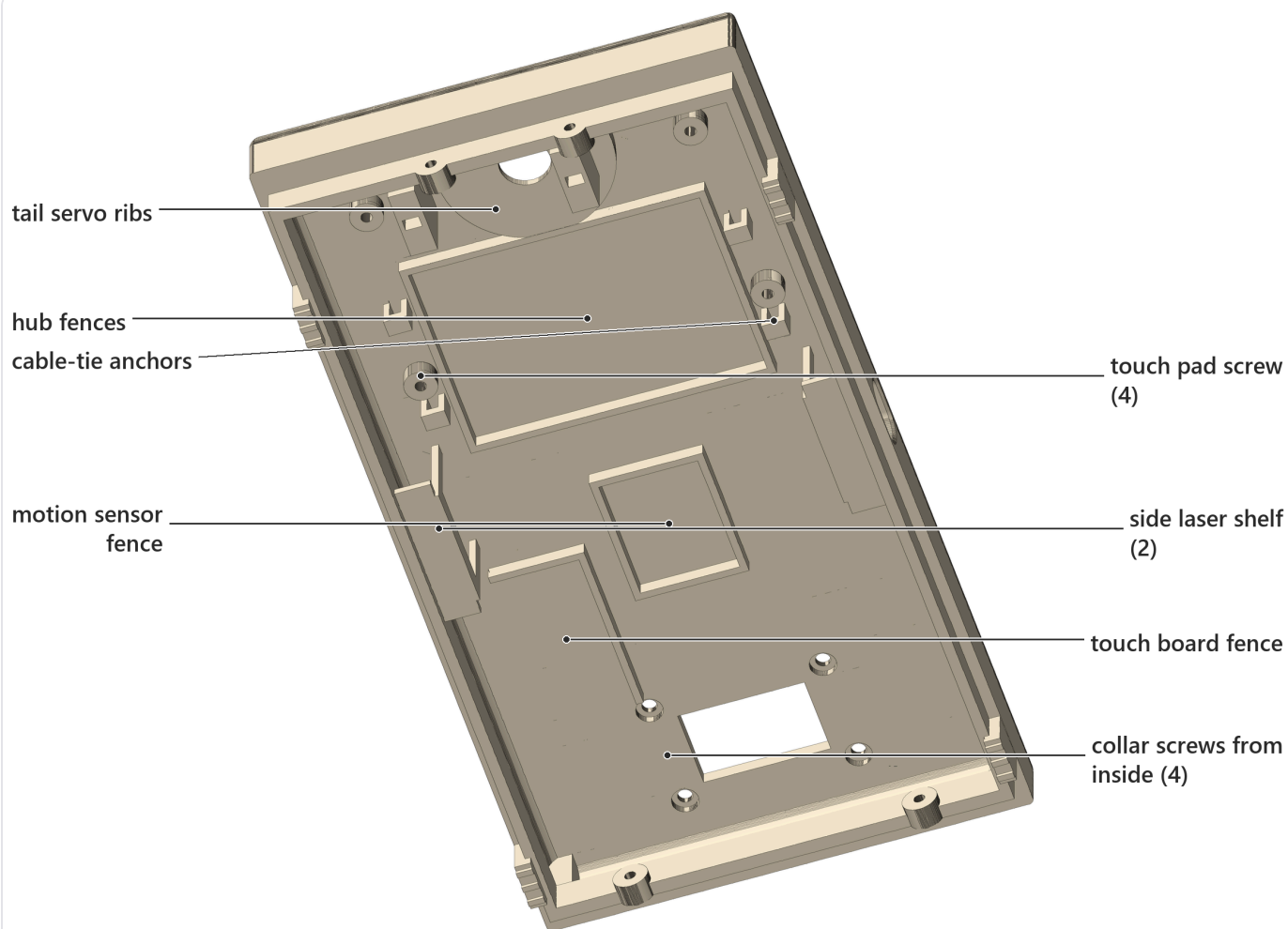
Pressing on one end only. Press both ends together, or a hook can break off.

B Lid

The lid is built on its own, upside down on the table. It carries the motion sensor, the touch board, the power hub, the two side lasers, the tail servo and four touch pads, and the collar screws on top of it.



B1 Get to know the lid (underside)



YOU NEED

the lid (Torso_Lid)

The lid carries the motion sensor, the touch board, the power hub, the tail servo, the two side lasers and four touch pads. Turn it upside down and find each place.

Clean the print and pre-thread the holes you will use in this chapter: the tail servo pilots, the four touch pad pilots and the four collar holes.

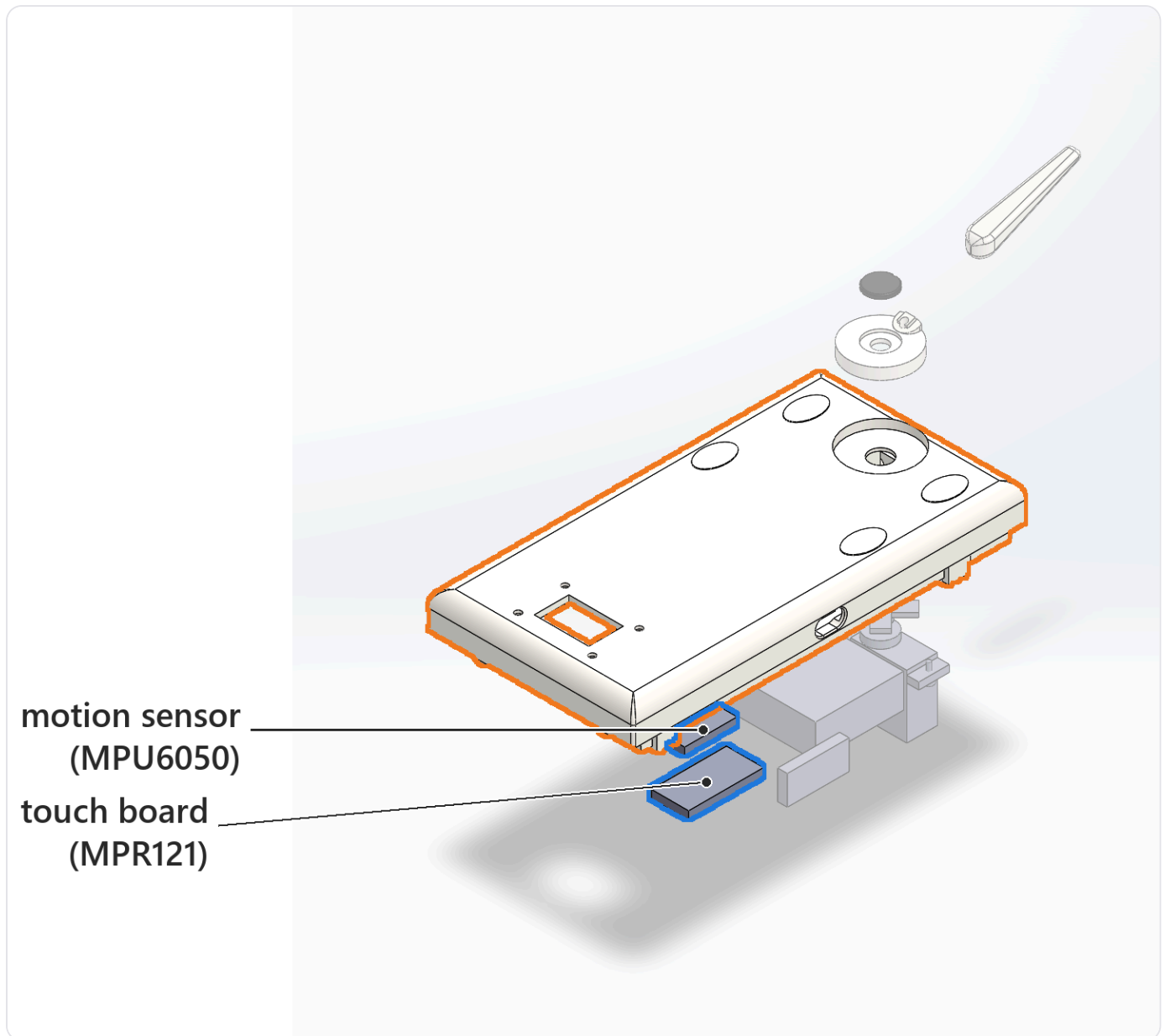
CHECK

The lid's top edge is smooth. If the first millimetre drooped while printing, sand it lightly (Printing, note A).

COMMON MISTAKE

Pre-threading the touch pad holes with an M2x6. It is too long and comes out through the top. Use an M2x4.

B2 Motion sensor and touch board



YOU NEED

MPU6050 motion sensor
MPR121 touch board
foam tape

Both boards sit in their own fences on the underside of the lid, components facing down.

Stick each one in with foam tape. The motion sensor must sit flat and square in its fence, because the robot uses it to know which way is up.

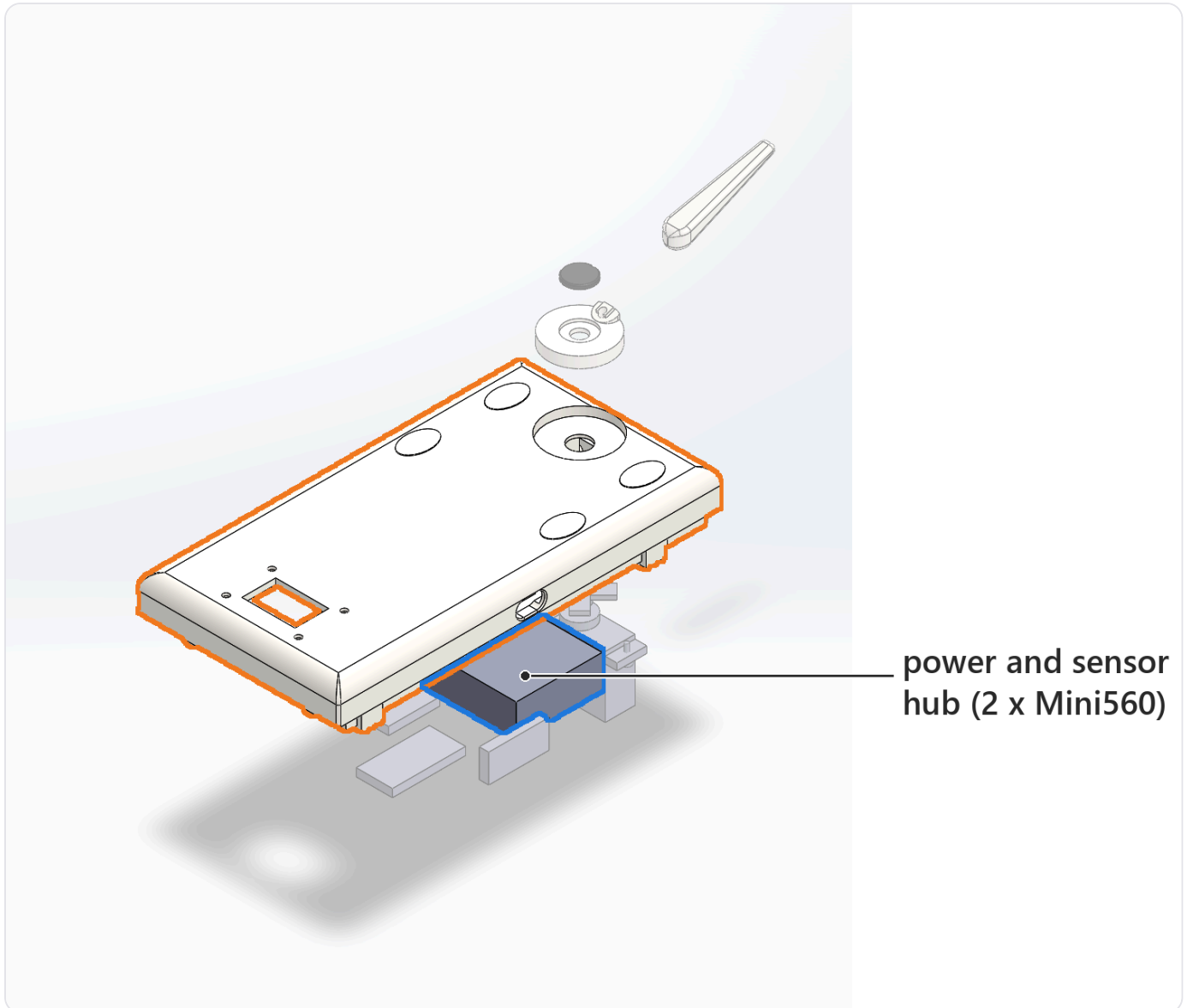
CHECK

Neither board can rock. The motion sensor's edges are parallel to the fence.

COMMON MISTAKE

Mounting the motion sensor at an angle. It then thinks the robot is always leaning.

B3 Power hub under the lid



YOU NEED

the hub you built in Wiring step 4.5
2 cable ties

The hub carries the two 5 V power modules, the two big capacitors and the small resistors. It fits in the large fence towards the back of the lid.

Press it in, components facing down, and pass one cable tie under each pair of anchors beside the fence. Pull them snug, not tight, and snip the tails.

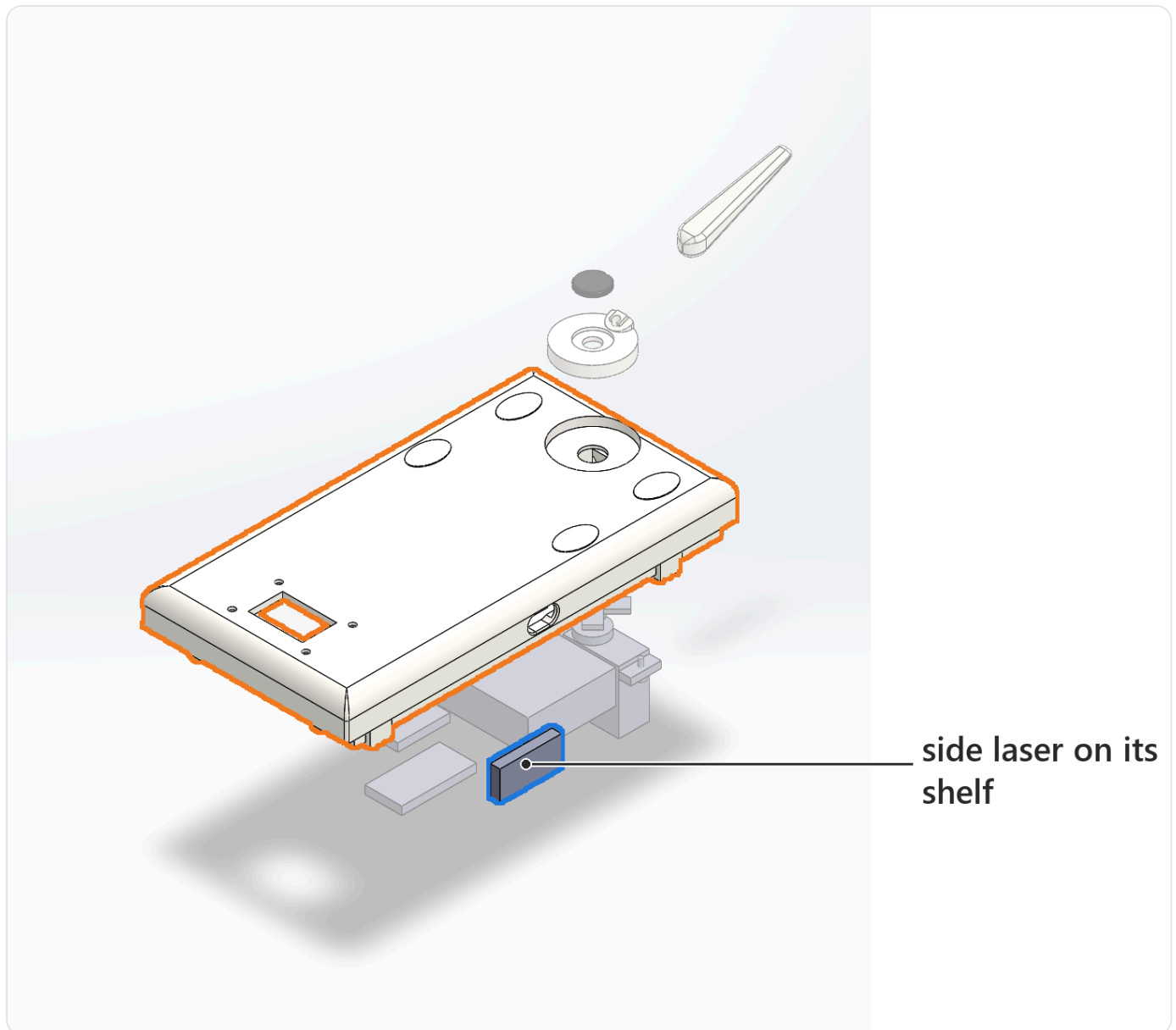
CHECK

The hub does not move, and no capacitor leg touches the lid's screw bosses.

COMMON MISTAKE

Standing the capacitors upright. Lay them on their side; there is only 13 mm of room.

B4 Two side lasers



YOU NEED

2 x VL53L0X laser sensor
foam tape

One laser stands on a shelf against each side wall of the lid, looking sideways through the small window.

Peel the film off the chip, slide the board down behind the lip on its two posts, chip facing the window, and fix it with a small square of foam tape.

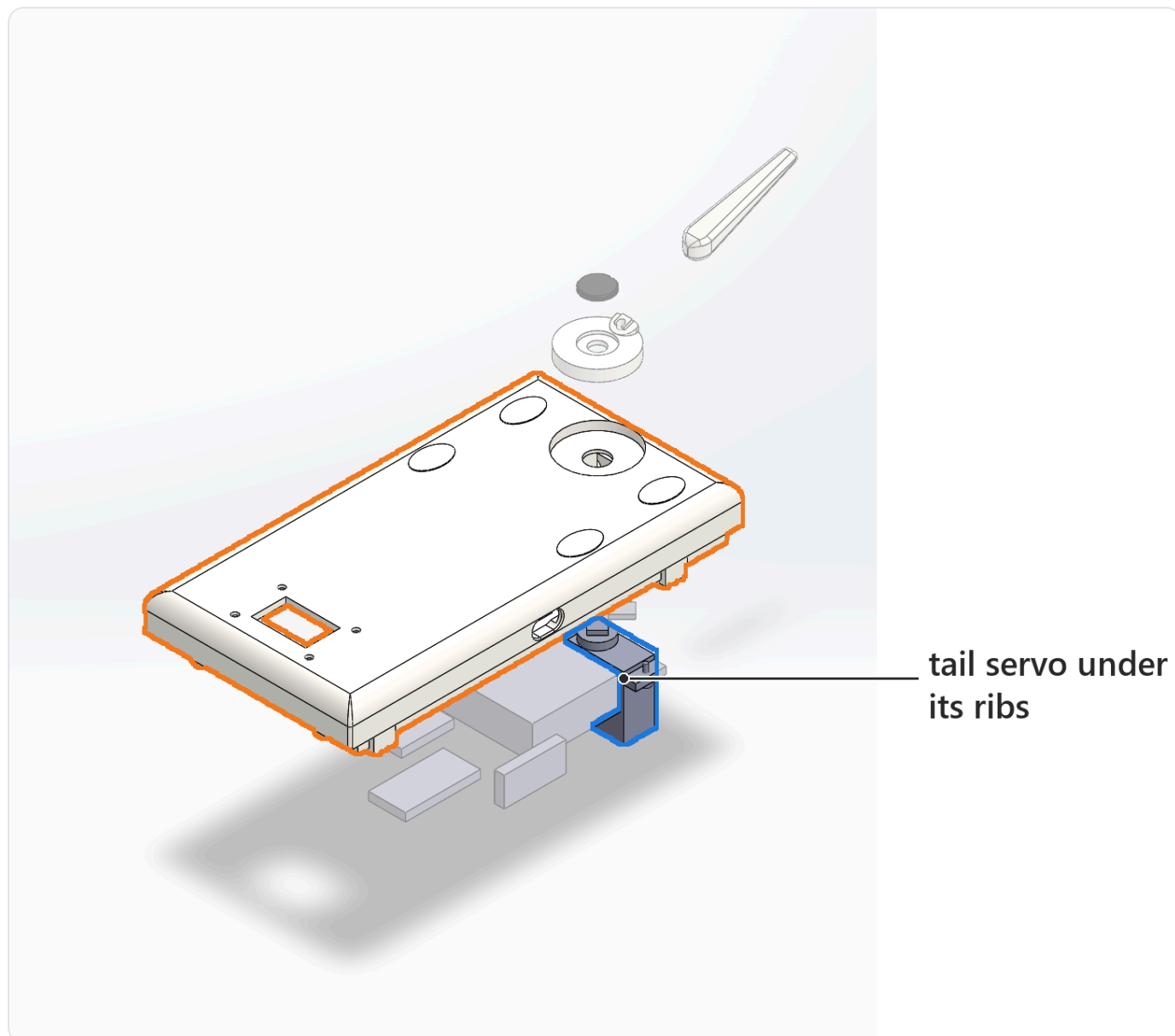
CHECK

From outside you can see the chip through the window.

COMMON MISTAKE

Fitting the board chip-inward. The chip must face the window.

B5 Tail servo



YOU NEED

MG90S servo, centred, lead
flagged TAIL



2 x M2 x 6

The tail servo hangs under two ribs near the back of the lid, with its output shaft pointing up into the round tail socket.

Hold the servo under the ribs with its tabs against them and drive 2 x M2x6 up through the tabs from inside the lid.

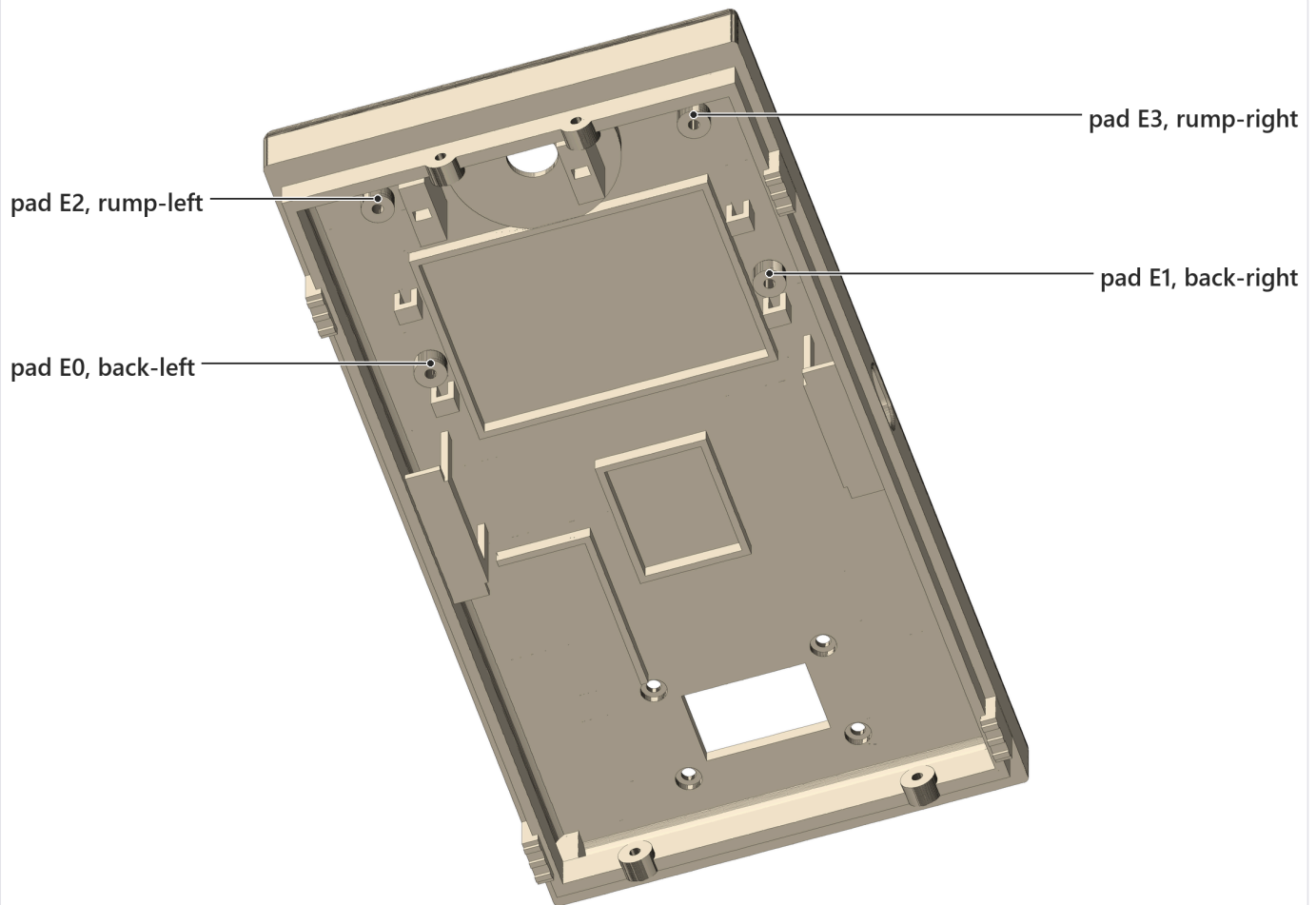
CHECK

Look at the lid from the top: the servo spline sits in the middle of the tail socket.

COMMON MISTAKE

Fitting the servo with its shaft at the wrong end. The shaft must be under the socket, not under the lid.

B6 Four foil touch pads



YOU NEED

kitchen foil, 4 pieces about 25 x 20 mm

4 grey wires from the touch board



4 x M2 x 4

The robot feels petting through four foil pads on the inside of the lid, under the four raised ovals on top.

Cut a foil piece with a small tab, poke a hole in the tab, and lay it on the inside of the lid under an oval. Strip 5 mm of its wire, wrap the bare end once round an M2x4 screw, and drive the screw through the foil tab into the pad hole so the wire and foil are clamped together.

A dot of superglue at each corner keeps the foil flat. Wire E0 to E3 as the labels show.

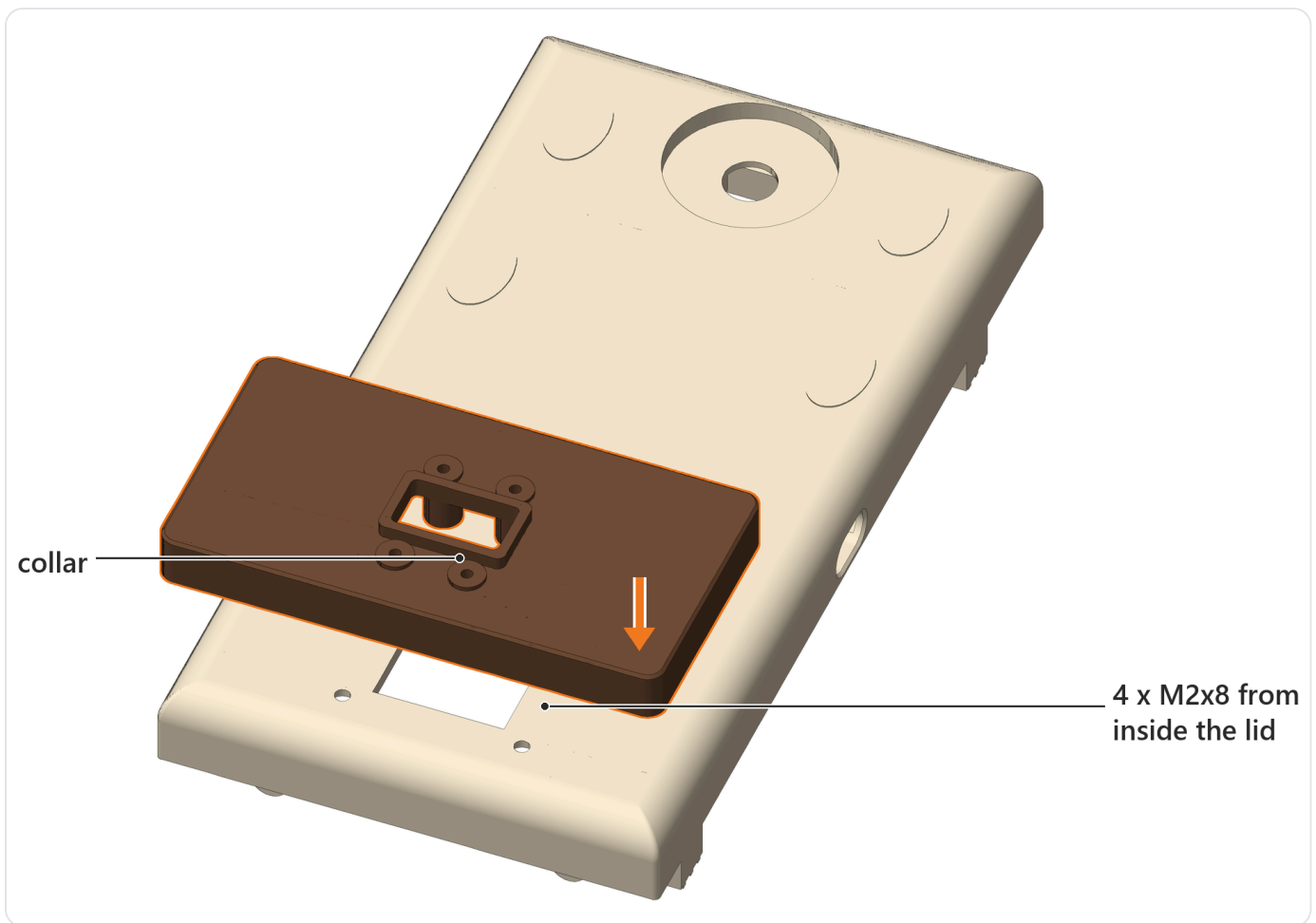
CHECK

With the multimeter on beeper mode, each wire beeps to its foil.

COMMON MISTAKE

Using an M2x6 here. It comes out through the top of the lid.

B7 Collar onto the lid



YOU NEED

collar



4 x M2 x 8

Before you fit the collar, feed the head's leads up from the hub through the lid's neck opening: the screen power lead, the screen sensor lead (port P4), the camera power lead, the gesture sensor lead and the head touch pad wire. Leave about 15 cm sticking out.

Set the collar on the lid, leads through its middle, and screw it on from inside the lid with 4 x M2x8.

CHECK

The collar sits flat all round, and every head lead comes out of its top.

COMMON MISTAKE

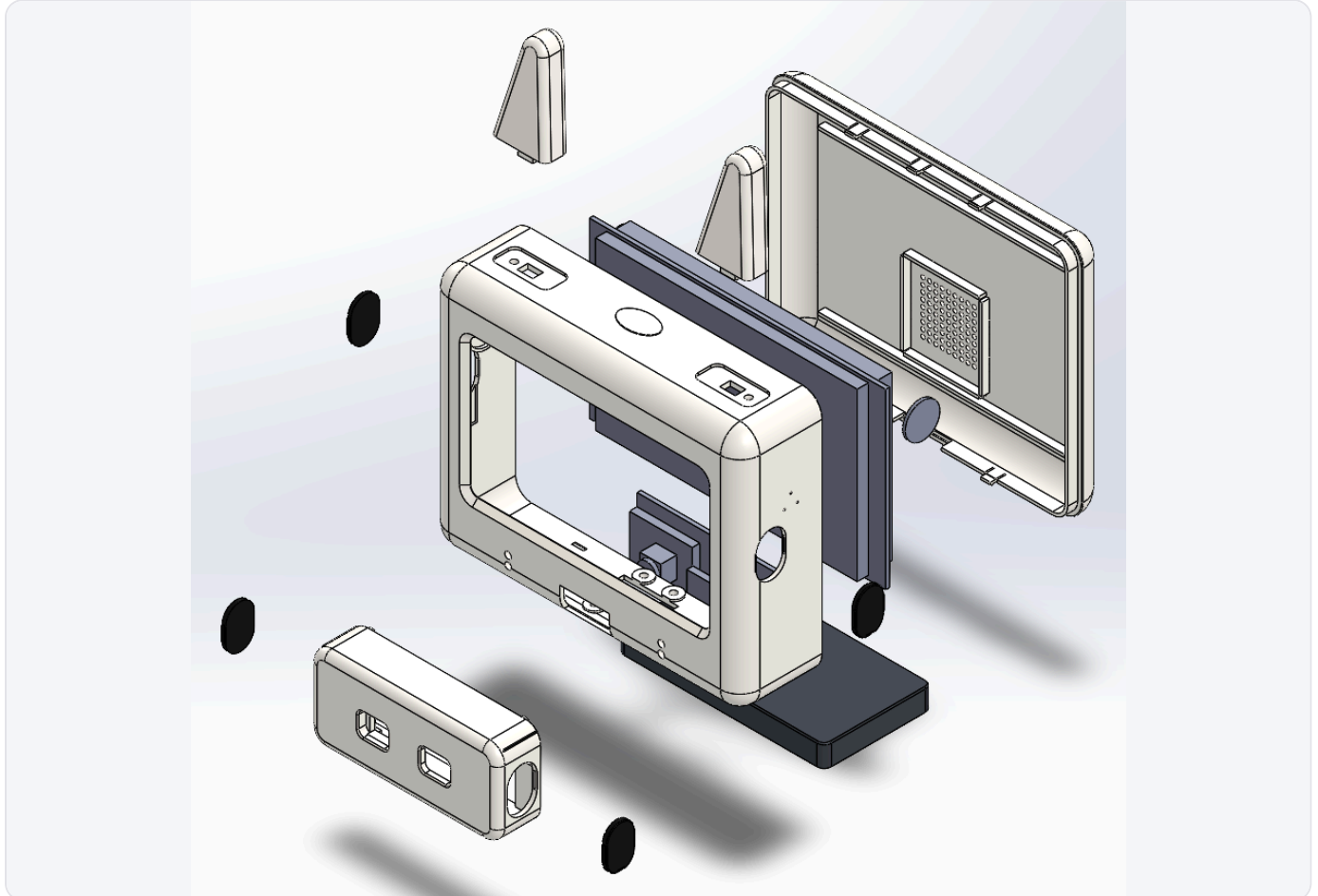
Fitting the collar first and then trying to push plugs through it. Feed the leads first.

C Head

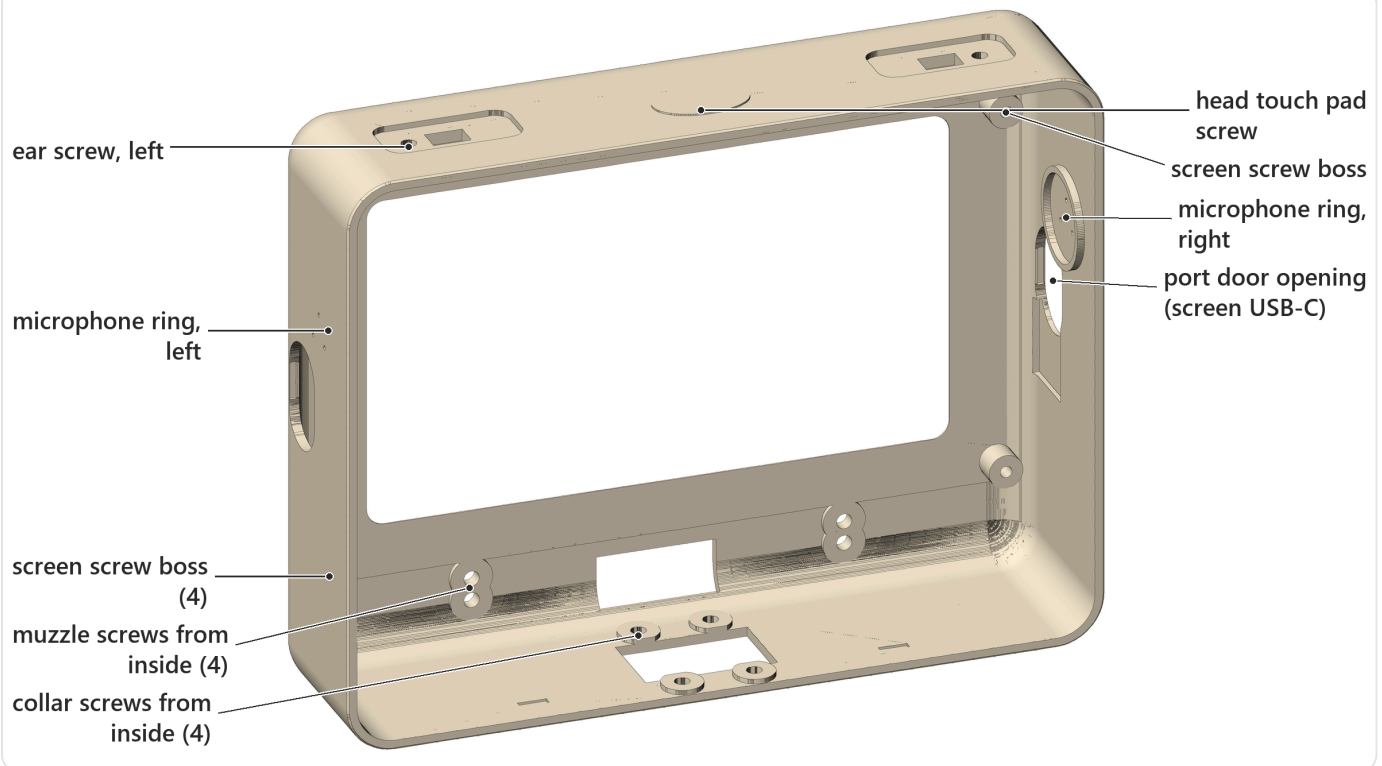
The head holds the screen, the microphones, the speaker, the camera and the gesture sensor. It is built on the table, then screwed onto the collar on the lid. The head back stays off until the very end.

NOTE

The muzzle (step C5) may change in the next version of the design. The rest of the head stays the same.



C1 Get to know the head front (inside)



YOU NEED

head front (Head_Front)

The head front holds the screen, both microphones, the head touch pad and the ears, and the muzzle and collar screw to it from inside. Look inside it and find each place.

Pre-thread the four screen bosses with an M2x6, then take the screw out again.

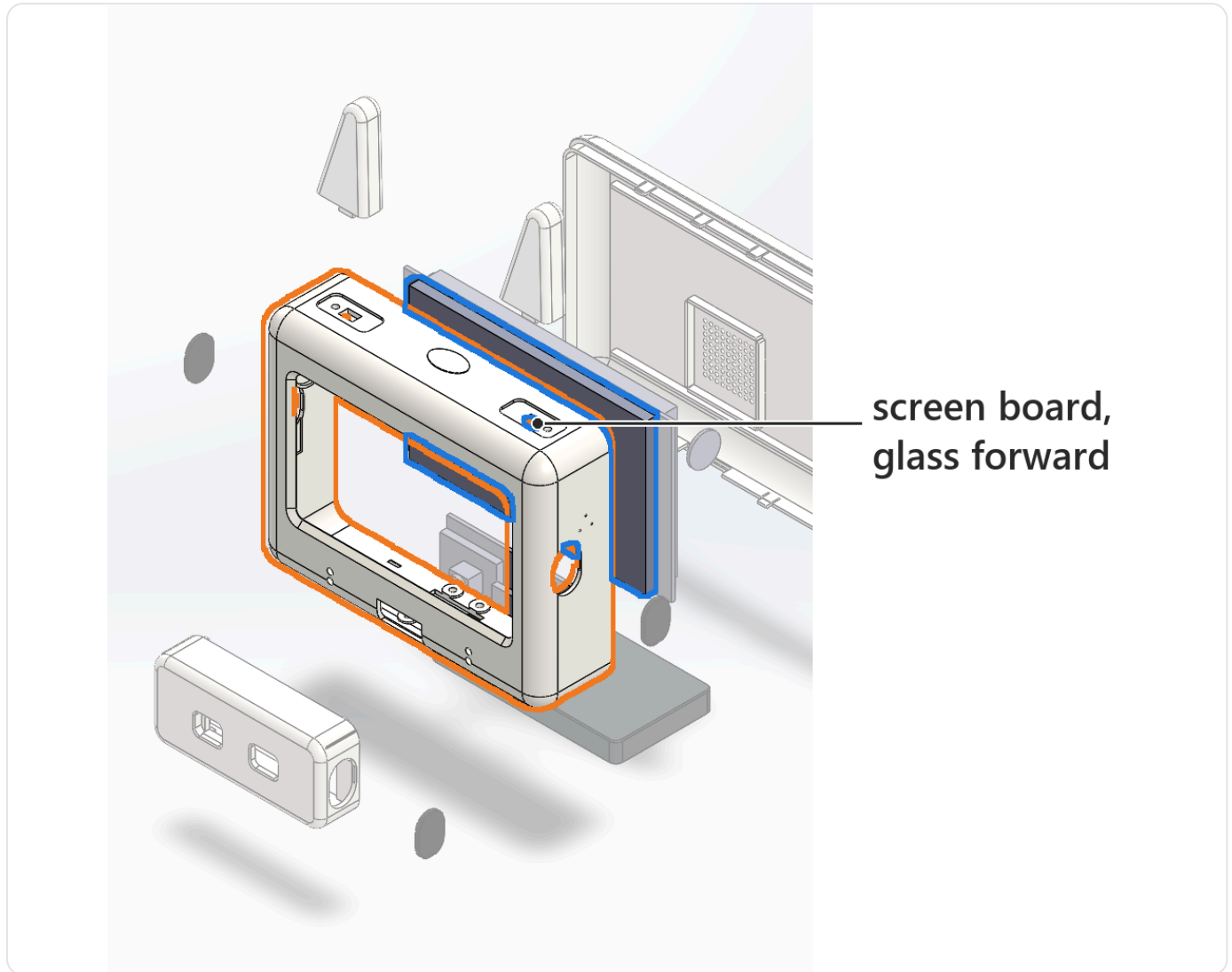
CHECK

The screen window's bevel is clean and the glass seat (the flat ledge round the window) has no blobs.

COMMON MISTAKE

Leaving support or brim blobs on the glass seat. The screen then sits crooked and can crack when you screw it down.

C2 Screen into the head front



YOU NEED

Guition 4.3 inch screen board (tested)

4 printed screen washers

 4 x M2 x 6

Lay the head front face down on a soft cloth. Put the screen board in, glass first, so the glass sits on the ledge round the window. The board's USB-C socket must face the robot's right side (your left as you look into the back of the head), next to the port door opening.

Put a printed washer over each of the four corner holes and drive 1 x M2x6 through each into the bosses. Tighten in a cross pattern, and stop as soon as each screw is snug.

Plug the 5 V lead into port P1 and the sensor lead into port P4 now, while you can still reach them.

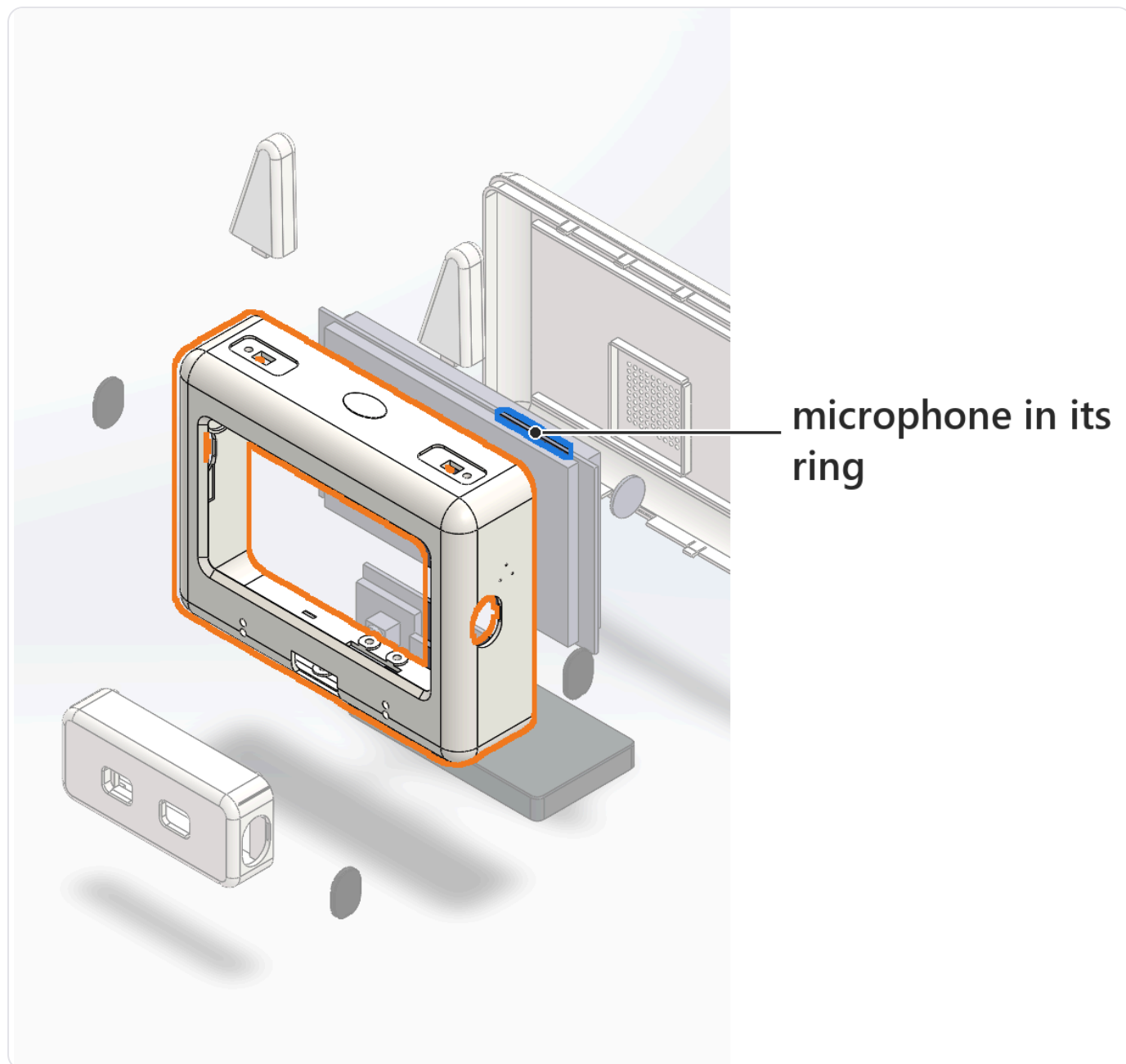
CHECK

Turn the head over: the picture area sits evenly in the window, with no gap at any edge.

COMMON MISTAKE

Over-tightening a corner screw. The glass is bonded to the board and cracks if the board bends.

C3 Microphones



YOU NEED

2 x MS3625 microphone, wired
(Wiring, S11 and S12)
superglue

Each microphone sits in the round ring on the inside of a side wall, with its sound hole facing the three tiny holes in the wall.

Put a very small drop of superglue on the ring's rim, not near the holes, and press the microphone in, sound hole outward. Hold it for 20 seconds.

The left microphone is the one with its L/R pin wired to GND.

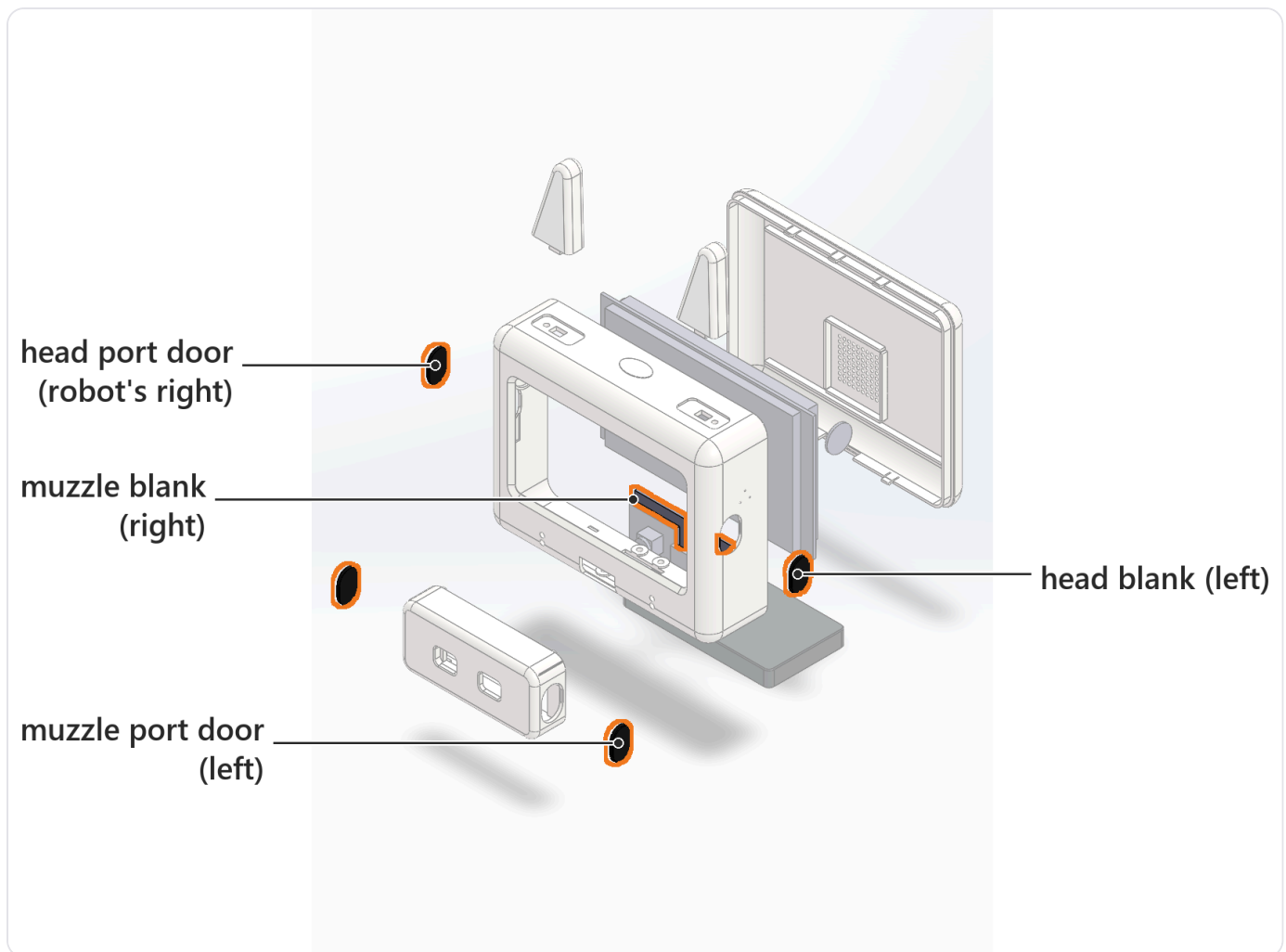
CHECK

Hold the head up to a light: you can see light through the three holes, and nothing blocks them from inside.

COMMON MISTAKE

Glue on the sound hole. The microphone is then deaf, and it cannot be cleaned.

C4 Head touch pad and port doors



YOU NEED

foil piece about 20 x 15 mm
and its grey wire (MPR121 E4)



1 x M2 x 4

4 black port doors and blanks

Head touch pad: lay the foil under the raised oval on the inside of the head's top, wrap the stripped wire end round an M2x4, and drive it through the foil tab into the pad hole, exactly as for the lid pads.

Port doors: the door with the finger notch covers a USB-C socket and the plain blank matches it on the other side, so the head looks the same from both sides. Press the head door into the opening on the robot's right side, and the head blank into the left. The muzzle door goes on the robot's left of the muzzle and its blank on the right.

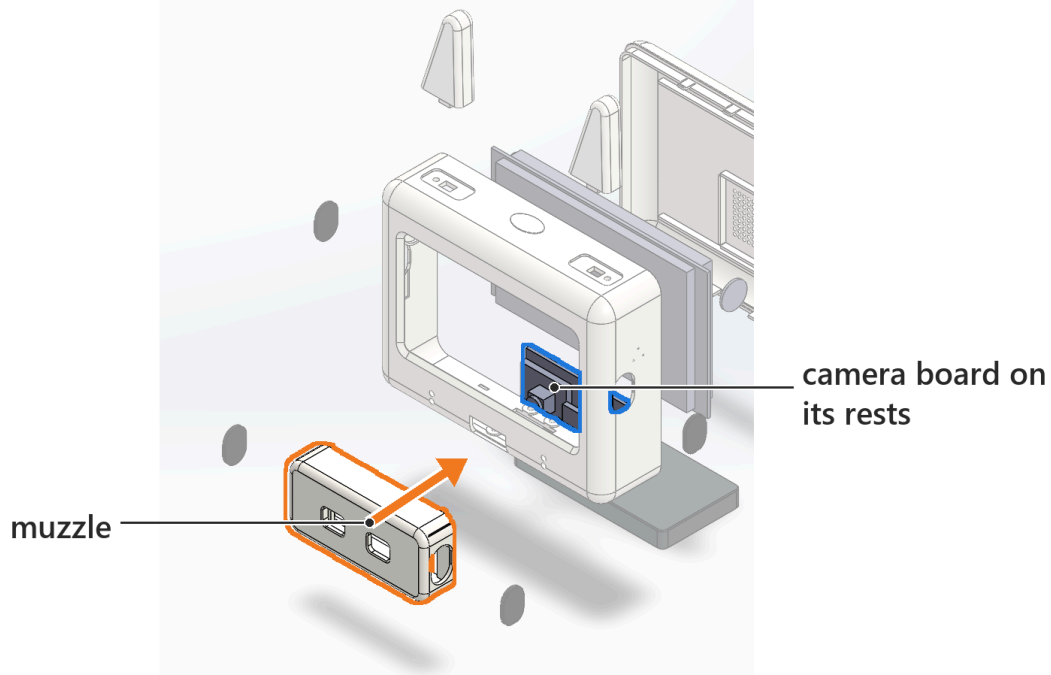
CHECK

Each door clicks in flush and comes out again with a fingernail in the notch.

COMMON MISTAKE

Using an M2x6 for the touch pad: it comes through the top of the head.

C5 Muzzle: camera and gesture sensor



YOU NEED

ESP32-S3 camera board with OV5640 (tested)

APDS-9960 gesture sensor
foam tape

 4 x M2 x 8

MAY CHANGE

This step may change in the next version. The muzzle design is still being decided (the current muzzle, a forehead camera above the screen, a camera in a collar tag, or a smooth chin, since the screen already draws a mouth). Build it as shown here for now.

Stand the camera board on the muzzle floor, against its rests, with the camera lens pushed into the square pocket behind the window on the robot's right (your left as you face the robot). Its USB-C socket then faces the muzzle port door.

Stand the gesture sensor on its shelf behind the other window, chip centred in the window, and fix it with a small square of foam tape.

Plug the camera board's 5 V lead and the gesture sensor lead, feed them through the hole in the head's face, then hold the muzzle against the head and drive 4 x M2x8 from inside the head into the muzzle.

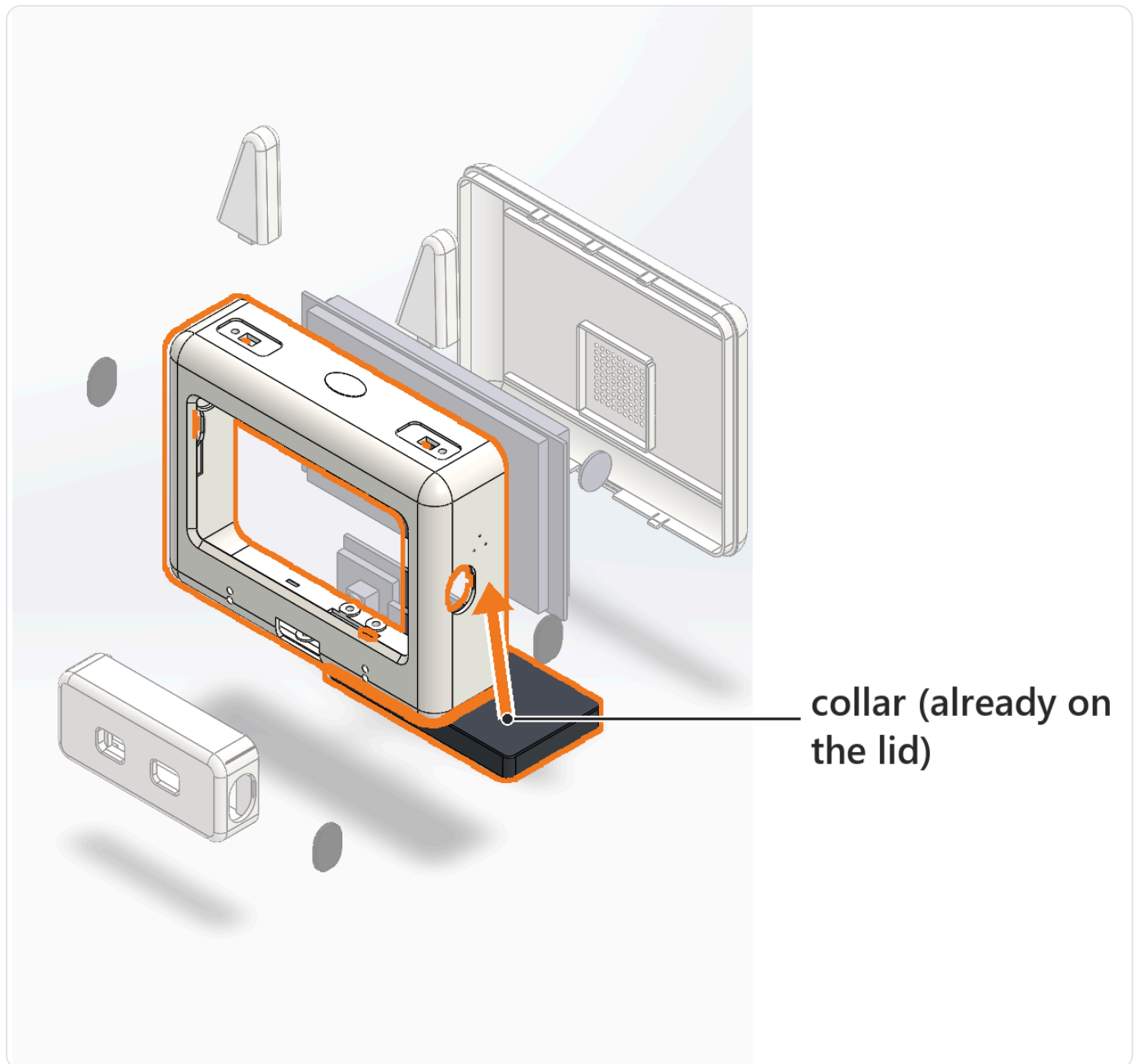
CHECK

Look into the muzzle windows from the front: you see the lens in one and the sensor chip in the other, both centred.

COMMON MISTAKE

Pinching the thin camera ribbon. Fold it gently; a sharp crease breaks it.

C6 Head onto the collar



YOU NEED

the lid with the collar fitted
(step B7)

 4 x M2 x 8

Feed the head leads that stick out of the collar up through the hole in the head's floor.

Sit the head on the collar and drive 4 x M2x8 down through the head floor into the collar, from inside the head.

Connect the leads: 5 V to P1 and the sensor lead to P4 if you have not already, the gesture sensor lead to its row, and the head touch pad wire to the touch board lead.

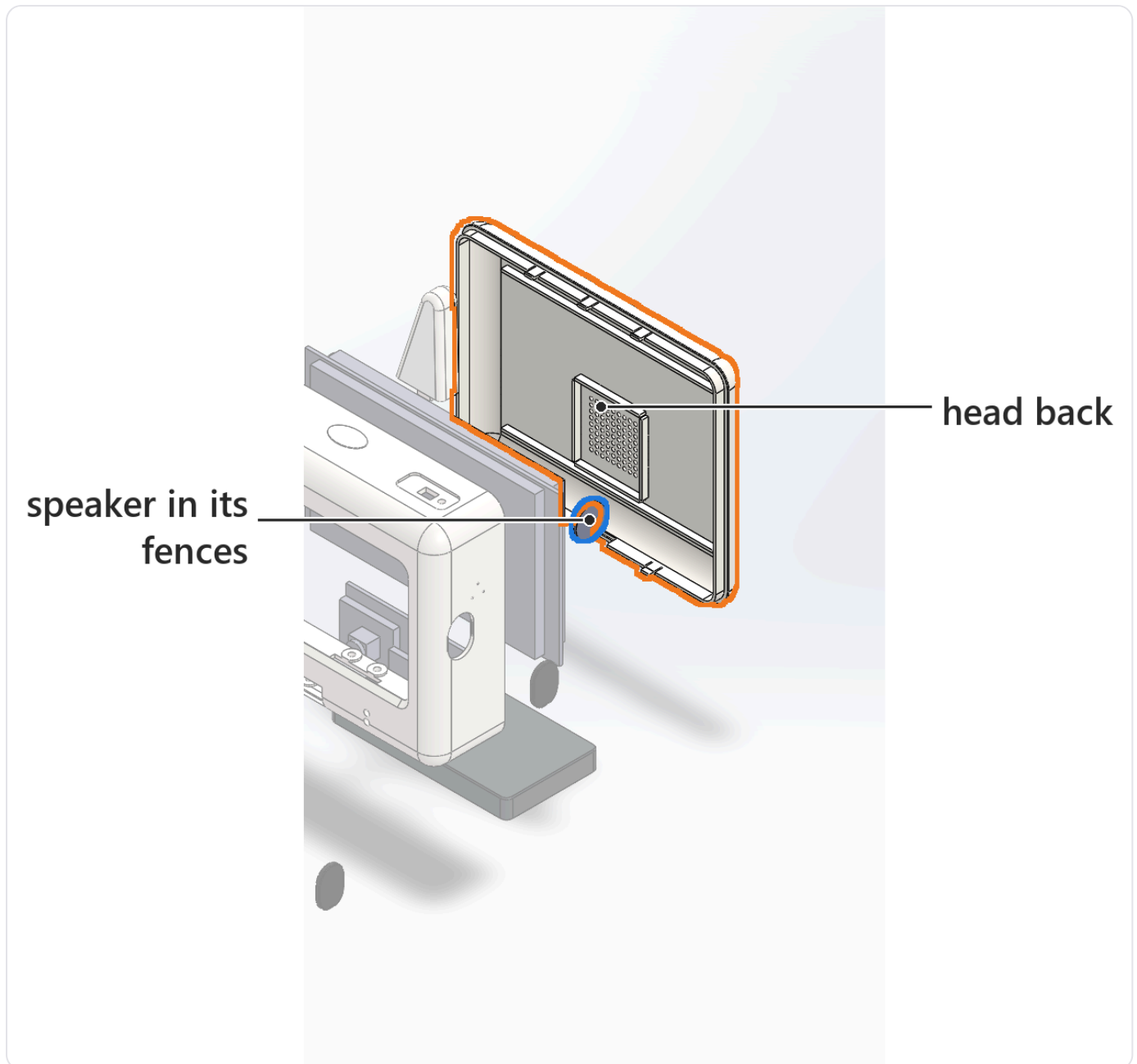
CHECK

The head sits square and does not wobble. The even gap under the head is meant to be there.

COMMON MISTAKE

Trapping a lead between the head floor and the collar.

C7 Speaker into the head back



YOU NEED

3 W speaker with its 1.25 mm plug

The speaker sits in the fences on the inside of the head back, facing the grille of small holes.

Press it in, plug its lead into the screen board's "Speak" port, and put the head back aside. It clips on at the very end (step G4), after the ears are fitted.

CHECK

The speaker cannot rattle in its fences.

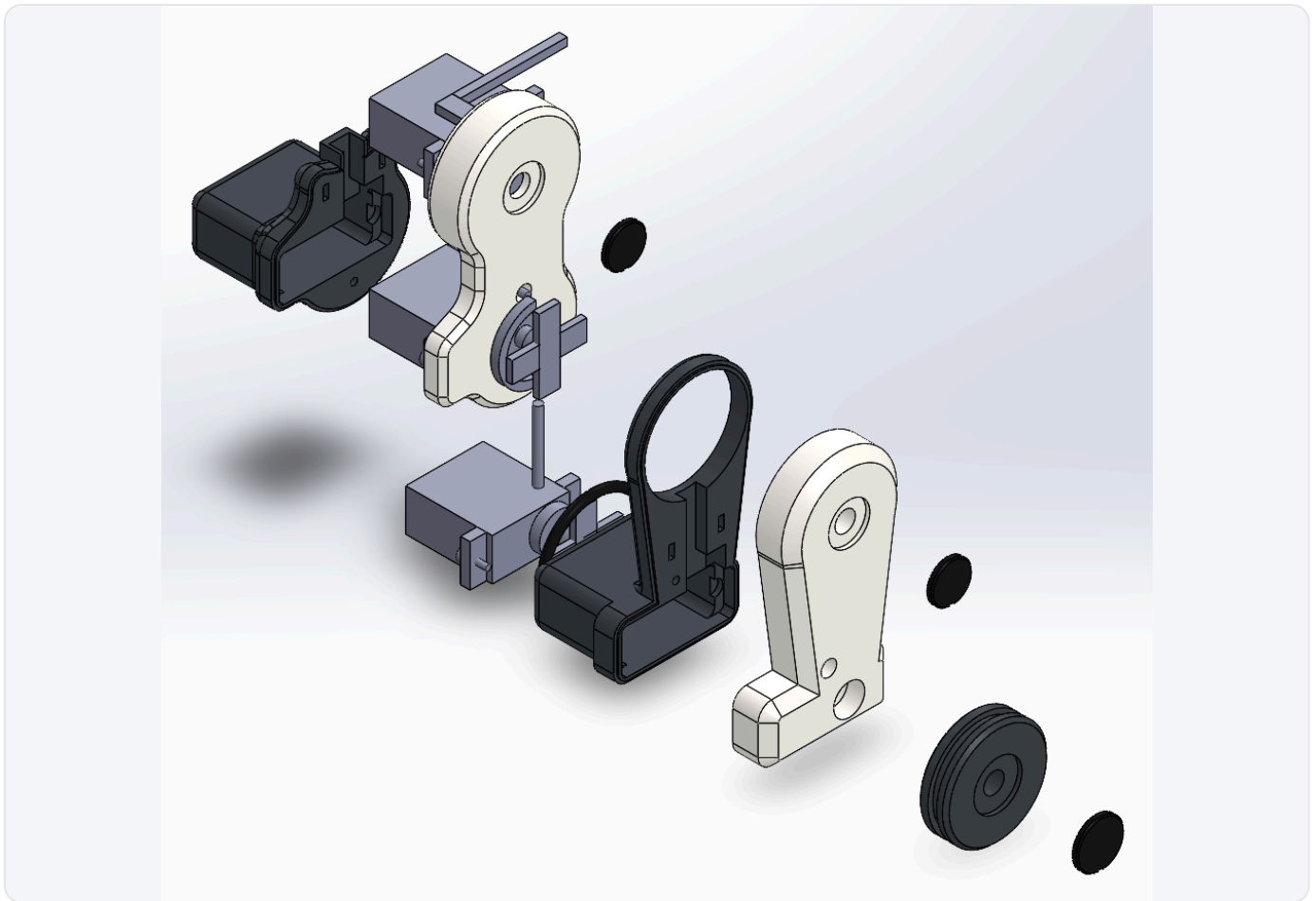
COMMON MISTAKE

Clipping the head back on now. You still need to reach inside the head for the ears.

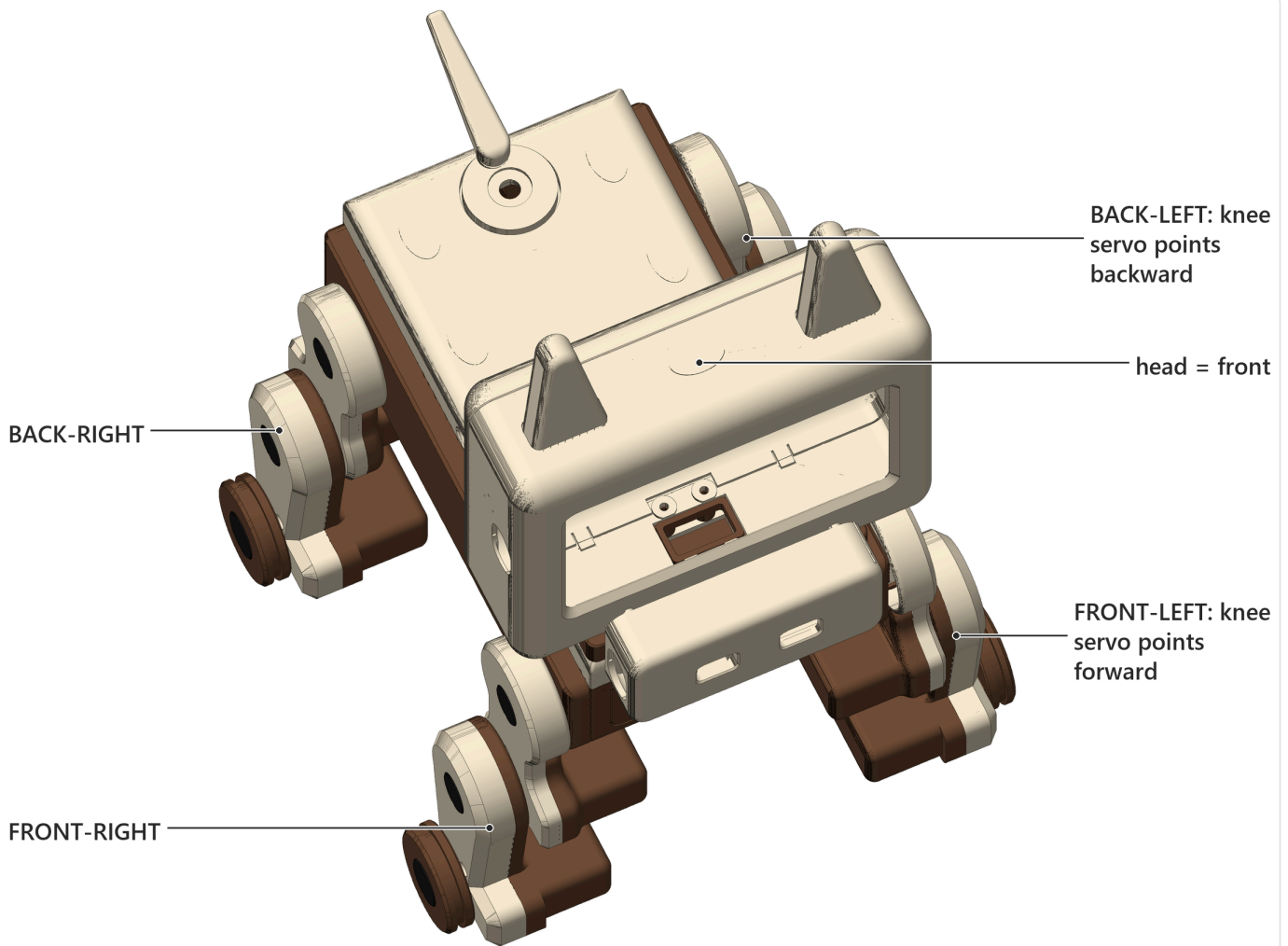
SECTION D

D Legs

Each leg is a thigh and a shin, each made of an outer half and a cover, plus a wheel. It carries the knee servo and the wheel servo, and hangs on the hip servo in the tub. All the leads run inside the leg. Build one leg all the way, then the other three.



D1 Which leg is which



YOU NEED

for each leg: thigh outer + cover, shin outer + cover, wheel, 3 caps

per leg: 2 MG90S 180 (knee) and 1 MG90S 360 (wheel), all centred

The four legs are built the same way. The pictures in this chapter show the front-left leg; build the other three the same.

Each printed file has its leg in the name, for example Thigh_FrontLeft_Outer. Some shapes are identical, so if you mix two up nothing is lost: the front-left and back-right thighs are the same shape turned round, and so are the front-right and back-left. The two left shins match, the two right shins match, and all four wheels match.

The knee servo body always points away from the middle of the robot: forward on the front legs, backward on the back legs. Every paw (the wheel servo body) points forward.

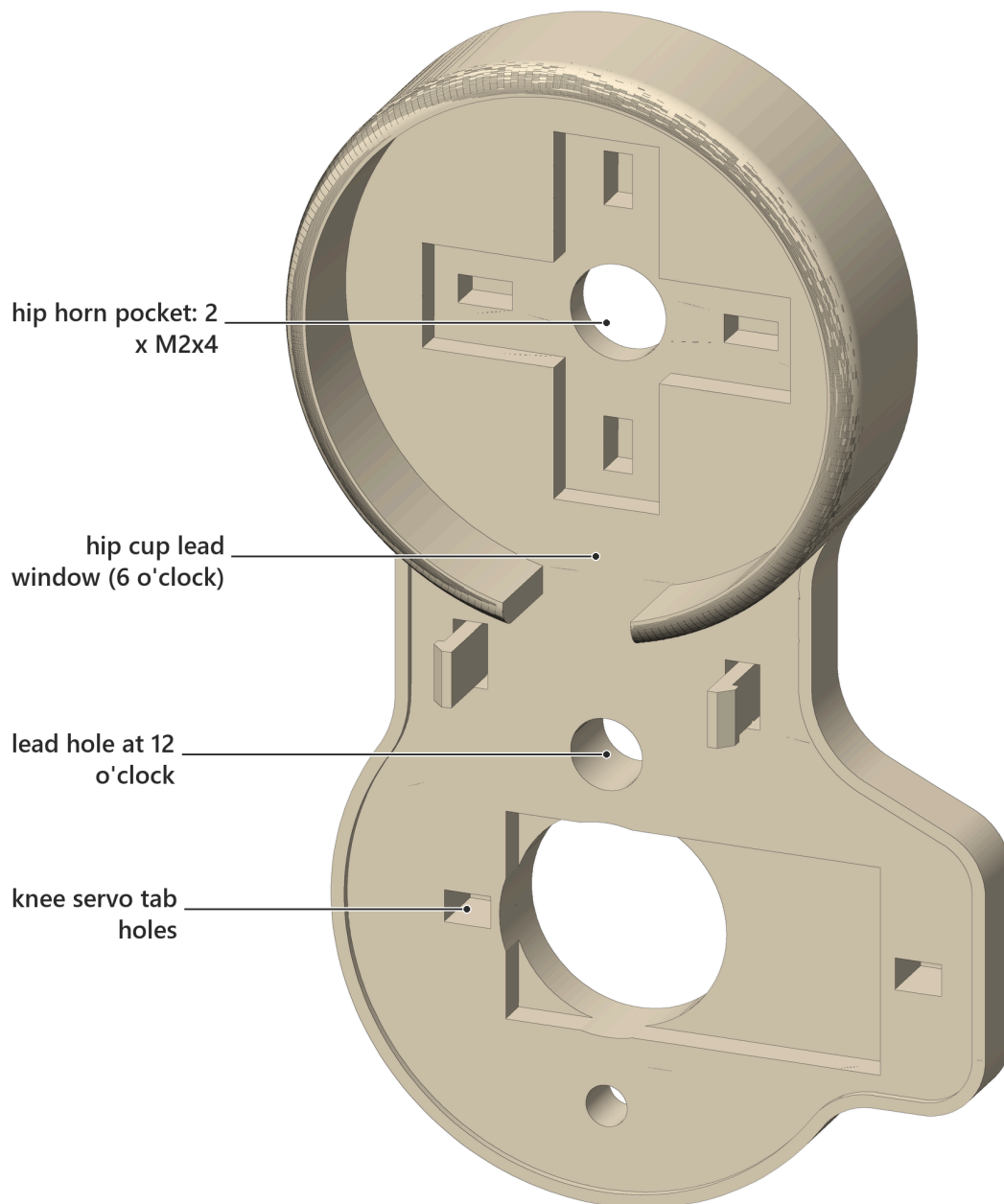
CHECK

Lay out the four sets on the table in the shape of the robot before you start.

COMMON MISTAKE

Mixing a left shin with a right leg. Left and right shins are mirror images and do not fit the other side.

D2 Hip horn into the thigh



YOU NEED

thigh outer half
a 4-arm cross horn from a servo bag
2.0 mm drill bit (turned by hand)

 2 x M2 x 4

The holes in the servo horns are about 1.5 mm, too small for an M2 screw. Pick the two opposite arms you will use and open one hole on each to 2 mm by turning the drill bit between your fingers. Use holes between 6.7 and 10 mm from the horn's centre.

Lay the horn in the cross-shaped pocket on the inside face of the thigh, at the hip end, with its hub facing out of the pocket, and drive 1 x M2x4 through each opened hole.

CHECK

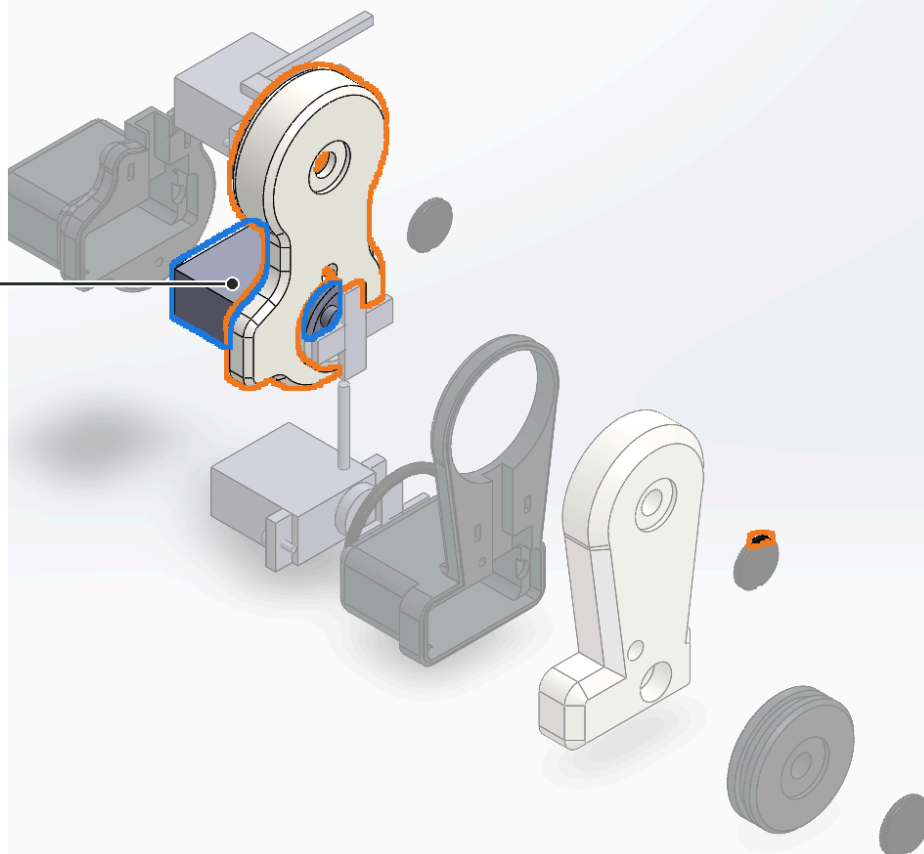
The horn sits flat in its pocket and cannot turn.

COMMON MISTAKE

Using M2x6 here. The thigh is thin at the hip, and a longer screw comes out through its outer face.

D3 Knee servo onto the thigh

knee servo, body
pointing away from
the body



YOU NEED

MG90S 180, centred, lead
flagged K + leg name



2 x M2 x 6

Set the knee servo onto the knee end of the thigh outer half, tabs flat on the face, with the round output boss in the round bore. Its body points away from the middle of the robot: forward on a front leg, backward on a back leg.

Fix it with 2 x M2x6 through the tabs into the blind holes. The knee servo's lead leaves the body at the end nearest the knee.

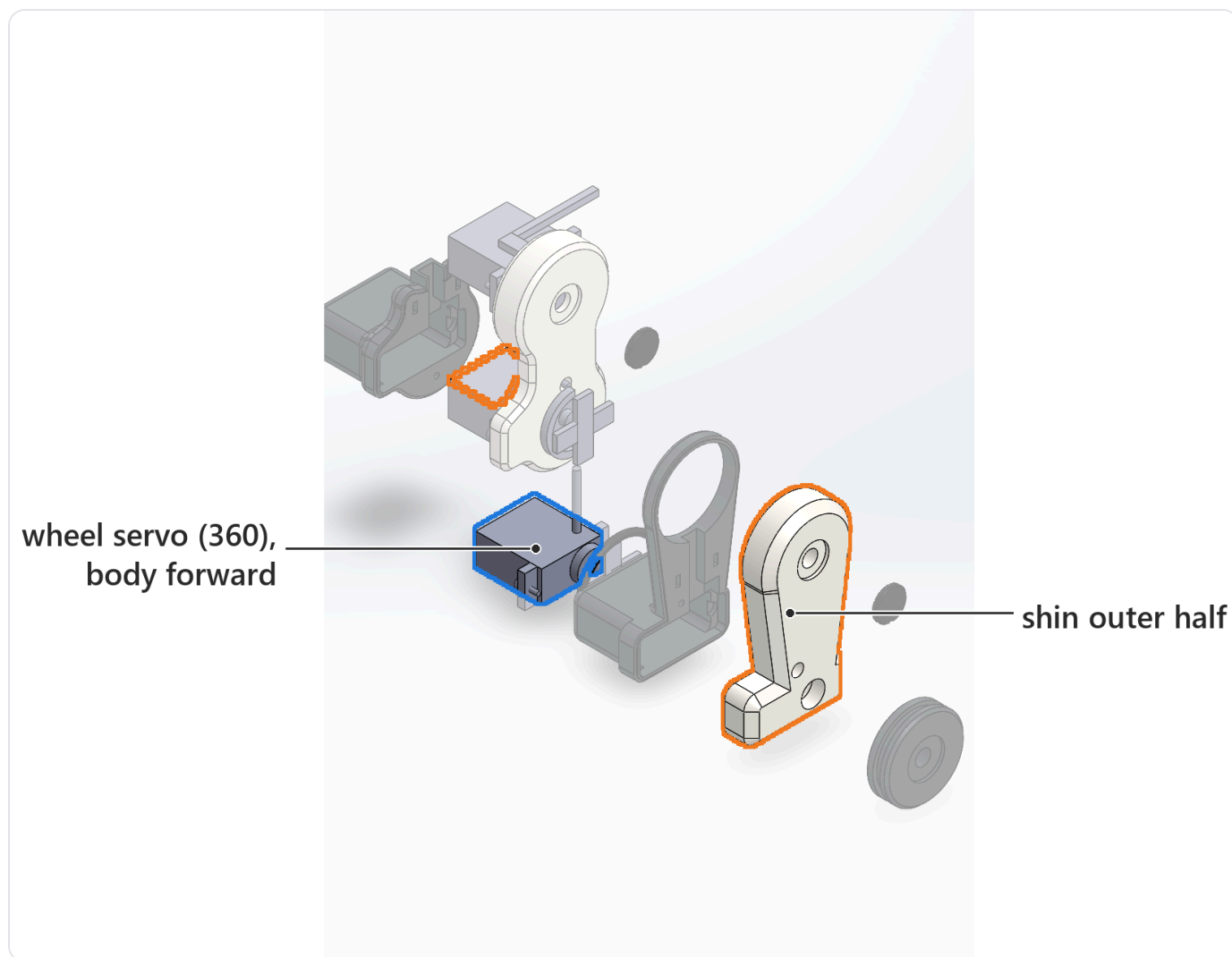
CHECK

The servo spline sits in the middle of the bore and the tabs touch the face all the way along.

COMMON MISTAKE

Turning the servo round so its body points toward the robot's middle. It will then hit the belly when the leg swings.

D4 Knee horn and wheel servo onto the shin



YOU NEED

shin outer half

cross horn (two holes opened to 2 mm)

MG90S 360, lead flagged W + leg name



4 x M2 x 6

At the knee end of the shin outer half there is a cross-shaped pocket, just like the thigh's. Lay a horn in it, hub facing out, and fix it with 2 x M2x6.

At the paw end, set the wheel servo (the 360 degree one) with its tabs on the face and its round boss in the bore, body pointing forward on every leg. Fix it with 2 x M2x6.

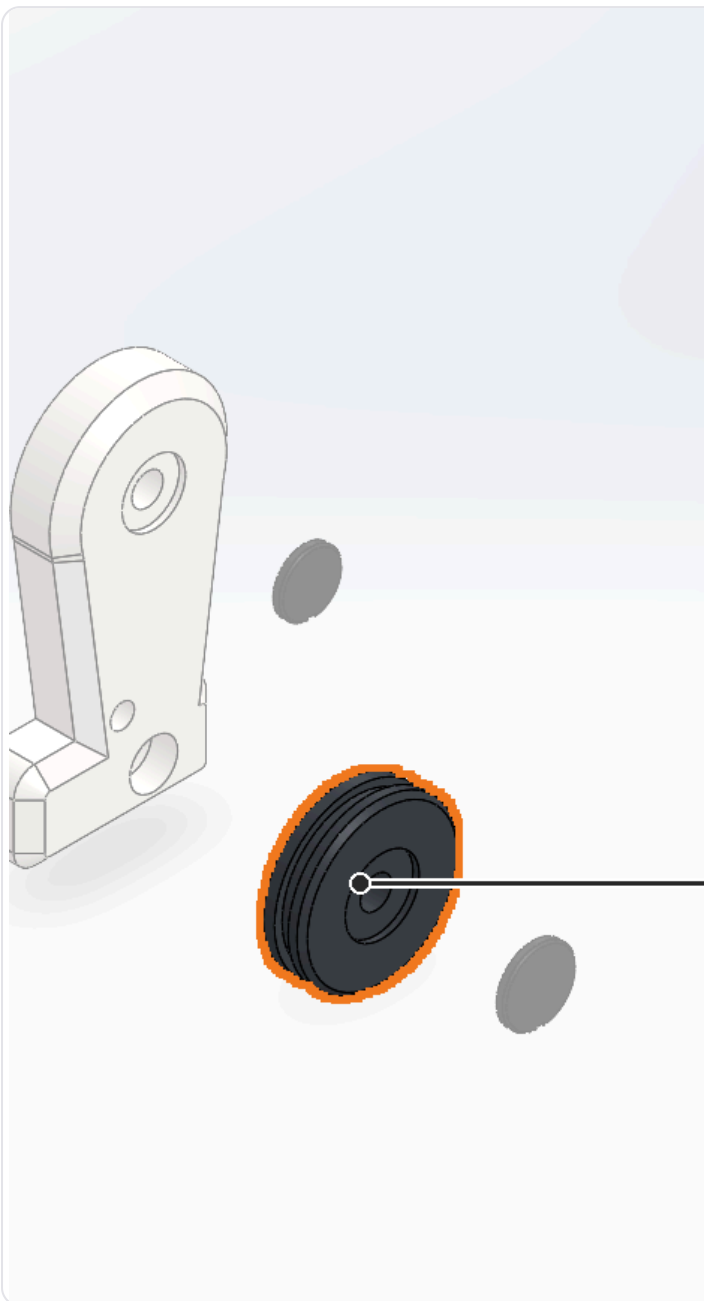
CHECK

Both screws of each pair are the same length. The knee horn screws here are M2x6, unlike the thigh.

COMMON MISTAKE

Fitting a 180 degree servo in the paw. The wheel servos are the 360 degree ones from the pack of four.

D5 Horn into the wheel



wheel: horn pocket
on the inner face

YOU NEED

wheel

cross horn (two OUTER holes
opened to 2 mm)



2 x M2 x 6

Lay a horn in the cross-shaped pocket on the inner face of the wheel, hub facing out, and fix it with 2 x M2x6 through the outer holes, at least 7.6 mm from the centre.

Put the wheel aside for now. It goes on after the shin cover.

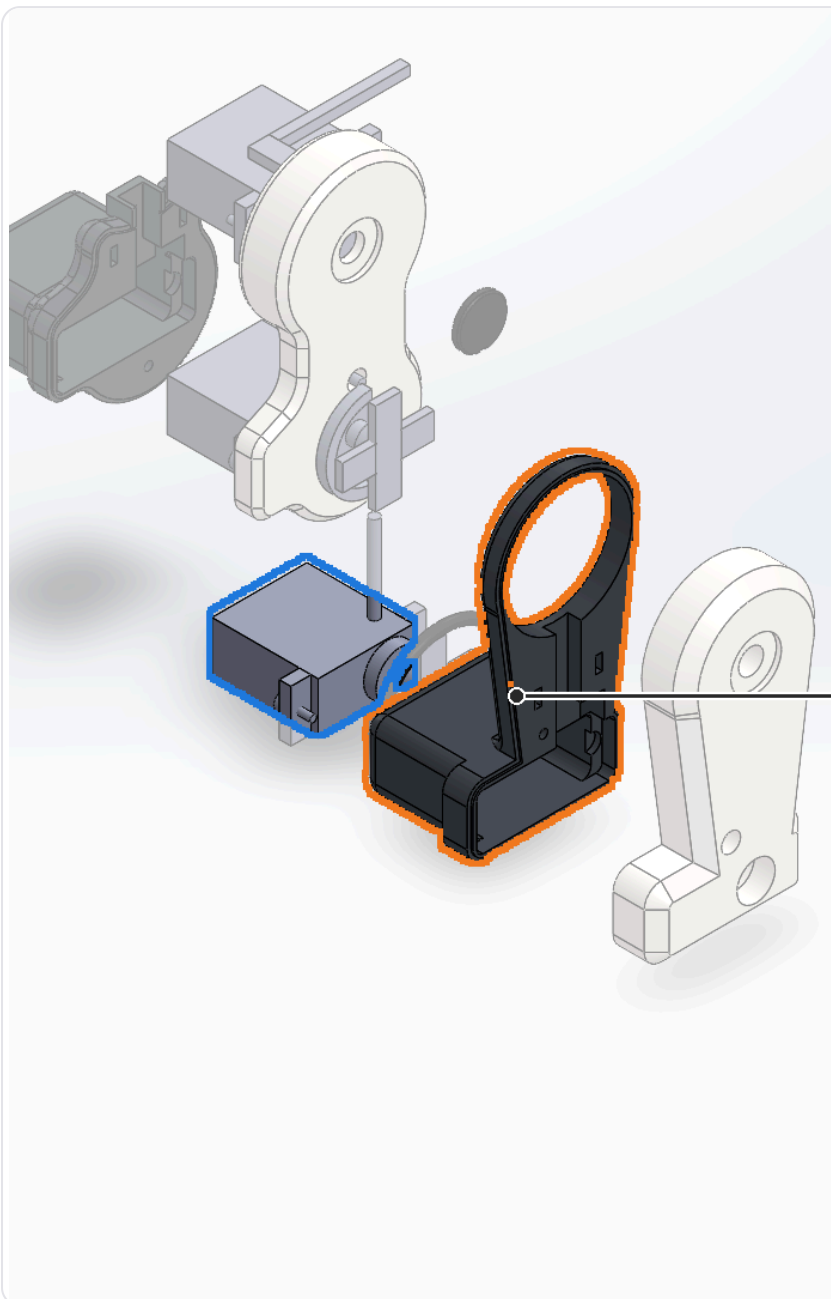
CHECK

The horn is flat in the pocket.

COMMON MISTAKE

Using the inner horn holes. The screw then misses the thicker part of the wheel.

D6 Shin cover, with the wheel lead inside



shin cover: tongue
at the knee ring,
1 x M2x8 under the
wheel

YOU NEED

shin cover



1 x M2 x 8

The wheel servo's lead runs up the inside of the shin to the knee. Lay it in the channel along the shin outer half, and bring it out through the window in the cover's knee ring at 6 o'clock (the bottom of the ring).

Slide the cover's tongue into the knee ring, then press the cover onto the shin until both snap hooks click. The paw pod closes round the wheel servo body.

Drive 1 x M2x8 into the hole in the outer face, under where the wheel will sit.

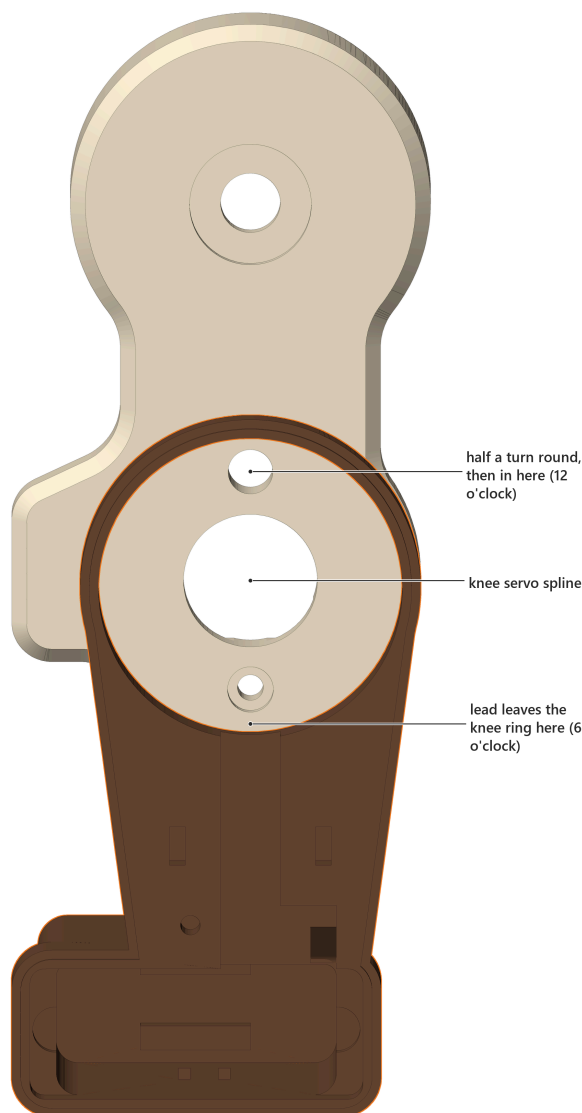
CHECK

No part of the lead shows along the shin, and about 20 cm of it comes out of the knee ring window.

COMMON MISTAKE

Trapping the lead under the cover's edge. If the cover does not close with a gentle press, open it and re-lay the lead.

D7 Join the knee (lead half a turn round)



YOU NEED

thigh with knee servo (D3), shin with cover (D6)

servo's own horn screw

12 mm black knee cap

Hold the thigh and the shin side by side. Pass the wheel lead from the knee ring window half a turn round the knee, then in through the small hole at 12 o'clock in the thigh face. This loose half turn is what lets the knee bend without pulling the lead.

Check the knee servo is still centred. Hold the leg straight, thigh and shin in one line, and push the shin's knee horn onto the knee servo spline.

Drive the servo's own horn screw through the 6 mm hole in the shin's outer face into the spline. Press the 12 mm cap into its recess, with the cap's small notch at the bottom so you can prise it out later.

CHECK

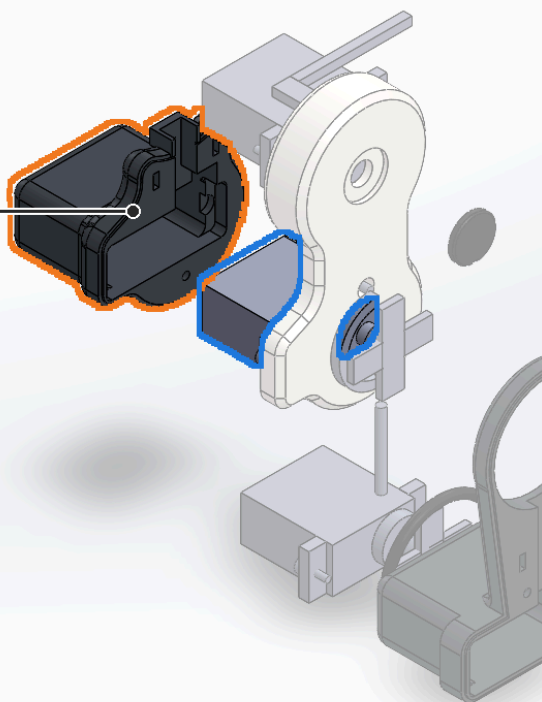
Bend the knee gently to about half way both ways: the lead slides round freely and never tightens.

COMMON MISTAKE

Fitting the horn one tooth off. If the leg is not straight with the servo centred, pull the shin off and move it one tooth.

D8 Thigh cover, both leads inside

thigh cover:
tongue into the
hip cup window, 1
x M2x8



YOU NEED

thigh cover



1 x M2 x 8

Now two leads run up the thigh: the knee servo's own lead, from the end of its body nearest the knee, and the wheel lead coming out of the 12 o'clock hole. Lay both in the channel up the thigh, into the hip cup, and out through the cup's window at 6 o'clock.

Slide the cover's tongue into the window at the bottom of the hip cup and press the cover on until both hooks click. The knee pod closes round the knee servo body.

Drive 1 x M2x8 into the hole in the outer face at the bottom of the knee disc. The shin hides it.

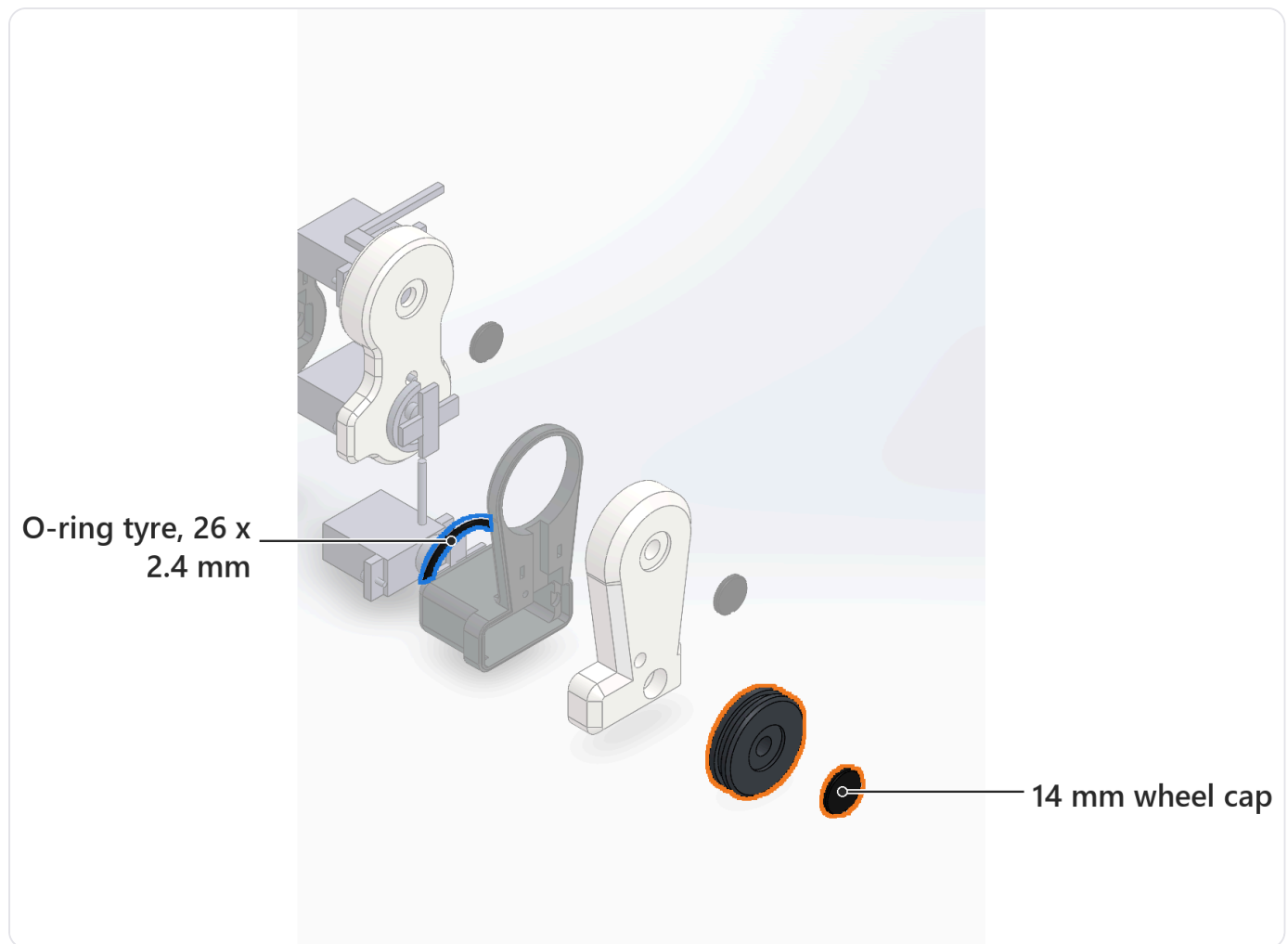
CHECK

Both leads come out of the hip cup, and each is at least 20 cm long from there.

COMMON MISTAKE

Cutting or shortening a lead. The wheel lead needs about 200 mm to reach the servo board and the knee lead about 135 mm.

D9 Wheel and tyre



YOU NEED

wheel with its horn (D5)

horn screw

14 mm black wheel cap

O-ring, the 31 mm outside-diameter size from the assortment (about 26.2 mm inside, 2.4 mm thick)

Push the wheel's horn onto the wheel servo spline. The wheel servo turns all the way round, so there is no "centre" to find here.

Drive the horn screw through the centre hole and press the 14 mm cap in, notch at the bottom.

Pick a 31 mm outside-diameter ring from the O-ring assortment (about 26.2 mm inside, 2.4 mm thick). Stretch it into the groove round the wheel. It should sit snug without twisting, and only about 1 mm of it should show above the rims, like a tread. Check the fit on the first wheel before you do the other three.

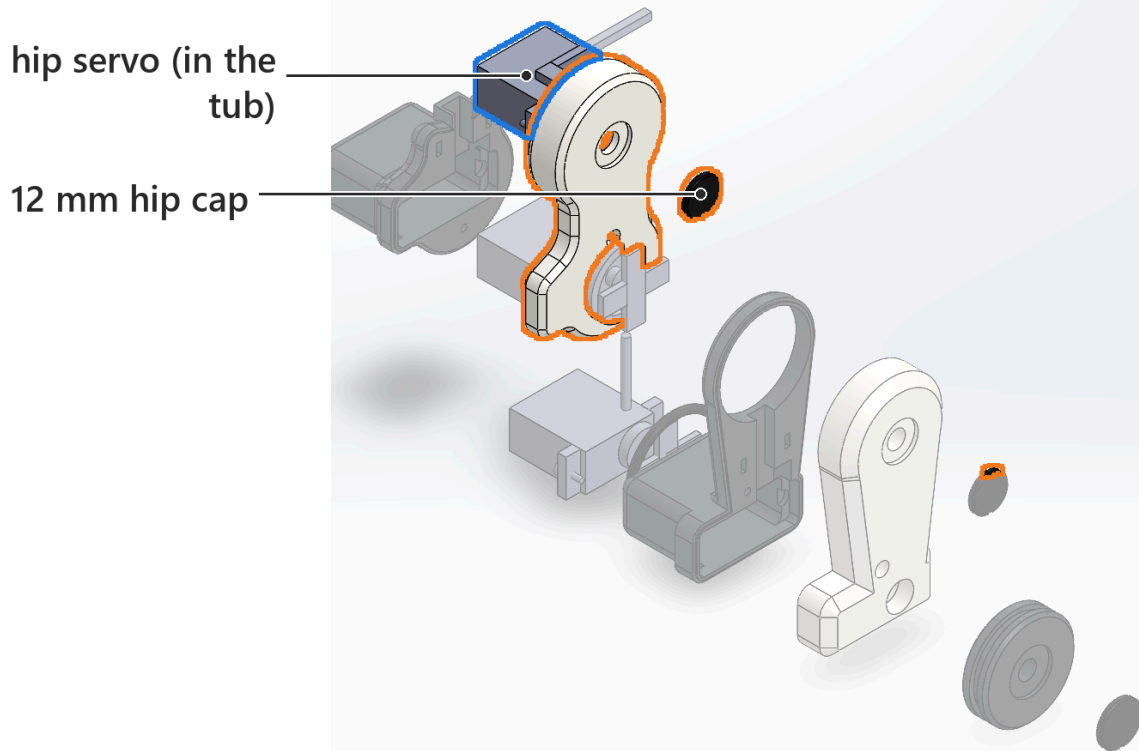
CHECK

Spin the wheel by hand: it turns smoothly, the tyre does not wobble, and no part of the rim touches the desk.

COMMON MISTAKE

Taking the wrong size from the assortment. A ring one size smaller is too tight and rolls; one size bigger falls off. Use the 31 mm outside-diameter rings.

D10 Leg onto the body



YOU NEED

the finished leg
horn screw
12 mm black hip cap

Pass both leads from the hip cup window half a turn round the hip, then out through the slot in the side panel at 12 o'clock (just above the round hole) and down into the notch in the tub wall behind it. Lay them in the notch; the lid clamps them later.

Check the hip servo is centred. Hold the leg straight down and push the thigh's hip horn onto the hip servo spline.

Drive the horn screw through the 6 mm hole in the thigh's outer face, and press the 12 mm cap in, notch at the bottom.

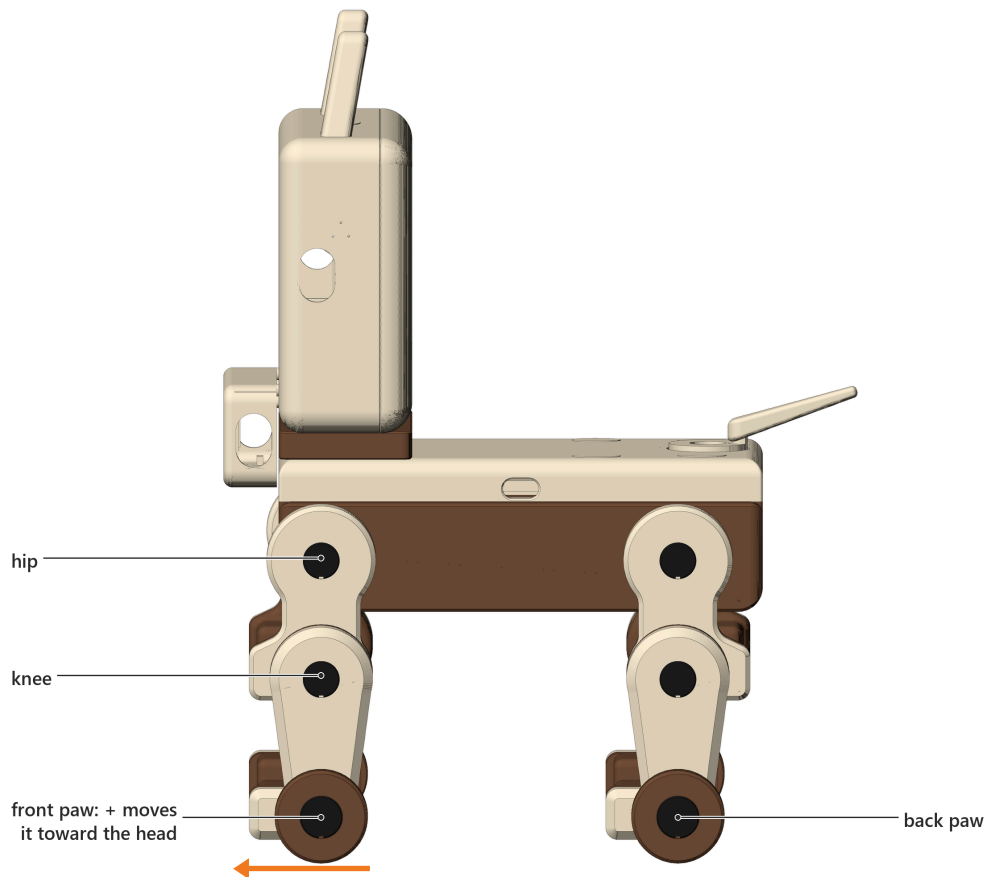
CHECK

Swing the leg gently forward and back by about 30 degrees: the leads move freely in the cup and nothing rubs.

COMMON MISTAKE

Threading the leads through a closed hole. The tub notch is open at the top on purpose: lay them in from above.

D11 Range check and joint limits



YOU NEED

all four legs fitted

Before the lid goes on, move every joint through its whole range by hand, slowly: each hip about 30 degrees forward and back, each knee about 50 degrees both ways. Watch the leads in each hip cup and at each knee. Nothing may pinch, rub or pull tight.

Angles count from the straight-down pose. Plus moves the foot forward, toward the head, and minus moves it back. The hip angle turns the thigh; the knee angle turns the shin relative to the thigh. The left and right legs are mirror images, so the firmware maps these angles to each servo's own direction.

Front legs need no limit inside hip plus or minus 30 and knee plus or minus 50 degrees. Back legs: when the hip is above plus 25 degrees, keep the knee at or below plus 45 degrees. At hip plus 30 and knee plus 50 the back paw would come within 0.9 mm of the belly; with this rule the gap never drops below 4.6 mm. Walking never gets near these poses. They only matter for tricks like sitting or folding a leg.

CHECK

Every joint moves by hand without catching, and every lead still lies in its channel.

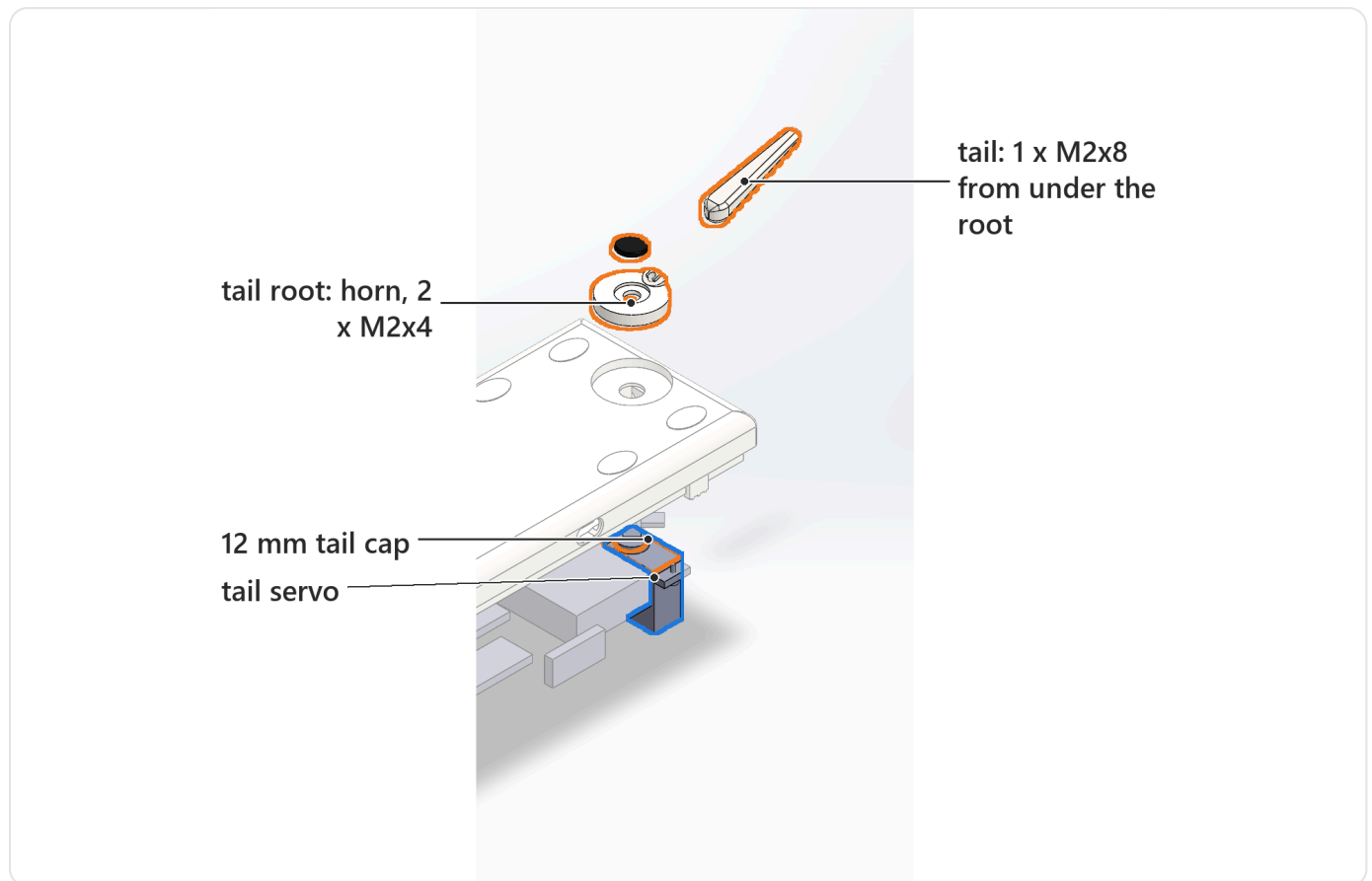
COMMON MISTAKE

Forcing a joint past a stop. Servo gears strip if you lean on them.

E Tail

The tail is a root that turns in the socket on top of the lid, with the tail keyed onto it.

E1 Fit the tail



YOU NEED

tail root, tail, 12 mm black tail cap

cross horn (two holes opened to 2 mm)

 2 x M2 x 4

 1 x M2 x 8

horn screw

Lay a horn in the pocket under the tail root and fix it with 2 x M2x4. The root is only 4.6 mm thick, so nothing longer.

Key the tail onto the root and fix it with 1 x M2x8 driven up from under the root. Do this before the root goes on the servo, because you cannot reach the screw afterwards.

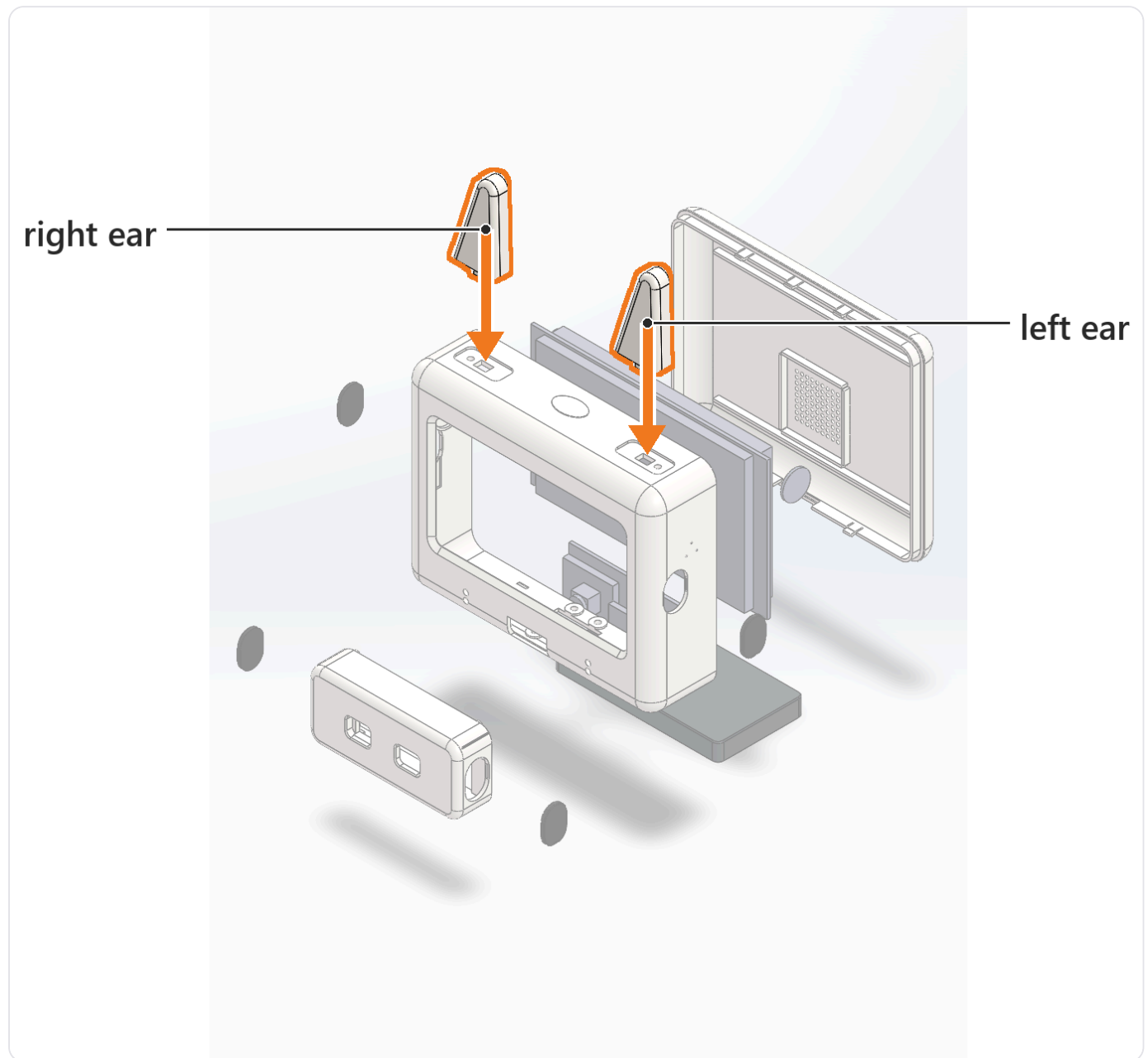
Check the tail servo is centred. Push the root into the round socket on top of the lid, horn onto the spline, with the tail pointing straight back. Drive the horn screw through the centre and press the cap in.

CHECK

The tail can swing a little both ways by hand and the root turns without rubbing the socket.

COMMON MISTAKE

Fitting the tail to the root after the root is on the servo. The tail screw is then out of reach.

F Ears**F1 Fit the ears****YOU NEED**

left and right ears



2 x M2 x 8

Each ear has a keyed peg that fits one way only into its pocket on top of the head. The ears lean back slightly.

Push each ear into its pocket, then reach inside the head and drive 1 x M2x8 up through the pad under the pocket into the ear.

CHECK

Neither ear moves when you wiggle it.

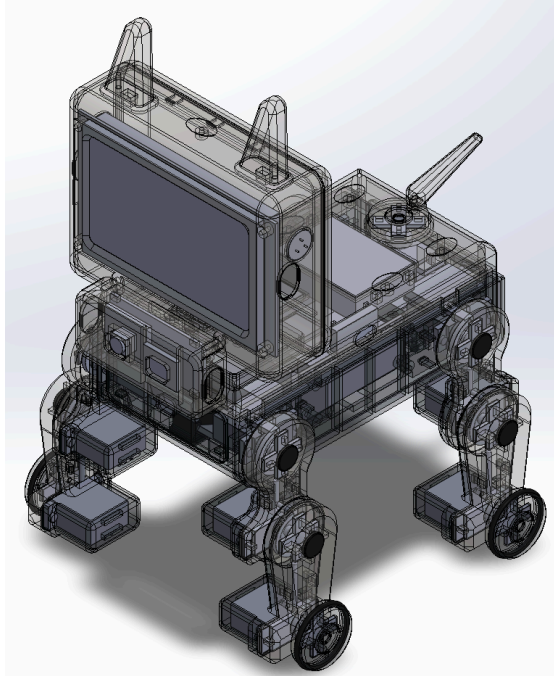
COMMON MISTAKE

Forcing an ear into the wrong pocket. If the peg does not drop in, try the other ear.

G Close up

Everything is built. Now every lead is connected, the lid with the head goes onto the tub, and the bands and the head back snap on.

G1 Connect every lead



YOU NEED

Wiring tables: power, signals,
servo channels, lasers
cable ties

With the lid standing on its side next to the tub, connect everything, one table row at a time, and tick each row off in the Wiring chapter.

Servo leads: each plugs into its channel on the servo board on its own side (servo channel table), with the brown wire on the GND row at the bottom of the header and the orange wire on the PWM row at the top. The tail servo goes to board A, channel 8.

Laser XSHUT wires go to board B, channels 9 to 15. Power and sensor leads from the tub go to the hub under the lid. Leave each tub-to-lid lead about 10 cm longer than it needs to be, so the lid can stand beside the tub while you work.

Tie slack loops with small cable ties, away from the servo horns and from the battery door opening.

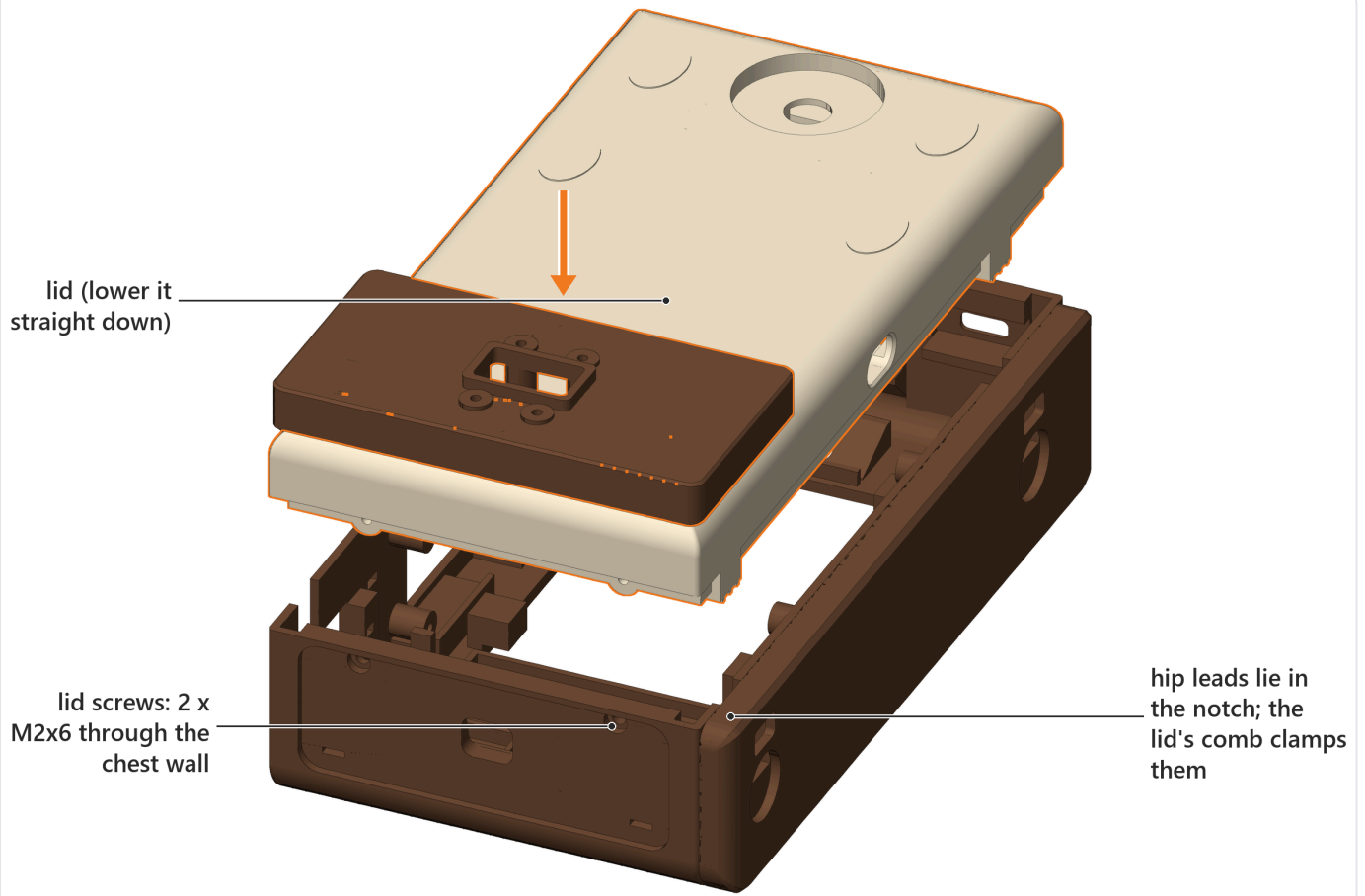
CHECK

Go through every table row once more with a finger on each plug. Every plug is fully home and no pin shows.

COMMON MISTAKE

Plugging a servo lead in upside down. The servo then does nothing; it is not damaged, just turn the plug round.

G2 Lid onto the tub



YOU NEED



4 x M2 x 6

Check that every hip lead lies flat in its notch at the top of the tub wall, and that no lead crosses the tub's top edge anywhere else.

Lower the lid straight down onto the tub. Its lip slides inside the tub wall and its combs press the leads into the notches.

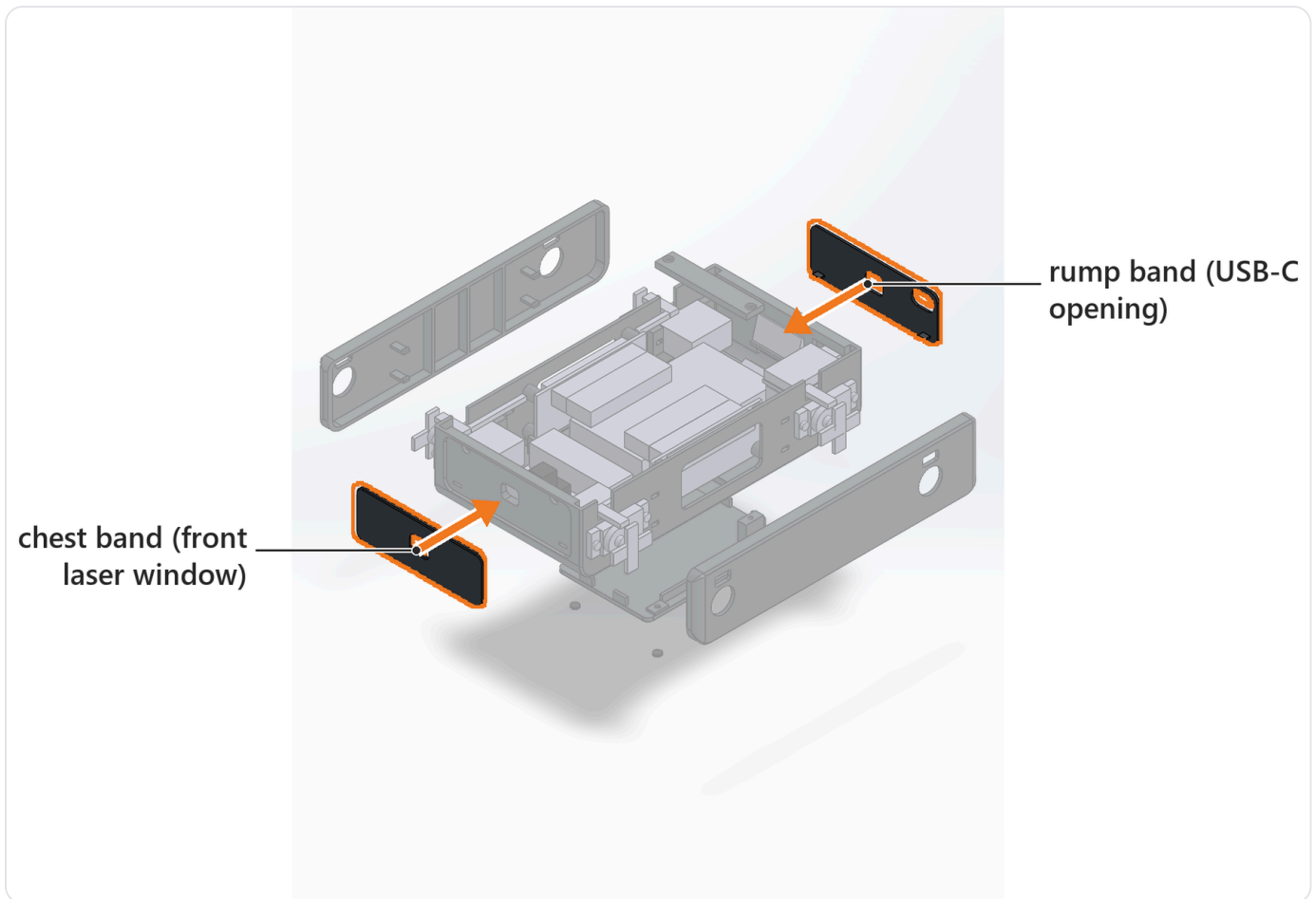
Drive 2 x M2x6 through the chest wall and 2 x M2x6 through the rump wall into the lid's bosses.

CHECK

The seam between lid and tub is even all round, and the lid does not rock.

COMMON MISTAKE

Closing the lid on a lead. If it does not sit down with a light press, lift it and find the lead.

**YOU NEED**

chest band and rump band (no screws)

The bands hide the four lid screws. Each has two snap hooks.

Press the chest band into its recess on the front of the tub, laser window over the front laser, until both hooks click. Do the same with the rump band at the back; its larger pill-shaped opening goes over the USB-C socket.

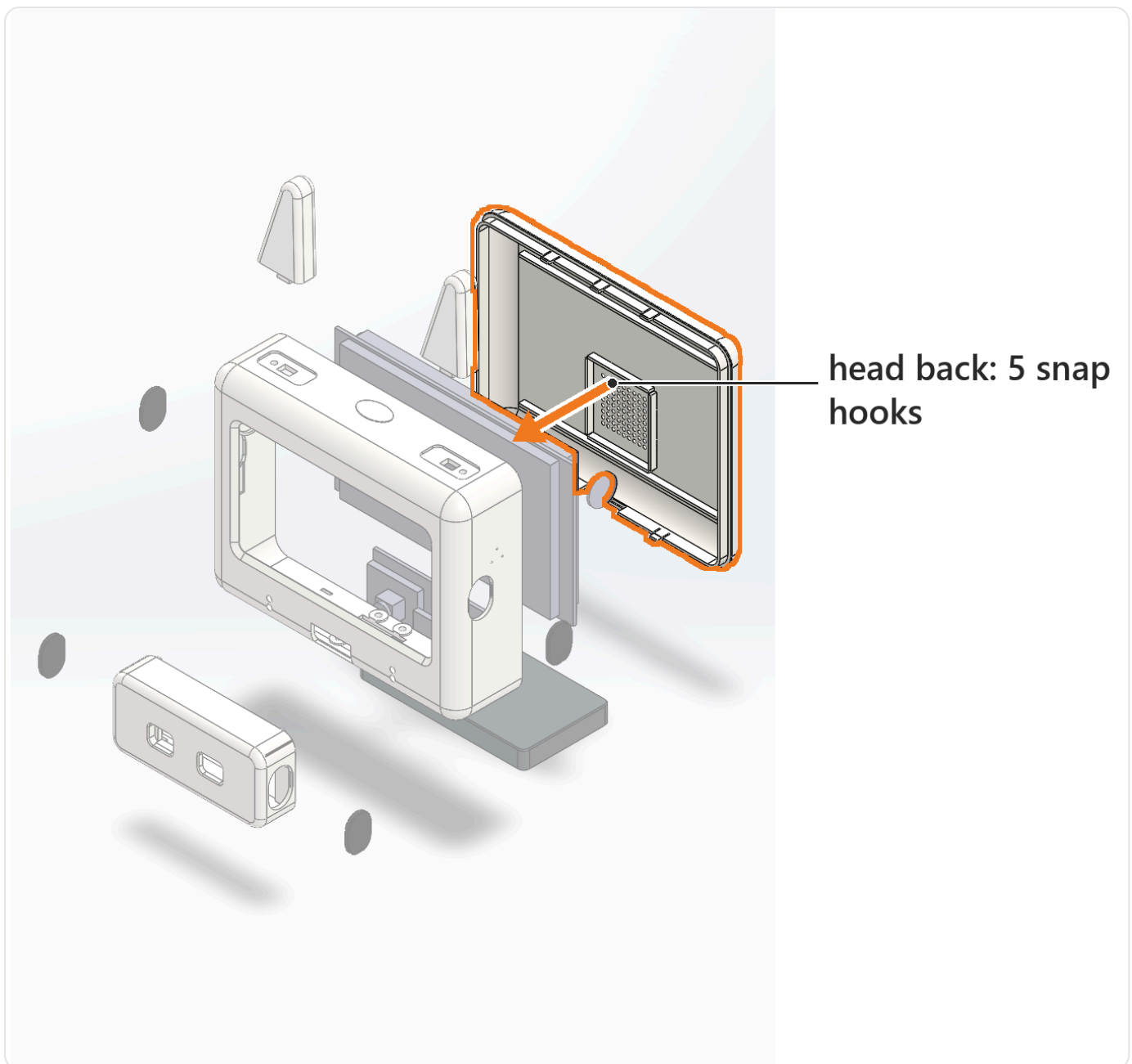
CHECK

Plug the USB-C cable in through the rump band: it goes fully home.

COMMON MISTAKE

Putting the bands on the wrong ends. The rump band is the one with the USB-C opening.

G4 Head back on



YOU NEED

head back with the speaker
(step C7)

Check the speaker is plugged into the "Speak" port and no lead lies across the rim of the head front.

Slide the head back's lip into the head front and press until all five hooks click: three along the top and two along the bottom.

CHECK

The groove between the two halves is even all round.

COMMON MISTAKE

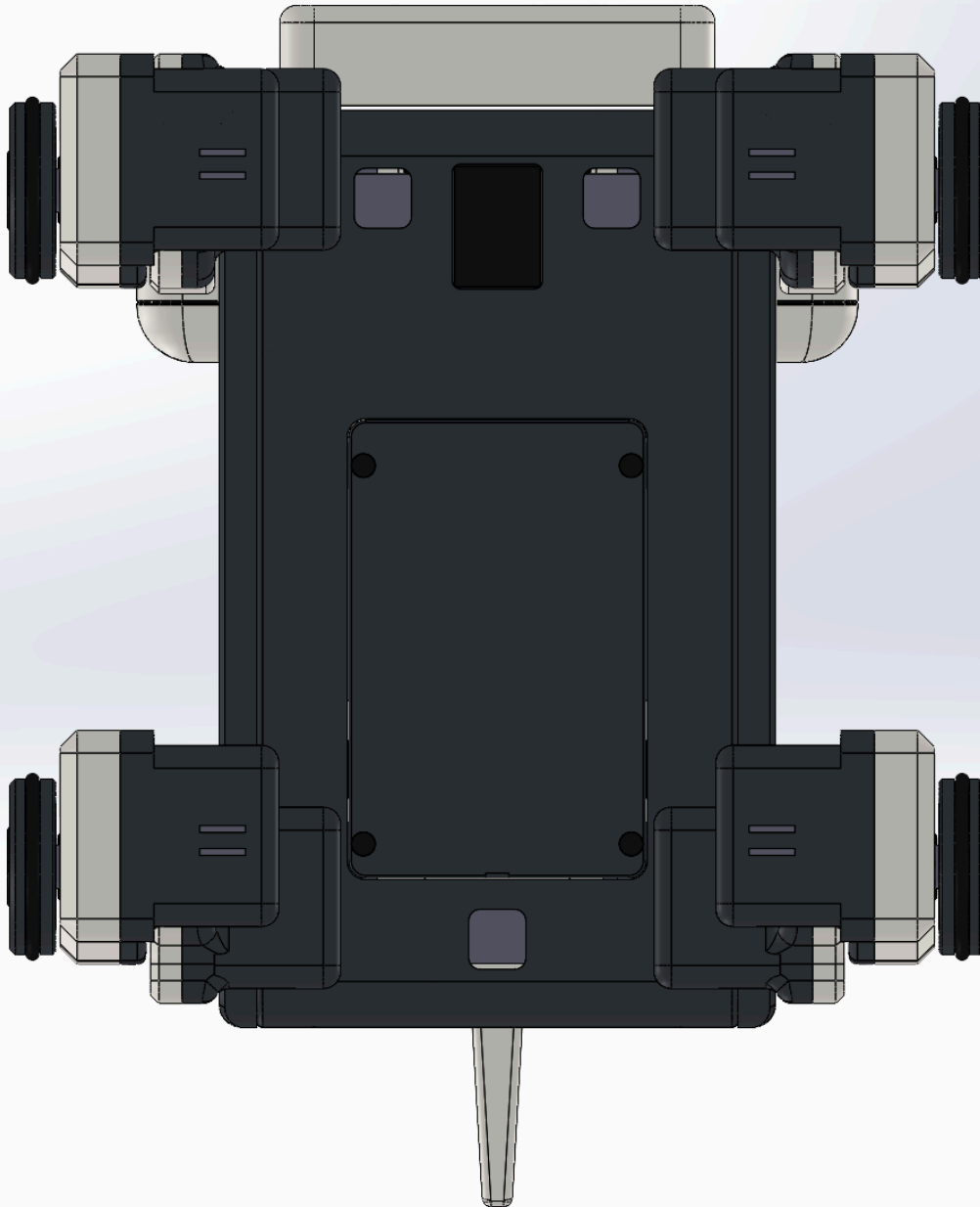
Squeezing a lead in the rim. The head back then will not sit flush; do not force it.

7 CHAPTER 7

First power-on and calibration

The cells go in, the battery door goes on, and the robot is switched on for the first time on a block, with the servo test sketch still in the screen board. Then every servo gets its small correction, written into the table at the end of this chapter.

7.1 Checks before the cells go in



YOU NEED

multimeter
a block or mug to prop the robot on

Set the power switch to OFF. Set the multimeter to its beeper (continuity) mode. Touch the probes to the protection board's P+ and P- pads: it must not beep. Then to the output of each 5 V module on the hub: it must not beep. A beep means a short circuit. Find it before you go on.

Stand the robot on a block so all four wheels are off the desk. It stays there for every first test.

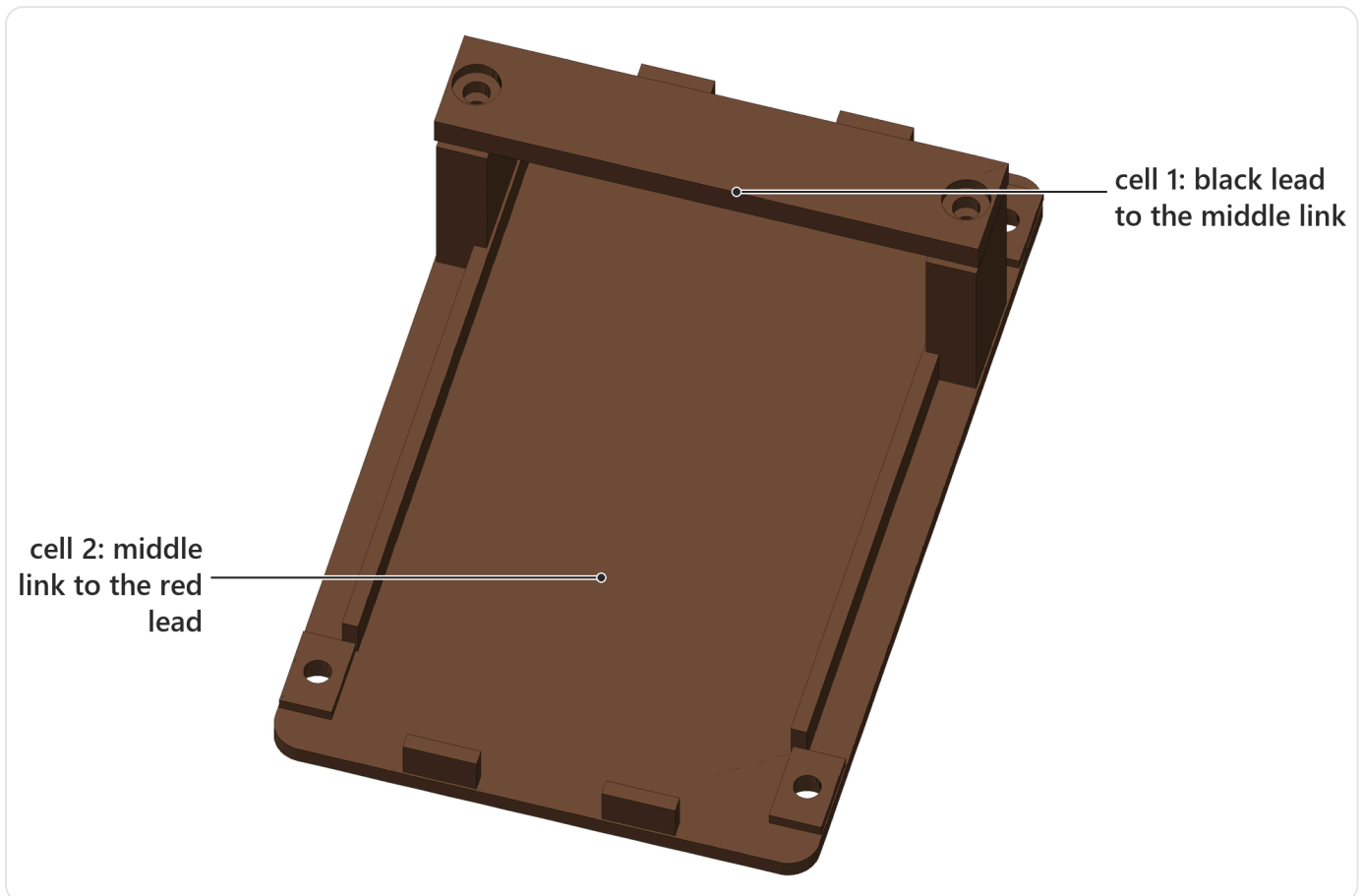
CHECK

No beep on any power pair, and the fuse is in its holder.

COMMON MISTAKE

Testing with the cells in. The beeper test is only safe with the power off and the cells out.

7.2 Cells in, protection board awake



YOU NEED

2 x Samsung 35E cells
USB-C charger and cable

Measure each cell with the multimeter on DC volts. Each should read between 3.0 and 4.2 V, and the two must be within 0.05 V of each other. If not, charge them separately or use the spare pair; never mix a Samsung with a Tinker cell.

Check the + and - marks in the holder twice. Put in first the cell that sits between the black lead and the middle wire, then the other one. This connects negative, then the middle, then positive, in the order the protection board needs.

Plug the USB-C charger into the rump socket for a few seconds and unplug it. This wakes the protection board. Now measure between P+ and P-: you should see the two cell voltages added together, between 6.0 and 8.4 V.

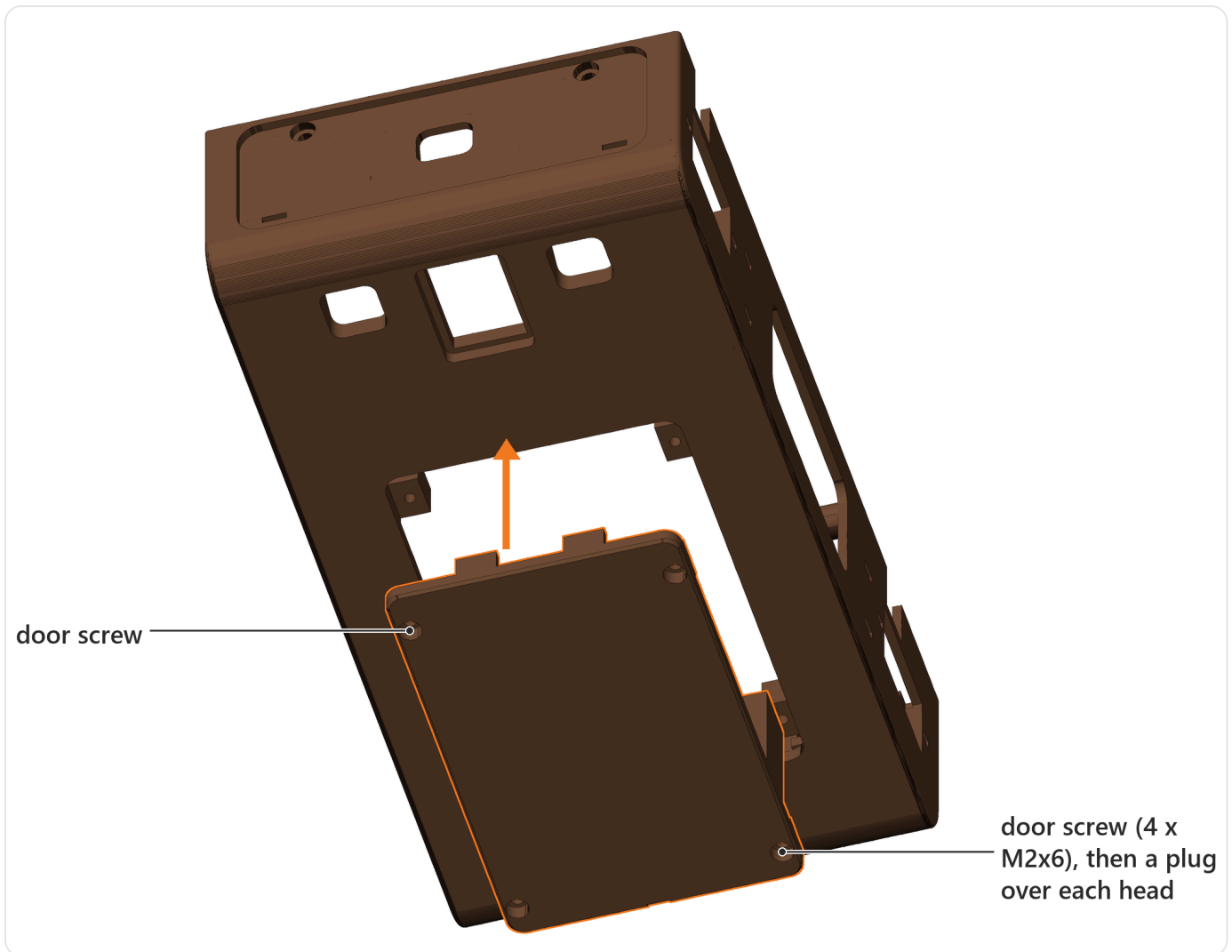
CHECK

The two cells are the same model, the same charge, and the right way round.

COMMON MISTAKE

Reading 0 V at P+ and P-. The protection board is still asleep: plug the charger in once more.

7.3 Battery door on



YOU NEED

 **4 x M2 x 6**
4 printed screw plugs

Tuck the battery leads in so nothing can be pinched, set the door into the belly opening, cells inside, and drive 4 x M2x6 into the lugs.

Press a black plug over each screw head. To open the door later, prise the plugs out with a fingernail.

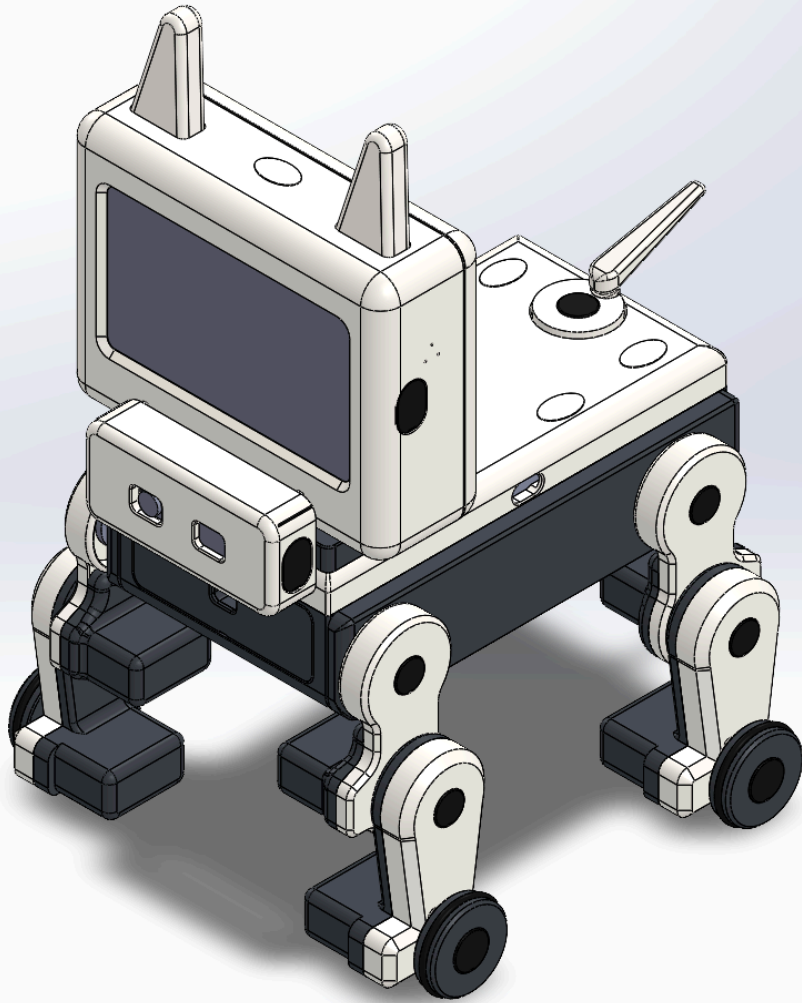
CHECK

The door sits flush with the belly.

COMMON MISTAKE

Pinching the middle wire under the door edge.

7.4 First switch-on



YOU NEED

laptop with the servo sketch
(tools/servo_center)

USB-C data cable

For the first switch-on the screen board runs the servo sketch from the Bench test chapter. Unplug the 5 V lead from port P1 (the rule: never two power sources on one board), then connect the laptop to the screen board's USB-C socket behind the head port door. The laptop now powers the screen board; the battery powers the servos.

Open the Serial Monitor (pio device monitor) and switch the robot on. Within two seconds every leg should stiffen with the legs straight and the tail straight back, and the wheels should stay still or creep slowly.

Type `s` and press Enter. The list must show 0x40 and 0x41 (both servo boards), 0x39 (gesture), 0x5A (touch) and 0x68 (motion). Type `l` to check all seven lasers answer one by one, and `v` to read the battery voltage.

CHECK

All thirteen servos hold still, nothing gets hot, and the battery reading matches the multimeter within about 0.2 V.

COMMON MISTAKE

Leaving P1 plugged in while the USB cable is in.

7.5 If something does not move

Switch off first, every time, before you touch a plug.

One servo limp: its plug is on the wrong channel, on the wrong board, or upside down. Check the servo channel table; the brown wire goes on the GND row.

Every servo limp: the servo boards are not getting 5 V, or the wheel-stop line is high. Measure between V+ and GND on each board's green terminal: it must read about 5 V. Then check the green OE wire goes to IO14 and the 2.2 kOhm resistor goes to 3.3 V, not to 5 V. Type c in the Serial Monitor to switch the outputs on again.

A servo board "NOT FOUND": check its VCC reads 3.3 V, its SDA and SCL go to the right rails, and, for board B, that the A0 pad is bridged.

A wheel keeps turning: it is normal for a 360 degree servo to creep at 1500 us. Find its real stop point in the next step.

7.6 Calibrate every servo

YOU NEED

the calibration table in this step
a pencil

A horn fits on its spline only in steps of one tooth, so a leg that looks almost straight is normal. The firmware corrects the last few degrees with a trim for each servo, and this step finds those trims.

In the Serial Monitor press n until the servo you want is selected (it wiggles). Press + or - to move it 10 us at a time until the leg is exactly straight, or the tail exactly straight back. Write the number in the table.

For each wheel, press + or - until the wheel stops completely, and write that number down as its stop point.

Finally, for each hip and knee, check which way plus moves it. Plus must move the foot forward, toward the head. If it moves back, write "reversed" in the table. The left and right legs are usually opposite.

Before long walks, check how hot the knee pods get with the multimeter's thermocouple, if it has one: stop and let the robot cool above 45 degrees C.

CHECK

Every row of the table has a number.

7.7 Charging and everyday care

YOU NEED

5 V, 2 A phone charger with a
USB-A to USB-C cable

Switch the robot off, then plug the charger into the USB-C socket in the rump band. The robot cannot run while it charges.

The firmware warns at 7.0 V and at 6.8 V parks the robot and switches the servos off. Charge it then; running the cells flatter shortens their life.

To take the cells out, switch off, prise out the four plugs under the belly, and undo the four door screws. Always swap both cells together.

7.8 Calibration table

Fill in a number for every servo. The robot firmware reads these values; keep this page with the robot.

Servo	Board / channel	Centre or straight (us)	Wheel stop (us)	+ moves the foot forward? (yes / reversed)
Hip, front-left	A / 0		—	
Knee, front-left	A / 1		—	
Wheel, front-left (360)	A / 2			—
Hip, back-left	A / 4		—	
Knee, back-left	A / 5		—	
Wheel, back-left (360)	A / 6			—
Tail	A / 8		—	
Hip, front-right	B / 0		—	
Knee, front-right	B / 1		—	
Wheel, front-right (360)	B / 2			—
Hip, back-right	B / 4		—	
Knee, back-right	B / 5		—	
Wheel, back-right (360)	B / 6			—

8 CHAPTER 8 Troubleshooting

Find the symptom, check the likely cause, then do the fix. Switch the robot off before you touch any wire, every time.

Nothing happens when I switch on.

Likely cause: The protection board is asleep, the fuse has blown, or a cell is in the wrong way round.

Fix: Switch off. Measure P+ to P- at the protection board: 0 V means it is asleep, so plug the USB-C charger in for a few seconds. If it still reads 0 V, check the fuse and the cells' + and - marks.

The fuse blows as soon as I switch on.

Likely cause: A short circuit after the fuse: a bare wire end, a solder blob, or a screw tip touching a board.

Fix: Take the fuse out. With the multimeter on beeper mode, test the switched battery wire to GND, then each 5 V module output to GND. Look for the joint that beeps and fix it before you fit a new fuse.

The screen board restarts when the legs move.

Likely cause: The servos pull the 5 V line down. Either the two 5 V modules are sharing a job, or a capacitor is missing.

Fix: Check that module #1 feeds only the servo boards and module #2 only the screen and camera boards, and that each has its 1000 uF capacitor, stripe to minus.

The screen board restarts when the lasers start up.

Likely cause: Seven lasers plus the other sensors ask too much of the screen board's 3.3 V supply.

Fix: Measure the 3.3 V rail while it happens. If it dips below 3.1 V, move the lasers' VIN to the 5 V boards rail, but only after checking that each laser's SDA and SCL pins still read 3.3 V or less. Otherwise add a separate 3.3 V module.

One servo does nothing.

Likely cause: Its plug is on the wrong channel or board, or upside down.

Fix: Check the servo channel table. The brown wire goes on the GND row. Use the servo sketch: press n until that servo is selected and watch whether it wiggles.

Every servo is limp.

Likely cause: No 5 V on the servo boards' green terminal, or the wheel-stop (OE) line is high.

Fix: Measure V+ to GND on each green terminal: about 5 V. Then check the green OE wire goes to IO14 and the 2.2 kOhm resistor goes to 3.3 V. In the servo sketch, press c.

A servo board is "NOT FOUND".

Likely cause: No 3.3 V on its VCC pin, SDA and SCL swapped or loose, or (board B) the A0 pad is not bridged.

Fix: Measure 3.3 V on VCC. Re-seat the SDA and SCL wires. Board B must show 0x41 in the scan; if both boards show as 0x40, bridge A0 on board B.

A wheel keeps creeping when it should stop.

Likely cause: Every 360 degree servo has its own stop point, a little away from 1500 us.

Fix: Find the stop point with + and - in the servo sketch and write it in the calibration table. The firmware uses that number.

A leg is not straight when its servo is centred.

Likely cause: The horn went onto the spline one tooth off, or the servo was turned by hand after centring.

Fix: Small errors are fixed with the trim in the calibration table. If the error is bigger than about half a tooth, take the horn screw out, lift the horn off, and refit it one tooth over.

A servo buzzes or gets warm while standing still.

Likely cause: It is pushing against a stop: a lead is pinched, a part rubs, or its trim asks for a pose it cannot reach.

Fix: Switch off, move the joint by hand to find what catches, and re-lay any lead. Check the joint limits for the back legs.

A laser reads a short distance all the time.

Likely cause: The protective film is still on the chip, or the chip is not centred in its window.

Fix: Peel the film off. Re-seat the board so the chip sits in the middle of the window, and keep the window clean.

Only one laser answers, or none.

Likely cause: All seven lasers start at the same address. The firmware wakes them one by one using the XSHUT wires.

Fix: Run I in the servo sketch. A laser that shows NO ANSWER has a loose XSHUT, SDA, SCL, VIN or GND wire, or its XSHUT is on the wrong channel of board B.

Touch does not register.

Likely cause: A foil pad is not clamped to its wire, or the MPR121 is not seen on the bus.

Fix: With the power off, check each pad wire beeps to its foil. Check 0x5A shows in the I2C scan.

The microphones hear nothing, or left and right are swapped.

Likely cause: Glue over the sound holes, a wrong pin on P3, or both mics with the same L/R setting.

Fix: Check the sound holes are open. One mic's L/R pin goes to GND and the other's to 3.3 V. Check the three data lines against the signal table.

No sound from the speaker.

Likely cause: The speaker is not in the "Speak" port, or a speaker wire was connected to ground.

Fix: Plug it into "Speak" only. Never connect either speaker wire to GND.

The laptop cannot find the board, or the upload fails.

Likely cause: A charge-only USB cable, or the board is also powered through P1.

Fix: Unplug the 5 V lead from P1 first. Try the phone cable test from the parts check: if the phone does not show up on the laptop, the cable is charge-only. Hold BOOT while you plug the cable in to force upload mode.

The Serial Monitor stays empty after an upload.

Likely cause: The screen board's USB-C goes to the chip's own USB port, not to a USB-serial chip.

Fix: In tools/servo_center/platformio.ini change ARDUINO_USB_CDC_ON_BOOT=0 to 1, upload again, and reopen the monitor.

A cover or panel will not snap shut.

Likely cause: A lead is trapped under its edge, or a print blob sits in the hook window.

Fix: Never force it. Open it, re-lay the lead in its channel, and clean the hook windows with the flush cutters.

A screw spins without tightening.

Likely cause: The pilot hole is stripped.

Fix: Take the screw out, put a drop of superglue in the hole, let it set for ten minutes, and drive the screw in gently. A heat-set insert is the long-term fix for covers you open often.

The knee pods get hot.

Likely cause: The knees work hard when the robot stands for a long time, and the pods are closed plastic.

Fix: Check with a thermocouple. Stop and let the robot cool above 45 degrees C. Let it sit or lie down between walks.

The robot tips when it lifts a foot.

Likely cause: The weight is not where the design expects: a heavy item moved, or the cells are different.

Fix: Check the hub and the battery sit in their fences. The firmware shifts the body about 8 mm before lifting a front foot; do not lift a foot without that shift.

The battery runs down fast.

Likely cause: Old or mismatched cells, or the servos hold a pose against a stop.

Fix: Charge both cells fully, check they are within 0.05 V, and look for a buzzing servo. Use the spare pair to compare.

9

CHAPTER 9 Appendix

9.1 Bought parts, with store listings

What was bought for this build (shopping list of 26 September 2026). The links go to the listing each part was ordered from; seller photos are not reproduced here.

Part	Qty	Where it goes	Store and listing
☐ Guition ESP32-S3 4.3" touchscreen board (JC4827W543C)	1	Head front, behind the screen window	AliExpress aliexpress.com/item/1005006729377800.html
☐ ESP32-S3 N16R8 camera board + OV5640 camera	1	Muzzle (lens in the left window)	AliExpress aliexpress.com/item/1005009210809394.html
☐ ESP32-S3-DevKitC-1 N16R8	1	Not fitted - bench spare	AliExpress aliexpress.com/item/1005010373962338.html
☐ MG90S metal-gear servo, 180 degrees	10	4 hips, 4 knees, tail, 1 spare	AliExpress aliexpress.com/item/1005006860780474.html
☐ MG90S metal-gear servo, 360 degrees (continuous)	4	Wheels, one in each paw	AliExpress aliexpress.com/item/1005012187908480.html
☐ PCA9685 16-channel servo board	2	Tub, one on each side wall	AliExpress aliexpress.com/item/1005010706769492.html
☐ VL53L0X laser distance sensor	8	7 fitted (chest, rump, 3 desk-edge in the tub floor, 2 sides in the lid) + 1 spare	AliExpress aliexpress.com/item/1005008660383200.html
☐ MS3625 I2S microphone	2	Mic rings on each side of the head front	AliExpress aliexpress.com/item/1005009213592784.html
☐ Cavity speaker 3 W 4 ohm (1.25 mm plug)	1	Head back, between its fences	AliExpress aliexpress.com/item/1005009465741899.html
☐ MPU6050 motion sensor (GY-521)	1	Under the lid, in its fence	AliExpress aliexpress.com/item/1005007169577953.html
☐ APDS-9960 gesture sensor	1	Muzzle, on its shelf behind the right window	AliExpress aliexpress.com/item/1005008247999939.html
☐ MPR121 touch board	1	Under the lid, in its fence	AliExpress aliexpress.com/item/1005010759757874.html
☐ Type-C 2S boost charging board	1	Tub, raised ledge at the rump; socket in the rump opening	AliExpress aliexpress.com/item/1005010706867355.html
☐ 2S 20 A protection board (with balance)	1	Tub floor, between its fences	AliExpress aliexpress.com/item/1005006995181873.html
☐ 2 x 18650 battery holder (series, wired)	1	Battery door	AliExpress aliexpress.com/item/1005007170299017.html
☐ Mini560 buck converter, fixed 5 V 5 A	2	On the perfboard hub under the lid	AliExpress aliexpress.com/item/1005007937920693.html
☐ KCD1 rocker switch	1	Front of the belly, flush in its recess	AliExpress aliexpress.com/item/4000125268992.html
☐ Mini blade fuse holder, 7.5 A, 16 AWG	1	On top of the protection board	AliExpress aliexpress.com/item/1005005625637629.html
☐ M2 self-tapping screw kit (500)	1	Everywhere - see the screw chart	AliExpress aliexpress.com/item/1005006904470760.html
☐ Dupont jumper wires (120)	1	Signal wiring	AliExpress aliexpress.com/item/1005007791660521.html
☐ MX1.25 4-pin pre-wired plugs (10)	1	Plugs for the screen board ports P1, P3, P4	AliExpress aliexpress.com/item/1005007277110532.html

Part	Qty	Where it goes	Store and listing
☐ Perfboard 5 x 7 cm (5)	1	The power and sensor hub under the lid	AliExpress aliexpress.com/item/1005007024264426.html
☐ 3M double-sided foam tape, 20 mm	1	Sensor boards and the hub	AliExpress aliexpress.com/item/1005007010049062.html
☐ Heat-shrink tubing (127)	1	Every soldered joint	AliExpress aliexpress.com/item/1005008728485990.html
☐ 830-point breadboard	1	Tool: bench test	AliExpress aliexpress.com/item/1005007345339483.html
☐ ES02 soldering kit + digital multimeter	1	Tool	AliExpress aliexpress.com/item/1005007170830774.html
☐ PLATO 170 flush cutters	1	Tool	AliExpress aliexpress.com/item/1005004257411917.html
☐ Digital caliper 150 mm	1	Tool: measure the servos before printing	AliExpress aliexpress.com/item/1005005671598813.html
☐ Samsung 35E 18650 cell, 3500 mAh, button top	2	In the battery holder	TinkerTech AU tinkertechau.com.au/products/samsung-35e-18650
☐ Tinker 25R 18650 cell, 2500 mAh (spare pair)	2	Spare pair - never mix with the Samsungs	TinkerTech AU tinkertechau.com.au/products/tinker-25r-18650-
☐ Heavy-duty hook-up wire 7.5 A, red (WH3040)	2 m	Battery and servo power (+)	Jaycar jaycar.com.au/p/WH3040
☐ Heavy-duty hook-up wire 7.5 A, black (WH3041)	2 m	Battery and servo power (-)	Jaycar jaycar.com.au/p/WH3041
☐ 1000 uF 16 V capacitor (RE6220)	2	Hub: one across each Mini560 output	Jaycar jaycar.com.au/p/RE6220
☐ 2.2 kOhm resistors (RR0580)	1 pack	Hub: wheel-stop pull-up on the OE line	Jaycar jaycar.com.au/p/RR0580
☐ 10 kOhm resistors (RR0596)	1 pack	Hub: battery-level divider (top)	Jaycar jaycar.com.au/p/RR0596
☐ 4.7 kOhm resistors (RR0588)	1 pack	Hub: battery-level divider (bottom)	Jaycar jaycar.com.au/p/RR0588
☐ NBR O-ring assortment, OD 21-35 mm x 2.4 mm CS	1	Wheel tyres: use the 31 mm OD rings (about 26.2 mm inside), 4 + spares	AliExpress in store
☐ Superglue 3 g (6 pack)	1	Only where the guide says	Bunnings in store
☐ Cable ties 100 mm (25)	1	Hub anchors, slack loops	Bunnings in store

9.2 Printed parts, colours and plates

Part (file)	Colour	What it holds	Qty	g each	Plate
Torso_Bottom the tub	 brown	4 hip servos, 2 servo boards, protection board, fuse, USB-C charger, power switch, 5 laser sensors	1	49.7	G1
Band_Chest chest band	 brown	front laser window; hides 2 lid screws (snaps on)	1	3.5	G1
Band_Rump rump band	 brown	rear laser window and the USB-C opening; hides 2 lid screws	1	3.4	G1
Side_Panel_Left / _Right side panels	 brown	cover the 2 hip servos on each side (4 snap hooks, no screws)	2	14.9	G2
Battery_Door battery door	 brown	the 2 x 18650 holder and its clamp; 4 screws into the tub	1	11.1	G2

Part (file)	Colour	What it holds	Qty	g each	Plate
Battery_Clamp battery clamp	 brown	holds the cell holder down	1	1.3	G2
Collar collar (neck)	 brown	nothing - cables pass through it; joins the lid to the head	1	12.4	G2
Torso_Lid the lid	 cream	motion sensor, touch board, perfboard hub, tail servo, 2 side lasers, 4 touch pads	1	46.0	W1
Tail_Root tail root	 cream	the tail servo horn and the tail cap	1	1.6	W1
Tail tail	 cream	nothing (keyed onto the root, 1 hidden screw)	1	1.1	W1
Ear_Left / Ear_Right ears	 cream	nothing (1 screw each from inside the head)	2	2.5	W1
Head_Front head front	 cream	4.3 inch screen, 2 microphones, head touch pad, both ears	1	35.8	W2
Muzzle muzzle	 cream	camera board, camera, gesture sensor	1	11.4	W2
Head_Back head back	 cream	the speaker; clips onto the head front (5 snap hooks)	1	31.0	W3
Thigh_<leg>_Outer thigh outer halves	 cream	hip horn, knee servo (2 tab screws), 2 snap hooks	4	7.3	W3 + W4
Shin_<leg>_Outer shin outer halves	 cream	knee horn, wheel servo (2 tab screws), 2 snap hooks	4	9.0	W4
Thigh_<leg>_Cover thigh covers	 brown	the knee servo body (in the knee pod) and its leads	4	7.2	G3
Shin_<leg>_Cover shin covers	 brown	the wheel servo body (in the paw pod), leads, knee lead ring	4	8.1	G4
Wheel_<leg> wheels	 brown	the wheel horn and a 26 x 2.4 mm O-ring tyre	4	2.9	G3 + G4
Cap_Hip_<leg>, Cap_Knee_<leg> hip and knee caps (12 mm)	 black	hide the hip and knee horn screws	8	0.2	B1
Cap_Wheel_<leg> wheel caps (14 mm)	 black	hide the wheel horn screws	4	0.3	B1
Cap_Tail tail cap (12 mm)	 black	hides the tail horn screw	1	0.2	B1
Port_Door_Head, Port_Blank_Head head port door + blank	 black	cover for the screen board USB-C, and its matching blank	2	0.3	B1
Port_Door_Muzzle, Port_Blank_Muzzle muzzle port door + blank	 black	cover for the camera board USB-C, and its matching blank	2	0.4	B1
Screw_Plugs screw plugs (4 in one file)	 black	hide the 4 battery-door screws	1	0.1	B1
Screen_Washers screen washers (4 in one file)	 black	go under the 4 screen screws	1	0.1	B1

9.3 Screws, where each one goes

Every length was checked against the 3D model: each screw reaches at least 2 mm into its hole and stops before the far side of the part.

Where	Count	Screw	Note
Hip servo tabs into the tub wall (from outside, under the side panels)	8	M2x6	servo bag screws fit
Knee and wheel servo tabs into the leg outer halves	16	M2x6	from the tab side into blind holes
Tail servo tabs into the two ribs under the lid	2	M2x6	driven up from inside
PCA9685 boards onto the upper two standoffs each	4	M2x6	the lower standoffs are spacers only
Horn arms into the thighs (hip horns)	8	M2x4	use horn holes 6.7-10 mm from the centre
Horn arms into the shins (knee horns)	8	M2x6	
Horn arms into the wheels	8	M2x6	outer horn holes (7.6 mm or more from the centre)
Horn arms into the tail root	2	M2x4	the root is only 4.6 mm thick
Horn onto each servo spline (13 outputs)	13	horn screw	the servo's own horn screw, then press the cap in
Thigh and shin covers	8	M2x8	thigh: bottom of the knee disc; shin: under the wheel
Lid to tub, through the chest and rump walls	4	M2x6	hidden by the snap-on bands
Lid to collar, from inside the lid	4	M2x8	before the lid goes on the tub
Head to collar, from inside the head	4	M2x8	before the head back goes on
Muzzle to head, from inside the head	4	M2x8	
Screen to the head front, through the 4 printed washers	4	M2x6	
Ears, from inside the head	2	M2x8	
Battery door to tub (heads under the plugs)	4	M2x6	
Battery clamp onto the door posts	2	M2x8	
Touch pads: foil tab and wire clamped under the screw	5	M2x4	4 under the lid, 1 under the head top
Tail onto the tail root, from under the root	1	M2x8	before the root goes onto the servo
Total		M2x4: 15, M2x6: 58, M2x8: 25 + 13 horn screws	the side panels, bands, head back, port doors, caps and plugs have no screws

Actual-size outlines (print at 100 %)



9.4 Firmware pin and channel map

Every pin, address and channel used in this build, in one table. The robot firmware and the servo test sketch ([guide/tools/servo_center](#)) use exactly these values.

What	Pin / address / channel	Note
Brain	Guition JC4827W543C screen board (ESP32-S3)	drives both servo boards and every sensor
I2C SDA	IO17 (port P4)	P4 order not confirmed; the servo sketch also tries SDA=IO18/SCL=IO17
I2C SCL	IO18 (port P4)	100 kHz; one bus, star-wired at the hub
Wheel stop / servo enable	IO14 -> OE of PCA9685 A and B	LOW = outputs on; 2.2 kOhm pull-up to 3.3 V, so reset/crash = all servos off. Never IO46
Battery voltage	IO5	divider 10 kOhm (top) / 4.7 kOhm (bottom) after the switch: $V_{bat} = V_{pin} \times 14.7 / 4.7$; warn 7.0 V, park + servos off 6.8 V
Microphones (I2S, both mics)	SCK IO6, WS IO7, SD IO15 (port P3)	left mic L/R pin to GND (left channel), right mic L/R pin to 3.3 V (right channel)
Speaker	screen board "Speak" port (built-in amp)	no GPIO wiring
Screen board power	port P1: 5 V and GND only	from Mini560 #2; never the BAT port
PCA9685 A	I2C 0x40, left tub wall	no address pads bridged; 50 Hz
PCA9685 B	I2C 0x41, right tub wall	A0 pad bridged; 50 Hz
PCA9685 all-call	0x70	default; ignore (the sketch turns all-call off)
Servo A ch 0 / 1 / 2	hip / knee / wheel, FRONT-LEFT	wheel = MG90S 360 (continuous)
Servo A ch 4 / 5 / 6	hip / knee / wheel, BACK-LEFT	
Servo A ch 8	tail	A ch 3, 7, 9-15 unused
Servo B ch 0 / 1 / 2	hip / knee / wheel, FRONT-RIGHT	
Servo B ch 4 / 5 / 6	hip / knee / wheel, BACK-RIGHT	B ch 3, 7, 8 unused
Laser XSHUT on B ch 9-15	9 front obstacle (chest), 10 rear obstacle (rump), 11 desk-edge front-left, 12 desk-edge front-right, 13 desk-edge rear, 14 side left, 15 side right	PCA9685 full-on = XSHUT high; all start at 0x29; wake one at a time and re-address. OE high also pulls these LOW: re-init the lasers after any OE-high event
MPU6050	I2C 0x68	under the lid; INT not wired
MPR121	I2C 0x5A	E0 back-left, E1 back-right, E2 rump-left, E3 rump-right, E4 head; E5-E11 unused; IRQ not wired (poll)
APDS-9960	I2C 0x39	in the muzzle; INT not wired
Camera board	ESP32-S3 CAM + OV5640, own firmware	5 V + GND from Mini560 #2 only; talks to the laptop over Wi-Fi
Servo pulses	1500 us = centre (90 deg) / wheel stop nominal	per-servo trims and wheel stop points from the calibration table (guide 7.8)
Joint sign	+ = foot forward (toward the head)	left and right legs mirrored; back legs: hip above +25 deg -> knee at most +45 deg

9.5 Wiring tables

Power

	From	To	Wire	Length	Note
P1	Cell holder, black lead (-)	Protection board B-	holder's own lead	as supplied	long enough for the door to sit beside the robot

	From	To	Wire	Length	Note
P2	Metal link between the two cells	Protection board BM (middle)	thin red or white, soldered	10 cm	the "middle wire" - solder it with the cells OUT
P3	Cell holder, red lead (+)	Protection board B+	holder's own lead	as supplied	connect last (negative, middle, positive)
P4	Protection board P+	Fuse holder, lead 1	red, thick	5 cm	7.5 A blade fuse fitted
P5	Fuse holder, lead 2	USB-C charger BAT+ (OUT+)	red, thick	8 cm	the fuse protects the charger too
P6	Fuse holder, lead 2 (same joint as P5)	Power switch (KCD1) terminal 1	red, thick	12 cm	
P7	Protection board P-	USB-C charger BAT- (OUT-)	black, thick	8 cm	
P8	Protection board P-	Hub GND pad	black, thick	25 cm	slack so the lid lifts off
P9	Power switch (KCD1) terminal 2	Hub BATT-SW pad	red, thick	25 cm	slack so the lid lifts off
P10	Hub BATT-SW / GND	Mini560 #1 (servos) IN+ / IN-, Mini560 #2 (boards) IN+ / IN-	red / black, thick	on the hub	7.4 V in
P11	Mini560 #1 OUT+ / OUT-	PCA9685 A and B, green V+ / GND screw terminal	red / black, thick	20 cm each	5 V servo power only
P12	1000 uF capacitor #1	Across Mini560 #1 OUT (+ leg to OUT+)	capacitor legs	on the hub	the stripe marks the - leg
P13	Mini560 #2 OUT+ / OUT-	Screen board port P1: 5 V / GND only	red / black (MX1.25 plug)	30 cm	through the collar; P1 only, never BAT
P14	Mini560 #2 OUT+ / OUT-	Camera board 5V / GND pins	red / black (Dupont)	30 cm	through the collar
P15	1000 uF capacitor #2	Across Mini560 #2 OUT (+ leg to OUT+)	capacitor legs	on the hub	

Signals

	From	To	Wire	Length	Note
S1	Screen board port P4: 3.3 V, GND, IO17, IO18	Hub 3V3, GND, SDA, SCL rails	orange, black, blue, yellow (MX1.25 plug)	30 cm	through the collar; IO17 = SDA, IO18 = SCL (the servo sketch checks)
S2	Hub 3V3 / GND / SDA / SCL	PCA9685 A and B: VCC, GND, SDA, SCL	orange, black, blue, yellow	20 cm each	VCC is 3.3 V logic - NOT 5 V
S3	Hub 3V3 / GND / SDA / SCL	MPU6050 (VCC, GND, SDA, SCL)	orange, black, blue, yellow	10 cm	under the lid
S4	Hub 3V3 / GND / SDA / SCL	MPR121 (3.3V, GND, SDA, SCL)	orange, black, blue, yellow	10 cm	under the lid
S5	Hub 3V3 / GND / SDA / SCL	7 laser sensors (VIN, GND, SDA, SCL)	orange, black, blue, yellow	10-25 cm	2 in the lid, 5 in the tub
S6	Hub 3V3 / GND / SDA / SCL	APDS-9960 gesture sensor (VCC, GND, SDA, SCL)	orange, black, blue, yellow	30 cm	through the collar to the muzzle
S7	Screen board IO14	Hub OE pad -> PCA9685 A OE and B OE	green	30 cm	2.2 kOhm from the OE pad to 3V3 (wheel stop)
S8	Hub BATT-SW -> 10 kOhm -> VBAT pad -> 4.7 kOhm -> GND	Screen board IO5 from the VBAT pad	white	30 cm	reads the battery; after the switch, so it never drains the pack when off
S9	PCA9685 B channels 9-15 (PWM pin)	XSHUT pin of each laser (see the laser table)	purple	10-25 cm	turns the lasers on one at a time at boot
S10	MPR121 E0, E1, E2, E3, E4	Foil pads: back-left, back-right, rump-left, rump-right, head	grey	5-30 cm	the wire is clamped under the pad screw

	From	To	Wire	Length	Note
S11	Screen board port P3: IO6, IO7, IO15	Both mics: SCK (IO6), WS (IO7), SD (IO15); 3.3 V and GND	brown (MX1.25 plug)	10 cm	both mics share all three lines
S12	Left mic L/R pin	GND (right mic L/R pin to 3.3 V)	black / orange	3 cm	tells the two mics apart
S13	Speaker (1.25 mm plug)	Screen board "Speak" port	speaker's own lead	as supplied	never connect a speaker wire to ground
S14	13 servo leads	PCA9685 headers (see the servo table); brown wire on the GND row	servo's own lead	do not cut	wheel lead needs about 200 mm, knee lead about 135 mm

Servo channels

	Servo	Servo board	Channel	Lead route
☐	Hip, front-left	A (left wall, 0x40)	0	straight through the left tub wall
☐	Knee, front-left	A	1	front-left leg, through the side-panel slot
☐	Wheel, front-left (360)	A	2	front-left leg, through the side-panel slot
☐	Hip, back-left	A	4	straight through the left tub wall
☐	Knee, back-left	A	5	back-left leg, through the side-panel slot
☐	Wheel, back-left (360)	A	6	back-left leg, through the side-panel slot
☐	Tail	A	8	down from the lid
☐	Hip, front-right	B (right wall, 0x41)	0	straight through the right tub wall
☐	Knee, front-right	B	1	front-right leg, through the side-panel slot
☐	Wheel, front-right (360)	B	2	front-right leg, through the side-panel slot
☐	Hip, back-right	B	4	straight through the right tub wall
☐	Knee, back-right	B	5	back-right leg, through the side-panel slot
☐	Wheel, back-right (360)	B	6	back-right leg, through the side-panel slot

Laser channels

	Laser	Where	XSHUT to board B channel
☐	Front obstacle	behind the chest band window	9
☐	Rear obstacle	behind the rump band window	10
☐	Desk edge, front-left	tub floor incline, left of the switch	11
☐	Desk edge, front-right	tub floor incline, right of the switch	12
☐	Desk edge, rear	tub floor incline at the rump, centred	13
☐	Side, left	shelf under the lid, left wall	14
☐	Side, right	shelf under the lid, right wall	15

9.6 About this edition

Desk Buddy Assembly Guide v3.1, October 2026. Built from the v3.1 print model ([cad/print_v3_1_README.md](#)), the parts audit of 26 September 2026 and the shopping list of the same day. The pictures are the v3.1 SolidWorks renders and views drawn from the v3.1 STL files. This is the first draft of the official build manual; send corrections to the design lead.